# Dr. Denys Dutykh

CURRICULUM VITÆ

mathematical modelling · free-surface flows · dispersive waves · scientific computing

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| 43 | 145 | 3 | 352 | 235 |
|---|---|---|---|---|
| H-INDEX | I10-INDEX | ERDŐS N$^{O}$ | PUBLICATIONS | COLLABORATORS |

**Keywords** mathematical modelling · free surface flows · hydrodynamics · variational principles · partial differential equations · finite volumes · spectral methods · high-performance scientific computing

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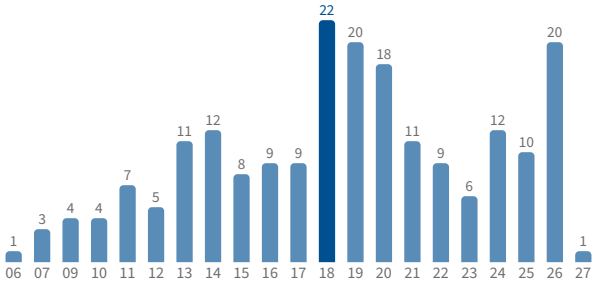
# Detailed Highlights

| 352 | 202 | 5 |
|---|---|---|
| PUBLICATIONS | JOURNAL ARTICLES | BOOKS |
| **235** | **114** | **55** |
| COLLABORATORS | TALKS DELIVERED | STUDENTS SUPERVISED |
| **10** | **43** | **145** |
| PATENTS | H-INDEX | I10-INDEX |

## Publications by type




## Journal articles per year




Active 2006–2027 · peak 2018 (22) · avg 9.6/yr

## Mentoring & academic service

| **Students supervised** (55) | |
|---|---|
| Post-doctoral fellows | **6** |
| PhD students | **15** |
| Master 2 students | **10** |
| Senior research projects | **4** |
| Work-study students | **2** |
| Master 1 students | **11** |
| Other students | **7** |
| Habilitation committees | **2** |
| PhD committees | **10** |

| **Talks delivered** (114) | |
|---|---|
| International conferences | **19** |
| Workshops | **24** |
| National conferences | **2** |
| Summer schools | **4** |
| Seminars | **54** |
| Short courses | **1** |
| Posters | **6** |
| General-audience lectures | **4** |

| **Editorial & service** | |
|---|---|
| Journals refereed for | **112** |
| Conferences refereed for | **7** |
| Editorial boards | **4** |
| Special issues edited | **2** |
| Meetings organized | **16** |
| Software projects | **33** |
| Research visits abroad | **23** |
| Grant-call panels | **10** |

# 1 Curriculum vitæ

## 1.1 Personal information


| | |
|---|---|
| First name | Denys |
| Last name | DUTYKH |
| Date of birth | the 17$^{th}$ August 1982 |
| Place of birth | Pologui, Ukraine |
| Citizenships | French, Ukrainian |
| Family status | married, two children (2017, 2020) |
| Driving licence | Category B (France, UAE) |
| Professional address | Mathematics Department<br>Khalifa University of Science and Technology<br>P O Box 127788<br>AD, Abu Dhabi, UAE |
| Personal address 1 | Saraya Tower, Jabal Murayshid street,<br>Appt. 1601, Al Sa'Adah area, Abu Dhabi<br>AD, United Arab Emirates |
| Personal address 2 | 116 rue de la Digue,<br>73360 Les Échelles<br>France |
| Tel 1 | +971 2 312 45 42 |
| Mobile 1 | +971 5 455 09 167 |
| Mobile 2 | +33 6 52 27 72 53 |
| Professional e-mail | Denys.Dutykh@ku.ac.ae |
| Personal e-mail 1 | Denys.Dutykh@gmail.com |
| Personal e-mail 2 | Denys.Dutykh@crans.org |
| Home page | https://www.denys-dutykh.com/ |
| Public repository | https://github.com/dutykh/ |
| ResearchGate | https://www.researchgate.net/profile/Denys-Dutykh |
| Google Scholar | https://scholar.google.com/citations?user=cvOVca4AAAAJ |
| ORCID | https://orcid.org/0000-0001-5247-2788 |
| ResearcherID | **R-8861-2019** |

# 2 Awards

- Prime d'Excellence Scientifique (PES) attributed by INSMI (C.N.R.S.), 2014 – 2018
- Le Prix La Recherche 2007, nomination "Environment" sponsored by Veolia, research: "Extreme waves: from physics to the effective prevention". The ceremony was held on the 27th of November 2007 at Luxembourg Palace, Paris, France
- Best Student Paper Award at "*The Fifth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory*", Athens, GA, USA, April 16 – 19, 2007

## 2.1 Professional societies memberships

- Society for Industrial and Applied Mathematics (SIAM) Member N° 020992425
  - SIAG on Nonlinear Waves and Coherent Structures membership
- American Mathematical Society (AMS) Member N° DTDNXH
- European Mathematical Society (EMS) Member N° 22680
- European Geosciences Union (EGU) Member N° 525335
- Société de Mathématiques Appliquées et Industrielles (SMAI) Member N° 6908
- Société Mathématique de France (SMF) Member N° 18091
- International Association for Hydro-Environment Engineering and Research (IAHR) Member N° 92055
- International Solar Energy Society (ISES) Professional Silver membership
- GDR CNRS 2948: Groupement de Recherche MOAD (2005 – 2009):
  *MOdélisation, Asymptotique, Dynamique non-linéaire*

# 3 Patents

1. "*Procédé d'estimation d'une tension de claquage d'une cellule photovoltaïque*".
   Patent number and publication date: **WO2024110535 – 2024-05-30**
   https://data.inpi.fr/brevets/WO2024110535/
   Long Ha Duy (CEA), Carlos Cardenas (CEA), Mohamed Amhal and Denys Dutykh (CNRS/USMB)
2. "*Procédé d'estimation d'une tension de claquage d'une cellule photovoltaïque*".
   Patent number and publication date: **EP4376296 – 2024-05-29**
   https://data.inpi.fr/brevets/EP4376296/
   Long Ha Duy (CEA), Carlos Cardenas (CEA), Mohamed Amhal and Denys Dutykh (CNRS/USMB)
3. "*Procédé d'estimation de l'état de charge d'une batterie au plomb en situation d'autodécharge*".
   Patent number and publication date: **EP4198535 – 2023-06-21**
   https://data.inpi.fr/brevets/EP4198535/
   Mikaël Cugnet (CEA), Denys Dutykh (CNRS/USMB), Angel Kirchev (CEA) and Florian Gallois (USMB).
4. "*Procédé d'estimation de l'état de charge d'une batterie au plomb en situation d'autodécharge*".
   Patent number and publication date: **FR3130999 – 2023-06-23**

https://data.inpi.fr/brevets/FR3130999/
Mikaël Cugnet (CEA), Denys Dutykh (CNRS/USMB), Angel Kirchev (CEA) and Florian Gallois (USMB).

5. "*Method for determining the state of a system and device implementing said methods*".
US Patent number and publication date: **2022/0405346 A1 – 2022-12-22**
https://patentimages.storage.googleapis.com/53/b9/14/f5d40fbcc8116f/US20220405346A1.pdf
Sylvain Lespinats (CEA), Benoît Colange (CEA), Denys Dutykh (CNRS/USMB) and Laurent Vuillon (USMB).
6. "*Procédé de détermination de l'état d'un système et dispositif mettant en oeuvre lesdits procédés*".
Patent number and publication date: **CN114902214A – 2022-08-12**
https://worldwide.espacenet.com/publicationDetails/biblio?CC=CN&NR=114902214&KC=A#
Sylvain Lespinats (CEA), Benoît Colange (CEA), Denys Dutykh (CNRS/USMB) and Laurent Vuillon (USMB).
7. "*Procédé de détermination de l'état d'un système et dispositif mettant en oeuvre lesdits procédés*".
Patent number and publication date: **EP4049148A1 – 2022-08-12**
https://data.inpi.fr/brevets/EP4049148
Sylvain Lespinats (CEA), Benoît Colange (CEA), Denys Dutykh (CNRS/USMB) and Laurent Vuillon (USMB).
8. "*Procédé d'analyse et procédé de détermination et de prédiction du régime de fonctionnement d'un système énergétique*".
Patent number and publication date: **FR3099596 – 2021-02-05 (BOPI 2021-05)**
https://data.inpi.fr/brevets/FR3099596
Sylvain Lespinats (CEA), Benoît Colange (CEA), Julien Berger (USMB), Denys Dutykh (CNRS/USMB) and Hugo Geoffroy (USMB).
9. "*Procédé de détermination de l'état d'un système et dispositif mettant en oeuvre lesdits procédés*".
Patent number and publication date: **WO2021078712A1 – 2021-04-29**
https://data.inpi.fr/brevets/WO2021078712
Sylvain Lespinats (CEA), Benoît Colange (CEA), Denys Dutykh (CNRS/USMB) and Laurent Vuillon (USMB).
10. "*Procédé de détermination de l'état d'un système et dispositif mettant en oeuvre lesdits procédés*".
Patent number and publication date: **FR3102263A1 – 2021-04-23 (BOPI 2021-16)**
https://data.inpi.fr/brevets/FR3102263
Sylvain Lespinats (CEA), Benoît Colange (CEA), Denys Dutykh (CNRS/USMB) and Laurent Vuillon (USMB).

# 4 Entrepreneurship

Chief Scientific Advisor at Causal Dynamics, a startup founded by Drs. Ashkan Rafiee and Nitin Repalle at Nedlands (Perth area), Western Australia, Australia — *Sep 2023 – Aug 2024*

# 5 Professional experience

| | |
|---|---|
| Associate Dean of Graduate Studies, College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates | *Jan 2026 – present* |
| Associate Professor at the Mathematics Department, Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates | *Aug 2022 – present* |
| On leave (*en détachement*) from C.N.R.S. to Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates | *Aug 2022 – Jan 2028* |
| Chargé de recherche C.N.R.S. of the 1$^{st}$ class (formerly CR1, presently CN – Classe Normale) at INSMI affiliated with the Laboratory of Mathematics (LAMA – UMR 5127), University Savoie Mont Blanc, France | *Nov 2012 – Jul 2022* |
| Senior Research Fellow (on temporal leave from C.N.R.S.) at the School of Mathematics and Statistics, University College Dublin, Ireland (working on ERC AdGr "MULTIWAVE" project) | *Sep 2012 – Dec 2013* |
| Chargé de recherche C.N.R.S. of the 2$^{nd}$ class (CR2, titulaire) at INSMI affiliated with the Laboratory of Mathematics (LAMA – UMR 5127), University of Savoie, France | *Oct 2009 – Oct 2012* |
| Chargé de recherche C.N.R.S. stagiaire at INSMI affiliated with the Laboratory of Mathematics (LAMA – UMR 5127), University of Savoie[1], France | *Oct 2008 – Sep 2009* |
| Post-doctoral fellow[2] at LRC Méso CEA DAM/CMLA under the direction of Frédéric Dias and Jean-Michel Ghidaglia | *Dec 2007 – Sep 2008* |
| PhD student and Teaching Assistant (*Moniteur*) at Centre de Mathématiques et de Leurs Applications (CMLA UMR 8536), École Normale Supérieure de Cachan, France | *Oct 2005 – Dec 2007* |

# 6 Education and Training

| | |
|---|---|
| Habilitation à Diriger des Recherches defended at the Laboratory of Mathematics (LAMA), University of Savoie[3]. Title: "*Mathematical modeling in the Environment*" | *Dec 2010* |
| PhD in Applied Mathematics at CMLA, Ecole Normale Supérieure de Cachan. Advisor: Professor Frédéric Dias. Title: "*Mathematical modeling of tsunami waves*" | *Oct 2005 – Dec 2007* |

[1] Since the 27$^{th}$ of May 2014, University of Savoie changed the name to University Savoie Mont Blanc

[2] This fellowship was basically funded by the 3$^{rd}$ year of my PhD thesis scholarship since the thesis was defended within two years.

[3] Nowadays, University of Savoie Mont Blanc

Master Degree in "*Numerical methods for continuum mechanics models*", Ecole Normale Supérieure de Cachan, rank: **1**/10 *Oct 2004 – Jul 2005*

- Research dissertation: "*Moving load on a layered floating ice sheet*" under the supervision of Frédéric Dias. Grade: **19**/20

Master's Degree in Mathematical Modelling, Faculty of Applied Mathematics, Oles Honchar National University, Dnepropetrovsk, Ukraine *Sep 2003 – Jun 2004*

- Research dissertation: "*Harmonic oscillations of an inhomogeneous elastic layer under the action of a stamp*" under the supervision of Vladimir Lamzyuk. Grade: **5**/5

Bachelor's Degree in Applied Mathematics at the Faculty of Applied Mathematics, Oles Honchar National University, Dnepropetrovsk, Ukraine *Sep 1999 – Jun 2003*

- Research dissertation: "*Harmonic oscillations of an inhomogeneous elastic layer*" under the supervision of Vladimir Lamzyuk. Grade: **5**/5

School N° 23, class specialized in physics, Dnepropetrovsk, Ukraine *Sep 1997 – May 1999*

School N° 83, Dnepropetrovsk, Ukraine *Sep 1989 – May 1997*

## 6.1 Academic qualifications

**Remark 1** *Qualification in France is an official permission to apply for Professor and Assistant Professor positions in the national education system. All demands are examined once per year by "Conseil National des Universités" (CNU).*

| *Section* | *Grade* | *Validity* | *Field* |
|---|---|---|---|
| 26 | Professor | 2011 – 2015 | Applied Mathematics |
| 26 | Professor | 2015 – 2019 | Applied Mathematics |
| 37 | Professor | 2011 – 2015 | Physical Oceanography |
| 26 | Assistant Professor | 2008 – 2012 | Applied Mathematics |
| 60 | Assistant Professor | 2008 – 2012 | Mechanics |

## 6.2 Research visits abroad

Visitor at Marine Systems Institute, Tallinn University of Technology, Tallinn, Estonia *Nov 2018*

Visitor at Marine Systems Institute, Tallinn University of Technology, Tallinn, Estonia *Oct 2017*

Visitor at Victoria University of Wellington, School of Mathematics and Statistics, New Zealand *Mar 2017*

Visitor at Al-Farabi Kazakh National University, Faculty of Mechanics and Mathematics, Almaty, Kazakhstan *Feb 2017*

Visitor at Pontifical Catholic University of Paraná, Laboratório de Sistemas Térmicos (LST), Curitiba, Brazil *Apr 2016*

Visitor at Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania *Feb 2016*

Visitor at Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania *Nov 2015*

| | |
|---|---|
| Visitor at Institute of Computational Technologies, Siberian Branch of RAS, Novosibirsk, Russian Federation | *Oct 2015* |
| Visiting research fellow at the Basque Center for Applied Mathematics (BCAM), Bilbao, Spain | *Feb 2015* |
| Visitor at Al-Farabi Kazakh National University, Faculty of Mechanics and Mathematics, Almaty, Kazakhstan | *Dec 2014* |
| Visitor at the Johannes Kepler Universität Linz, Institut für Analysis, Austria | *Oct – Nov 2014* |
| Visitor at RIMS (Kyoto University) and Keio University, Japan | *Jul 2014* |
| Visitor at the Johannes Kepler Universität Linz, Institut für Analysis, Austria | *Apr 2014* |
| Visitor at the Fields Institute (Toronto, Canada) in the framework of the Thematic Program on the Mathematics of Oceans | *May 2013* |
| Visitor at the Georgia Institute of Technology, School of Electrical and Computer Engineering, Atlanta, Georgia | *Apr 2012* |
| Visiting research fellow at the Basque Center for Applied Mathematics (BCAM), Bilbao, Spain | *Mar 2012* |
| Applied Mathematics Department, University of Valladolid, Spain | *Feb 2012* |
| Fields Institute, University of Toronto, Canada | *Jun 2011* |
| Department of Mathematics, University of Bergen, Norway | *Oct 2011* |
| Applied Mathematics Department, University of Valladolid, Spain | *Jul 2011* |
| Wolfgang Pauli Institute, Vienna, Austria | *May 2010* |
| School of Mathematical Sciences, University College Dublin, Ireland | *Mar 2010* |
| Wolfgang Pauli Institute, Vienna, Austria | *Sep 2009* |

# 7 Other qualifications and skills

## 7.1 Continuous professional education

| | |
|---|---|
| Completion of the CITI Program on “Responsible Conduct of Research in Physical Sciences” | *Jan 2023* |

## 7.2 Computer skills

| | |
|---|---|
| Programming languages | C/C++, Fortran, Pascal |
| Script languages | Python, Matlab |
| Operating systems | Linux/Unix, Windows, Dos |
| Math software | Maple, MatLab, Mathematica, Scilab, Octave, Maxima |
| FEM | FreeFem++, FreeFEM3D |

| | |
|---|---|
| Scientific libraries | OpenFOAM, Deal.II, libMesh, gmm++, blitz++, gsl, FFTW |
| Meshers | GiD, GMSH |
| Visualisation | ParaView, OpenDX, MatLab, gnuplot |
| Office | LaTeX, OpenOffice, AbiWord |

## 7.3 Foreign language skills

| | |
|---|---|
| Ukrainian 🇺🇦 | native language |
| Russian 🇷🇺 | native language |
| French 🇫🇷 | almost native language |
| English 🇬🇧 | fluent |
| Italian 🇮🇹 | basic knowledge |
| Arabic 🇦🇪 | basic knowledge (Emarat dialect) |

# 8 Research activities

## 8.1 Scientific interests

- Broadly, my scientific interests can be described by the following categories:
  - Machine Learning
    - dimensionality reduction methods
    - visual data exploration
  - Fluid mechanics
    - free surface flows
    - models in shallow and deep waters
    - variational methods and geometric mechanics
    - water wave run-up
    - compressible and two-phase flows
  - Heat and Mass Transfer in porous materials
  - Solid mechanics
    - co-seismic displacements computation
    - theory and dynamics of dislocations
    - sources and propagation of seismic waves
  - Numerical methods and scientific computing
    - finite volumes
    - finite elements

- pseudo-spectral methods
- geometric integration methods

- More specifically, here are some current areas of my active research:

  **Shallow waters** Quest for improved shallow water models (dispersive effects, large bathymetry variations). Focusing and resonant effects during wave/wall and wave/beach interactions

  **Deep waters** Quest for integrable models. Computation of breathers in higher-order models.

  **Full Euler** Development of fast and arbitrarily accurate algorithms for the computation of regular and singular travelling gravity and capillary-gravity wave solutions. Direct simulation of the free surface Euler dynamics

  **Tsunami generation** Study of the energy transfer from the bottom motion to water waves. Construction of realistic co-seismic bottom displacements during underwater earthquakes

  **Geometric integration** Design and practical assessment of symplectic and multi-symplectic schemes performance in the long time integration of dispersive PDEs

  **Numerics** Development of higher order finite volume, finite element and spectral methods for dispersive wave equations

  **Solitonic gases** Direct simulation of solitonic gases. Verification and validation of the kinetic approach to solitonic gas modelling

- The present list can evolve depending on new contacts that I will make in the future.

## 8.2 Theses

### 8.2.1 Habilitation thesis

**Habilitation à Diriger des Recherches** in Applied Mathematics

| | |
|---|---|
| TITLE: | "*Mathematical modeling in the Environment*" |
| ADVISOR: | Didier BRESCH (DR CNRS, University of Savoie) |
| MANUSCRIPT: | http://tel.archives-ouvertes.fr/tel-00542937/ |

**Habilitation was defended on 3rd December 2010 at the University of Savoie after a review by**:

- Benoît DESJARDINS (Associated Professor, ENS Ulm),
- Florian DE VUYST (Professor, ENS de Cachan),
- Christian KHARIF (Professor, École Centrale de Marseille),
- Paul MILEWSKI (Professor, University of Wisconsin, Madison)

**Committee composition**:

| | | |
|---|---|---|
| Didier BRESCH | DR CNRS, Univ. Savoie | Examinator |
| Thierry COLIN | Professor, Univ. Bordeaux | President |
| Benoît DESJARDINS | Associated Professor, ENS Ulm | Referee |
| Florian DE VUYST | Professor, ENS de Cachan | Referee |
| Frédéric DIAS | Professor, Univ. College Dublin | Examinator |
| Christian KHARIF | Professor, Centrale Marseille | Referee |
| David LANNES | DR CNRS, ENS Ulm | Examinator |
| Paul MILEWSKI | Professor, Univ. Wisconsin | Referee |

**Abstract.** The present manuscript is devoted to the mathematical modelling of several environmental

problems ranging from water waves to powder-snow avalanches. This Habilitation is organized globally in three parts. The first part is essentially introductive and also contains a complete description of my scientific activities.

Scientific works dealing with water waves are regrouped in Part II. The spectrum of covered topics is large. We start by proposing in Chapter 3 a generalized Lagrangian for the water wave problem. This generalization allows for easy and flexible derivation of approximate models in shallow, deep and intermediate waters. Some questions of viscous wave damping are also investigated in the same chapter. Chapter 4 is entirely devoted to various aspects of tsunami wave modelling. We investigate the complete range of physical processes from generation through energy transformations and propagation up to the run-up onto coasts. The next Chapter 5 is devoted specifically to the numerical simulation and mathematical modelling of the inundation phenomena. This question is studied using various approaches: Nonlinear Shallow Water Equations (NSWE) solved analytically and numerically, Boussinesq-type systems, and two-fluid Navier-Stokes equations.

In Part III, we investigate two important questions belonging to the field of multi-fluid flows. Chapter 6 is essentially devoted to the formal justification of the single-velocity two-phase model proposed earlier for aerated flow modelling. Several numerical results are presented as well. Moreover, similar analytical computations performed in a simpler barotropic setting are provided in Appendix A. These results could apply, for example, to the simulation of violent wave breaking.

Finally, in Chapter 7, we propose a novel model for powder-snow avalanche flows. This system is derived from classical bi-fluid Navier-Stokes equations and has several nice properties. Numerical simulations of the avalanche interaction with obstacles are also presented.

**Keywords**: free surface flows, variational methods, finite volumes, dispersive waves, runup, two-phase flows, snow avalanches

### 8.2.2 PhD thesis

**PhD degree from École Normale Supérieure de Cachan in Applied Mathematics**

Title: "*Mathematical modeling of tsunami waves*"
Advisor: Frédéric Dias (Professor, ENS de Cachan)
Manuscript: http://tel.archives-ouvertes.fr/tel-00194763/

**Dissertation defended on 3rd December 2007 at École Normale Supérieure de Cachan after a review by**:

- Jean-Claude Saut (Professor, University Paris-Sud, Orsay),
- Didier Bresch (DR CNRS, University of Savoie),

**Committee composition**:

| | | |
|---|---|---|
| Jean-Michel Ghidaglia | Professor, ENS de Cachan | Examinator |
| Jean-Claude Saut | Professor, Paris-Sud | Referee & President |
| Didier Bresch | DR CNRS, University of Savoie | Referee |
| Costas Synolakis | Professor, USC | Examinator |
| Vassilios Dougalis | Professor, University of Athens | Examinator |
| Daniel Bouche | HDR, CEA/DAM IdF | Invited member |
| Frédéric Dias | Professor, ENS de Cachan | Advisor |

USC = University of Southern California

**Abstract.** This thesis is devoted to tsunami wave modelling. The life of tsunami waves can be conditionally divided into three parts: generation, propagation and inundation (or run-up). In the first part of the manuscript, we consider the generation process of such extreme waves. We examine various existing approaches to its modelling. Then, we propose a few alternatives. The main conclusion is that the seismology/hydrodynamics coupling is poorly understood at the present time.

The second chapter essentially deals with Boussinesq equations, which are often used to model tsunami propagation and sometimes even run-up. More precisely, we discuss the importance, nature and inclusion of dissipative effects in long-wave models.

In the third chapter, we slightly change the subject and turn to two-phase flows. The main purpose of this chapter is to propose an operational and simple set of equations in order to model wave impacts on coastal structures. Another important application includes wave sloshing in liquified natural gas carriers. Then, we discuss the numerical discretization of governing equations in the finite volume framework on unstructured meshes.

Finally, this thesis deals with a topic which should be present in any textbook on hydrodynamics, but it is not. We mean visco-potential flows. We propose a novel and sufficiently simple approach for weakly viscous flow modelling. We succeeded in keeping the simplicity of the classical potential flow formulation with the addition of viscous effects. In the case of finite depth, we derive a correction term due to the presence of the bottom boundary layer. This term is nonlocal in time. Hence, the bottom boundary layer introduces a memory effect to the governing equations.

**Keywords**: Water waves, tsunami generation, Boussinesq equations, two-phase flows, visco-potential flows, finite volumes

## 8.3 List of present and past collaborators:

The total number: **235** *(in alphabetical order)*:

The dagger † marks colleagues who have passed away (*in memoriam*).

**Obinna Abah** School of Mathematics, Statistics and Physics, Newcastle University, Newcastle upon Tyne, United Kingdom

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## 8.4 Editorial boards membership

- Journal of Ocean Engineering and Marine Energy
- Gulf Journal of Mathematics
- Geosciences
- Computational Technologies

## 8.5 Publications

- My ERDŐS number: **3**
- My $h$ —index[4]: **43**
- My $i10$ —index: **145**
- My ORCID number: 0000-0001-5247-2788
- Citations statistics: https://scholar.google.com/citations?user=cvOVca4AAAAJ

### 8.5.1 Preprints under review

22. **D. Dutykh** and L. Vuillon. *Euclidean minimality of Markov representations on thirteen Farey rows*. Submitted, 2026
21. **D. Dutykh** and L. Vuillon. *Rational reflections and palindromic words in the Cohn monoid*. Submitted, 2026
20. D. Batic and **D. Dutykh**. *Negative-energy bounds and outward communication near Schwarzschild black holes*. Submitted, 2026
19. D. Batic and **D. Dutykh**. *Perfect Wave Transfer beyond Conformal Invariance*. Submitted, 2026
18. **D. Dutykh** and L. Vuillon. *Effective trace bounds and rigidity for simple curves on the modular torus*. Submitted, 2026
17. **D. Dutykh**. *One exponent, two rates: the universality and the non-identifiability of Zipf laws*. Submitted, 2026
16. D. Batic, **D. Dutykh** and J.C.M. Hughes. *Gravitational instantons under nonlinear curvature actions*. Submitted, 2026
15. **D. Dutykh** and L. Vuillon. *Endpoint costs and balanced near-minimizers under Gaussian reflections*. Submitted, 2026
14. **D. Dutykh** and L. Vuillon. *Fibonacci–Pell nearest gaps: an all-exponent classification and quadratic-unit orbit rigidity*. Submitted, 2026
13. D. Batic and **D. Dutykh**. *Local realizability and global completion of constant-$\ell$ rotating dust*. Submitted, 2026
12. D. Batic, **D. Dutykh** and M. Sukaiti. *Schwarzschild spectral ladders on the negative imaginary axis: Endpoint nonselection, branch-cut phase, and Jost classification*. Submitted, 2026 https://arxiv.org/abs/2608.29095/
11. D. Batic and **D. Dutykh**. *Perfect Wave Transfer in Nonconformal Tomonaga–Luttinger Liquids*. Submitted, 2026
10. D. Batic and **D. Dutykh**. *Two-Channel Dollard Scattering and Partial-Wave Resonances for Massive Dirac Fields on Kerr Black Holes*. Submitted, 2026
9. D. Batic and **D. Dutykh**. *Timelike Growth and the Absence of Tempered Fundamental Solutions for a Lorentzian Noncommutative-Geometry-Inspired Wave Operator*. Submitted, 2026
8. S. Mukherjee, L. Bou Nassif, I. Tsanakas, H. Colin, S. Giroux-Julien, H. Pabiou, **D. Dutykh** and L. Vuillon. *Physics-Informed Graph Learning for Photovoltaic Power and Module Temperature Modelling*. Submitted, 2026
7. D. Batic and **D. Dutykh**. *Spectral and Metric Foundations for Cubic Ghost Quantisation*. Submitted, 2026

[4] The $h-$ and $i10$ —indices are retrieved from Google Scholar server.

6. D. Batic and **D. Dutykh**. *Global and dynamical viability of the T-duality-inspired Bardeen regular black hole*. Submitted, 2026
5. R. El Cheikh, A. Cheviakov, **D. Dutykh** and D. Mitsotakis. *Nonlinear Dynamics of Pulsatile Blood Flow in Viscoelastic Vessels: A Dispersive Wave Approach*. Submitted, 2026
4. D. Batic, A. Chrysostomou, A. Cornell and **D. Dutykh**. *Quasinormal modes of Schwarzschild de Sitter black holes: spectral branches and the Nariai limit*. Submitted, 2026
3. D. Batic, **D. Dutykh** and M. Sukaiti. *A no-go result for regular GR-connected isotropic Morris–Thorne wormholes in $f(R,T) = R + \sigma RT$ gravity*. Submitted, 2026
2. D. Batic, **D. Dutykh**. *Spectral Approximation Theory for Quasi-Normal Modes*. Submitted, 2026
1. D. Batic, **D. Dutykh** and J. Faraji. *Spectral Method for Eigenvalue Problems in Quantum Mechanics*. Submitted, 2026

### 8.5.2 Books

5. D. Clamond, **D. Dutykh** and D. Mitsotakis. *Variational approach to water wave modelling*, IAHR WATER MONOGRAPHS, International Association for Hydro-Environment Engineering and Research (IAHR), Spain, 2024. ISBN: 978-90-833476-6-0
   https://www.iahr.org/library/infor?pid=29802/
4. S. Lespinats, B. Colange and **D. Dutykh**. *Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach*, Springer, 2022. ISBN: 978-3-030-81025-2
   https://www.springer.com/gp/book/9783030810252/
3. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and O. Gusev. *Dispersive Shallow Water Waves: Theory, Modeling, and Numerical Methods*, 1$^{st}$ Ed., Birkhäuser Basel, Springer Nature Switzerland AG, 263 pp., 2020. ISBN: 978-3-030-46266-6
   https://www.springer.com/gp/book/9783030462666/
2. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Numerical methods for diffusion phenomena in building physics: a practical introduction*, 2$^{nd}$ Ed., Springer, Cham, Switzerland, 251 pp., 2020. ISBN: 978-3-030-31573-3
   https://www.springer.com/gp/book/9783030315733/
1. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Numerical methods for diffusion phenomena in building physics*, PUCPRESS, Curitiba, Brazil, 224 pp., 2017. ISBN: 978-8-568-32488-2
   https://books.google.com/books?id=KNcuDwAAQBAJ

### 8.5.3 International peer-reviewed journals

**— 2027 —**

202. A. Deeb and **D. Dutykh**. *Error estimation for numerical approximations of ODEs via composition techniques. Part I: One-step methods*. **Comput. Appl. Math.**, **46**, 33, 2027
   https://doi.org/10.1007/s40314-026-03864-5
   https://arxiv.org/abs/2409.10548/

**— 2026 —**

201. D. Batic, A. Cornell and **D. Dutykh**. *Scalar quasinormal modes of Schwarzschild–anti-de Sitter black holes: spectral analysis and a formal coefficient deformation*. Accepted to **Eur. Phys. J. C**, 2026
https://arxiv.org/abs/2607.15731/
200. D. Batic and **D. Dutykh**. *Quasinormal modes of Hayward black holes from Schwarzschild to extremality: A comparative spectral analysis*. **Phys. Dark Universe**, **54**, 102467, 2026
https://doi.org/10.1016/j.dark.2026.102467
https://arxiv.org/abs/2609.33267/
199. M. Zafar, P. Sabatier, **D. Dutykh**, W. Rapuc, R. Paris, A. Piccin, P. Vuillemin and F. Anselmetti. *Human-enhanced tsunami hazard in large lakes*. **Nat. Commun.**, **17**, 10531, 2026
https://doi.org/10.1038/s41467-026-78272-8
198. D. Batic and **D. Dutykh**. *Spectral Green functions for T-duality-regularised electrostatics in finite and bounded planar geometries*. Accepted to **Proc. R. Soc. A**, 2026
https://doi.org/10.1098/rspa.2026.0600
197. S. Mukherjee, L. Vuillon, **D. Dutykh** and I. Tsanakas. *Scalable weather data reduction for solar PV analysis using graph-based approach*. Accepted to **Energy Systems**, 2026
https://hal.science/hal-05235714/
196. D. Batic, A. Chrysostomou, A. Cornell and **D. Dutykh**. *Quasinormal-mode analysis of a massive scalar field in a Schwarzschild background via the Spectral Method*. **Phys. Rev. D**, **114**, 065020, 2026
https://doi.org/10.1103/kv2k-1d3m
https://arxiv.org/abs/2609.37942/
195. M. Sukaiti, D. Batic and **D. Dutykh**. *An exact Kerr-like rotating Morris–Thorne wormhole: Throat, ergoregion and equatorial shadow slice*. **Ann. Phys.**, **495**, Part 1, 170718, 2026
https://doi.org/10.1016/j.aop.2026.170718
https://arxiv.org/abs/2609.22823/
194. D. Batic and **D. Dutykh**. *Quasinormal Modes of Gauss–Bonnet Black Holes via the Spectral Method: Scalar, Vector, and Tensor Perturbations*. **Phys. Rev. D**, **114**, 044015, 2026
https://doi.org/10.1103/91q6-r3jd
https://arxiv.org/abs/2608.06083/
193. D. Batic, **D. Dutykh** and M. Sukaiti. *Quasinormal modes of the Kazakov–Solodukhin quantum-corrected black hole: a spectral analysis*. **Eur. Phys. J. C**, **86**, 847, 2026
https://doi.org/10.1140/epjc/s10052-026-16102-3
https://arxiv.org/abs/2607.19860/
192. D. Batic, **D. Dutykh** and F. Scardigli. *Quasinormal modes of Bonanno–Reuter black holes via the Spectral Method*. **Phys. Rev. D**, **114**, 024023, 2026
https://doi.org/10.1103/sgt6-kwfl
https://arxiv.org/abs/2607.10199/
191. M.K. Andaristiyan, A. Machmudah, R.E. Poetro, W. Hasbi and **D. Dutykh**. *Future Subsequent Ballistic Coefficient Estimation Using Deep Learning for Uncontrolled Reentry Object Prediction*. **Acta Astronaut.**, **248**, 736–749, 2026
https://doi.org/10.1016/j.actaastro.2026.06.022
190. A. Rashidi, **D. Dutykh** and M. Mokhtari. *Primarily tsunami modeling of the Mw 8.8 Kamchatka Peninsula earthquake on July 29, 2025*. **Nat. Hazards**, **122**, 450, 2026

189. D. Batic, C. Boehmer and **D. Dutykh**. *Energy conditions in consistent perfect fluid cosmology*. **Ann. Phys.**, **491**, 170517, 2026
http://arxiv.org/abs/2605.14479/
188. D. Batic, **D. Dutykh** and M. Sukaiti. *Exact solutions for slowly rotating wormholes in the presence of an anisotropic fluid*. **Class. Quantum Grav.**, **43**(9), 095007, 2026
http://arxiv.org/abs/2605.02555/
187. A. Deeb, **D. Dutykh** and M. Al Zohbi. *Error estimation for numerical approximations of ODEs via composition techniques. Part II: Backward difference formulas*. **Comput. Appl. Math.**, **45**, 355, 2026
http://arxiv.org/abs/2605.06735/
186. D. Batic, **D. Dutykh** and M. Sukaiti. *Noncommutative geometry-inspired wormholes supported by quasi-de Sitter and Chaplygin-like equations of state*. **Nucl. Phys. B**, **1025**, 117414, 2026
http://arxiv.org/abs/2603.26257/
185. M. Hunt and **D. Dutykh**. *Free Surface Waves in Electrohydrodynamics with Prescribed Vorticity Distribution*. **Math. Methods Appl. Sci.**, **49**(7), 7068–7081, 2026
184. D. Batic, **D. Dutykh** and M. Sukaiti. *Exact Spinning Morris–Thorne Wormhole: Causal Structure, Shadows, and Multipole Moments*. **Eur. Phys. J. C**, **86**, 179, 2026
http://arxiv.org/abs/2602.21906/
183. D. Batic, **D. Dutykh** and F. Scardigli. *Spectral Analysis of Quasinormal Modes of Planck Stars*. **Eur. Phys. J. C**, **86**, 165, 2026
http://arxiv.org/abs/2602.19833/
182. **D. Dutykh**, R. Escobedo and L. Spector. *Modeling How Hunting Strategies and Pack Size Shape Each Other*. **J. Theor. Biol.**, **623**, 112390, 2026
https://hal.archives-ouvertes.fr/hal-01182799/

**— 2025 —**

181. M. Zafar, **D. Dutykh**, R. Paris, R. Gastineau, J. De Sigoyer, E. Duarte, N. Ünsal, N. Harrichhausen, S. Falvard, M. Sahin and P. Sabatier. *Strike-Slip Fault-Generated Paleotsunamis in Lake Iznik (NW Türkiye): Numerical Modeling Corroborated by Coastal Deposits*. **Geophys. Res. Lett.**, **52**(18), e2025GL117422, 2025
https://hal.science/hal-05293993/
180. D. Batic, **D. Dutykh**, and Z. Babou. *Quasinormal modes of noncommutative geometry-inspired dirty black holes*. **Proc. R. Soc. A**, **481**(2318), 20250021, 2025
https://arxiv.org/abs/2507.19107/
179. S. Savvopoulos, A. Al-Aghbari, K. El Kadi, **D. Dutykh** and I. Janajreh. *Advancing Freeze Desalination through Ultrasound-Enhanced Modelling: Case Studies and Insights for Commercial Applications*. **Case Stud. Therm. Eng.**, **73**, 106518, 2025
178. M. Zafar, P. Sabatier, **D. Dutykh**, H. Jomard, W. Rapuc, P. Lajeunesse and E. Chapron. *Modelling earthquake-induced seiche process and subsequent homogenite deposits in lacustrine setting*. **Earth Planet. Sci. Lett.**, **660**, 119348, 2025
177. D. Batic, **D. Dutykh** and J. Beek. *A Spectral Approach for Quasinormal Frequencies of Noncommutative Geometry-inspired Wormholes*. **Class. Quantum Gravity**, **42**(8), 085003, 2025
https://arxiv.org/abs/2504.02370/

176. M. Amazioug, J.-X. Peng, **D. Dutykh** and M. Asjad. *Emergence and enhancement of feedback control induced quantum entanglement*. **Eur. Phys. J. Plus**, **140**, 132, 2025
https://arxiv.org/abs/2311.06578/
175. A. Deeb and **D. Dutykh**. *Numerical integration of Navier–Stokes equations by time series expansion and stabilized FEM*, **Math. Comput. Simul.**, **233**, 208–236, 2025
https://hal.science/hal-04932473/
174. D. Batic and **D. Dutykh**. *Instability Analysis of Massive Static Phantom Wormholes via the Spectral Method*. **Eur. Phys. J. C**, **85**, 144, 2025
https://arxiv.org/abs/2502.05486/
173. **D. Dutykh** and E. Leichtnam. *On complex algebraic singularities of some genuinely nonlinear PDEs*. **Rend. Mat. Appl.** (7), **46**(1-2), 63–150, 2025
https://hal.science/hal-03372673/
172. **D. Dutykh** and Ya. Prykarpatsky. *On integrability of a new dynamical system associated with the BBM-type hydrodynamic flow*, **Math. Methods Appl. Sci.**, **48**(1), 107–121, 2025
https://arxiv.org/abs/2407.07900/

**— 2024 —**

171. A. Rashidi, M. Mokhtari, **D. Dutykh**, M. Masoodi, P. Faridi and S. Kiani. *Researching Tsunami Hazards in Makran: Insights into Challenges and Complex Tsunamis*. **J. Seismol. Earthq. Eng.**, **26**(4), 71–85, 2024
170. D. Batic, **D. Dutykh** and B.L. Giacchini. *A unified spectral approach for quasinormal modes of Lee–Wick black holes*, **Phys. Rev. D**, **110**, 084032, 2024
https://arxiv.org/abs/2410.13245/
169. D. Batic and **D. Dutykh**. *A Unified Spectral Approach for Quasinormal Modes of Morris–Thorne Wormholes*. **Class. Quantum Gravity**, **41**(21), 215003, 2024
https://arxiv.org/abs/2410.05979/
168. P. Djorwé, M. Asjad, Y. Pennec, **D. Dutykh** and B. Djafari-Rouhani. *Parametrically enhancing sensor sensitivity at an exceptional point*. **Phys. Rev. Res.**, **6**, 033284, 2024
https://arxiv.org/abs/2312.05057/
167. F. Carbone and **D. Dutykh**. *Route to chaos and resonant triads interaction in a truncated Rotating Nonlinear shallow-water model*, **PLoS One**, **19**(8), e0305534, 2024
https://hal.science/hal-04670019/
166. C. Edet, M. Asjad, **D. Dutykh**, N. Ali and O. Abah. *Entropy production rate and correlations of cavity magnomechanical system*. **Phys. Rev. Res.**, **6**, 033037, 2024
https://arxiv.org/abs/2401.16857/
165. A. Deeb and **D. Dutykh**. *Stabilized Time Series Expansions for High-Order Finite Element Solutions of Partial Differential Equations*. **Stud. Appl. Math.**, **153**(2), e12708, 2024
https://hal.science/hal-04582473/
164. C. Cárdenas-Bravo, **D. Dutykh** and S. Lespinats. *On the parameters domain of the single-diode model*, **Solar Energy**, **277**, 112718, 2024
https://hal.science/hal-04622649/

163. D. Batic and **D. Dutykh**. *Quasinormal Modes in Noncommutative Schwarzschild Black Holes: A Spectral Analysis*. **Eur. Phys. J. C**, **84**, 622, 2024
https://arxiv.org/abs/2406.03353/

162. M. Zafar, **D. Dutykh**, P. Sabatier, M. Banjan and J. Kim. *Numerical Reconstruction of Landslide Paleotsunami Using Geological Records in Alpine Lake Aiguebelette*. **J. Geophys. Res. Solid Earth**, **129**(5), e2023JB028629, 2024
https://hal.science/hal-04592536/

161. M. Amazioug, **D. Dutykh**, B. Teklu and M. Asjad. *Achieving Strong Magnon Blockade through Magnon Squeezing in a Cavity Magnetomechanical System*, **Annalen der Physik**, **536**(4), 2300357, 2024
https://arxiv.org/abs/2308.06367/

160. M. Bowen, P. Samozino, M. Vonderscher, **D. Dutykh** and B. Morel. *Mathematical modeling of exercise fatigability in the severe domain: A unifying integrative framework in isokinetic condition*, **Journal of Theoretical Biology**, **578**, 111696, 2024
https://hal.science/hal-04330250/

**— 2023 —**

159. A. Rashidi, **D. Dutykh** and Ch. Beck. *Correction to: Modeling the potential genesis of tsunamis from below an accretionary prism and their potential impact: a case study along the eastern boundary of the Caribbean Plate*, **Natural Hazards**, **118**, 1765, 2023

158. A. Rashidi, **D. Dutykh** and Ch. Beck. *Modeling the potential genesis of tsunamis from below an accretionary prism, and their potential impact: a case study along the eastern boundary of the Caribbean Plate*, **Natural Hazards**, **118**, 307–329, 2023

157. A. Cheviakov and **D. Dutykh**. *Galilei-invariant and energy-preserving extensions of the Benjamin–Bona–Mahony-type equations*, **Partial Differ. Equations Appl. Math.**, **7**, 100519, 2023

156. M. Cugnet, F. Gallois, A. Kirchev and **D. Dutykh**. *NEOLAB: A Scilab tool to simulate the Negative Electrode of Lead-Acid Batteries*. **SoftwareX**, **22**, 101394, 2023

155. H. Geoffroy, J. Berger, B. Colange, S. Lespinats and **D. Dutykh**. *The use of dimensionality reduction techniques for fault detection and diagnosis in a AHU unit: critical assessment of its reliability*, **J. Build. Perform. Simul.**, **16**(3), 249–267, 2023

154. S. Gasparin, J. Berger, R. Belarbi, **D. Dutykh** and N. Mendes. *Solving parametric problems in building renovation with a spectral reduced-order method*, **J. Build. Perform. Simul.**, **16**(2), 211–230, 2023

**— 2022 —**

153. M. Loukili, **D. Dutykh**, S. Pincemin, K. Kotrasova and N. Abcha. *Theoretical investigation applied to scattering water wave by rectangular submerged obstacle/and submarine trench*. **Geosciences**, **12** (10), 379, 2022
https://hal.science/hal-03855065/

152. A. Machmudah, **D. Dutykh** and S. Parman. *Coupled and synchronization models of rhythmic arm movement in planar plane*. **Bioengineering**, **9**(2), 385, 2022

151. G. Khakimzyanov, Z. Fedotova and **D. Dutykh**. *Two-dimensional models of wave hydrodynamics with high accuracy dispersion relation. III. Linear analysis for an uneven bottom*. **Computational Technologies**, **27**(2), 37–53, 2022

150. B. Colange, L. Vuillon, S. Lespinats and **D. Dutykh**. *MING: an interpretative support method for visual exploration of multidimensional data*. **Information Visualization**, **21**(3), 246–269, 2022
149. D. Blackmore, Ya. Prykarpatsky, M. Prytula, **D. Dutykh** and A. Prykarpatsky. *On the integrability of a new generalized Gurevich–Zybin dynamical system, its Hunter–Saxton type reduction and related mysterious symmetries*, **Analysis and Mathematical Physics**, **12**, 66, 2022
148. **D. Dutykh** and J.-L. Verger-Gaugry. *On a class of lacunary almost Newman polynomials modulo $p$ and density theorems*, **Uniform Distribution Theory**, **17**(1), 29–54, 2022
https://hal.archives-ouvertes.fr/hal-03517173/
147. A. Machmudah, M. Shanmugavel, S. Parman, T.S.B. Abd Manan, **D. Dutykh**, S. Beddu and A. Rajabi. *Flight Trajectories Optimization of fixed-wing UAV by Bank-Turn Mechanism*, **Drones**, **6**(3), 69, 2022
146. T. Bueler-Faudree, S. Delamere, **D. Dutykh**, A. Rybkin and A. Suleimani. *Fast shallow water-wave solver for plane inclined beaches*. **SoftwareX**, **17**, 100983, 2022
145. A. Rashidi, **D. Dutykh**, N.K. Farajkhah and L. Audin. *Regional tsunami hazard from splay faults in the Gulf of Oman*. **Ocean Engineering**, **243**, 110169, 2022

**— 2021 —**

144. O. Gusev, G. Khakimzyanov, L. Chubarov and **D. Dutykh**. *Assessing the frequency dispersion influence on the solitary-wave interaction with a constant sloping beach*. **Journal of Applied Mechanics and Technical Physics**, **62**, 624–632, 2021
143. M. Loukili, **D. Dutykh**, C. Nadjib and K. Kotrasova. *Analytical and numerical investigations applied to study the reflection and transmission coefficients of wave-rectangular breakwater cited at the bottom of tank*. **Geosciences**, **11**(10), 430, 2021
142. **D. Dutykh** and J.-L. Verger-Gaugry. *Alphabets, rewriting trails and periodic representations in algebraic bases*. **Res. number theory**, **7**, 64, 2021
https://hal.archives-ouvertes.fr/hal-03090468/
141. T.S.B. Abd Manan, S. Beddu, N.L. Mohd Kamal, D. Mohamad, Z. Itam, T. Khan, A. Machmudah, **D. Dutykh**, W.H.M. Wan Mohtar, H. Jusoh, F. Mohamed Nazri, M.F.K. Mohd Yapandi, S. Pati, A. Ahmad and N. Wan Rasdi. *Ecological risk indicators for leached heavy metals from coal ash generated at a Malaysian power plant*, **Sustainability**, **13**(18), 10222, 2021
140. G. Khakimzyanov, Z. Fedotova and **D. Dutykh**. *Two-dimensional model of wave hydrodynamics with high accuracy dispersion relation. II. Fourth, sixth and eighth orders*, **Computational Technologies**, **26**(3), 4–25, 2021
139. M. Hunt and **D. Dutykh**. *Free surface flows in electrohydrodynamics with a constant vorticity distribution*, **Water Waves**, **3**, 297–317, 2021
https://hal.archives-ouvertes.fr/hal-02344438/
138. M. Loukili, **D. Dutykh**, K. Kotrasova and D. Ning. *Numerical Stability Investigations of the Method of Fundamental Solutions Applied to Wave-Current Interactions Using Generating-Absorbing Boundary Conditions*. **Symmetry**, **13**(7), 1153, 2021
137. A. Cheviakov, **D. Dutykh** and A. Assylbekuly. *On Galilean invariant and energy preserving BBM-type equations*. **Symmetry**, **13**(5), 878, 2021

136. **D. Dutykh** and H. Le Meur. *Derivation of a viscous Serre–Green–Naghdi equation: an impasse?* **Fluids**, **6**(4), 135, 2021
https://hal.archives-ouvertes.fr/hal-02414312/
135. M. Loukili, K. Kotrasova and **D. Dutykh**. *Numerical modeling of jet at the bottom of tank at moderate Reynolds number using compact Hermitian finite differences method*, **Fluids**, **6**(2), 63, 2021
134. P. Kundu, S.M. Sarkar, A. Rashidi and **D. Dutykh**. *Comparison of ground deformation by two infinite faults in different medium*, **GEM — Int. J. Geomath.**, **12**(3), 2021

**— 2020 —**

133. A. Rashidi, **D. Dutykh** and Z.H. Shomali. *Horizontal displacements effect in tsunami wave generation in the Western Makran region*, **J. Ocean Eng. Mar. Energy**, **6**, 427–439, 2020
132. G.S. Khakimzyanov, Z.I. Fedotova and **D. Dutykh**. *Two-dimensional model of wave hydrodynamics with high accuracy dispersion relation*, **Computational Technologies**, **25**(5), 17–41, 2020
131. D. Mj (Sarkar) and **D. Dutykh**. *Learning extreme wave run-up conditions*. **Applied Ocean Research**, **105**, 102400, 2020
130. S.H. Shamsnia and **D. Dutykh**. *An analytical study on wave-current-mud interaction*, **Water**, **12**(10), 2899, 2020
129. A. Rashidi, **D. Dutykh**, Z.H. Shomali, N.K. Farajkhah and M. Nouri. *A review of tsunami hazard in the Makran subduction zone*, **Geosciences**, **10**(9), 372, 2020
128. A. Abdalazeez, I. Didenkulova, **D. Dutykh** and P. Denissenko. *Comparison of dispersive and nondispersive models for modelling long wave run-up on a beach*, **Izvestiya**, Atmospheric and Oceanic Physics, **56**(5), 494–501, 2020
127. G. Sadaka and **D. Dutykh**. *Adaptive numerical modelling of tsunami wave generation and propagation with FreeFem++*, **Geosciences**, **10**(9), 351, 2020
https://hal.archives-ouvertes.fr/hal-02912526/
126. A. Abdalazeez, I.I. Didenkulov, **D. Dutykh** and P. Denissenko. *Comparison of Dispersive and Nondispersive Models for Wave Run-Up on a Beach*, **Izv. Atmos. Ocean. Phys.**, **56**, 494–501, 2020
125. **D. Dutykh** and E. Tobisch (Kartashova). *Formation of the dynamic energy cascades in quartic and quintic generalized KdV equations*, **Symmetry**, **12**(8), 1254, 2020
http://hal.archives-ouvertes.fr/hal-01070799/
124. A. Abdalazeez, I. Didenkulova, D. **Dutykh** and C. Labart. *Extreme inundation statistics on a composite beach*, **Water**, **12**(6), 1573, 2020
https://hal.archives-ouvertes.fr/hal-02657270/
123. **D. Dutykh** and E. Tobisch (Kartashova). *Resonance enhancement by suitably chosen frequency detuning*, **Mathematics**, **8**(3), 450, 2020
http://hal.archives-ouvertes.fr/hal-01084796/
122. J. Berger, **D. Dutykh**, N. Mendes and L. Gosse. *An efficient numerical model for the simulation of coupled heat, air and moisture transfer in porous media*, **Engineering Reports**, **2**(2), e12099, 2020
https://hal.archives-ouvertes.fr/hal-02413029/
121. J. Berger, T. Busser, T. Colinart and **D. Dutykh**. *Critical assessment of a new mathematical model for hysteresis effects on heat and mass transfer in porous building material*, **Int. J. Therm. Sci.**, **151**,

106275, 2020
https://hal.archives-ouvertes.fr/hal-02485497/

120. A. Rashidi, Z.H. Shomali, **D. Dutykh** and N.K. Farajkhah. *Tsunami hazard assessment in the Makran subduction zone*, **Natural Hazards**, **100**, 861–875, 2020
https://hal.archives-ouvertes.fr/hal-01737767/

119. I. Abroug, N. Abcha, **D. Dutykh**, A. Jarno and F. Marin. *Experimental and numerical study of the propagation of focused wave groups in the nearshore zone*, **Phys. Lett. A**, **384**(6), 126144, 2020
https://hal.archives-ouvertes.fr/hal-02362472/

118. J. Berger, S. Gasparin, **D. Dutykh** and N. Mendes. *On the comparison of three numerical methods applied to building simulation: finite-differences, RC circuit approximation and a spectral method*, **Building Simulation**, **13**(1), 1–18, 2020
https://hal.archives-ouvertes.fr/hal-02140325/

117. G. Khakimzyanov and **D. Dutykh**. *Long wave interaction with a partially immersed body. Part I: Mathematical models*, **Commun. Comput. Phys.**, **27**(2), 321–378, 2020
https://hal.archives-ouvertes.fr/hal-02070055/

116. S. Rusconi, **D. Dutykh**, A. Zarnescu, D. Sokolovski and E. Akhmatskaya. *An optimal scaling to computationally tractable dimensionless models: Study of latex particles morphology formation*, **Comput. Phys. Commun.**, **247**, 106944, 2020
https://hal.archives-ouvertes.fr/hal-02301950/

**— 2019 —**

115. A. Abdalazeez, I. Didenkulova and **D. Dutykh**. *Nonlinear deformation and run-up of single tsunami waves of positive polarity: numerical simulations and analytical predictions*, **Nat. Hazards Earth Syst. Sci.**, **19**, 2905–2913, 2019
https://hal.archives-ouvertes.fr/hal-02422302/

114. **D. Dutykh**. *Numerical simulation of Feller's diffusion equation*, **Mathematics**, **7**(11), 1067, 2019
https://hal.archives-ouvertes.fr/hal-02290284/

113. D. Clamond, **D. Dutykh** and D. Mitsotakis. *Hamiltonian regularisation of shallow water equations with uneven bottom*, **J. Phys. A: Math. Theor.**, **52**(42), 42LT01, 2019
https://hal.archives-ouvertes.fr/hal-02140161/

112. J. Berger and **D. Dutykh**. *Evaluation of the reliability of building energy performance models for parameter estimation*, **Comput. Technologies**, **24**(3), 4–32, 2019
https://hal.archives-ouvertes.fr/hal-02140452/

111. M. Abdykarim, J. Berger, **D. Dutykh**, L. Soudani and A. Agbossou. *Critical assessment of efficient numerical methods for a long-term simulation of heat and moisture transfer in porous materials*, **Int. J. Therm. Sci.**, **145**, 105982, 2019
https://hal.archives-ouvertes.fr/hal-02162803/

110. D. Mitsotakis, Q. Li, E. Peach and **D. Dutykh**. *On some model equations for pulsatile flow in viscoelastic vessels*, **Wave Motion**, **90**, 139–151, 2019
https://hal.archives-ouvertes.fr/hal-02135521/

109. A. Durán, **D. Dutykh** and D. Mitsotakis. *On the multi-symplectic structure of Boussinesq-type systems. II: Geometric discretization*, **Phys. D**, **397**, 1–16, 2019
https://hal.archives-ouvertes.fr/hal-02123559/

108. **D. Dutykh** and D. Ionescu-Kruse. *Effects of vorticity on the travelling waves of some shallow water two-component systems*, **Discrete & Continuous Dynamical Systems — A**, **39**(9), 5521–5541, 2019
https://hal.archives-ouvertes.fr/hal-02094922/

107. G. Khakimzyanov, **D. Dutykh**, D. Mitsotakis and N. Shokina. *Numerical simulation of conservation laws with moving grid nodes: Application to tsunami wave modelling*, **Geosciences**, **9**(5), 197, 2019
https://hal.archives-ouvertes.fr/hal-01223510/

106. E. Dinvay, **D. Dutykh** and H. Kalisch. *A comparative study of bi-directional Whitham systems*, **Appl. Numer. Math.**, **141**, 248–262, 2019
https://hal.archives-ouvertes.fr/hal-02020212/

105. J.-G. Caputo, **D. Dutykh** and B. Gleyse. *Coupling conditions for water waves at forks*, **Symmetry**, **11** (3), 434, 2019
https://hal.archives-ouvertes.fr/hal-01206504/

104. J. Berger, T. Busser, **D. Dutykh** and N. Mendes. *An efficient method to estimate sorption isotherm curve coefficients*, **Inv. Prob. Sci. and Eng.**, **27**(6), 735–772, 2019
https://hal.archives-ouvertes.fr/hal-01509113/

103. A. Jumabekova, J. Berger, **D. Dutykh**, H. Le Meur, A. Foucquier, M. Pailha and Ch. Ménézo. *An efficient numerical model for liquid water uptake in porous material and its parameter estimation*, **Numerical Heat Transfer**, Part A: Applications, **75**(2), 110–136, 2019
https://hal.archives-ouvertes.fr/hal-02047087/

102. J.-P. Chehab and **D. Dutykh**. *On time relaxed schemes and formulations for dispersive wave equations*, **AIMS Mathematics**, **4**(2), 254–278, 2019
https://hal.archives-ouvertes.fr/hal-02057433/

101. V. Liapidevskii and **D. Dutykh**. *On the velocity of turbidity currents over moderate slopes*, **Fluid Dyn. Res.**, **51**, 035501, 2019
https://hal.archives-ouvertes.fr/hal-01719313/

100. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *An innovative method to determine optimum insulation thickness based on non-uniform adaptive moving grid*, **Journal of the Brazilian Society of Mechanical Sciences and Engineering**, **41**:173, 2019
https://hal.archives-ouvertes.fr/hal-02045015/

99. S. Gasparin, **D. Dutykh** and N. Mendes. *A spectral method for solving heat and moisture transfer through consolidated porous media*, **Int. J. Numer. Meth. Eng.**, **117**(11), 1143–1170, 2019
https://hal.archives-ouvertes.fr/hal-02023462/

98. J. Berger, **D. Dutykh**, N. Mendes and B. Rysbaiuly. *A new model for simulating heat, air and moisture transport in porous building materials*, **Int. J. Heat Mass Transf.**, **134**, 1041–1060, 2019
https://hal.archives-ouvertes.fr/hal-02052022/

97. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *Solving nonlinear diffusive problems in buildings by means of a reduced spectral method*, **J. of Build. Perf. Sim.**, **12**(1), 17–36, 2019
https://hal.archives-ouvertes.fr/hal-01513304/

96. A. Durán, **D. Dutykh** and D. Mitsotakis. *On the multi-symplectic structure of Boussinesq-type systems. I: Derivation and mathematical properties*, **Phys. D**, **388**, 10–21, 2019
https://hal.archives-ouvertes.fr/hal-01930748/

**— 2018 —**

95. **D. Dutykh** and J.-L. Verger-Gaugry. *On the reducibility and the lenticular sets of zeroes of almost Newman lacunary polynomials*, **Arnold Mathematical Journal**, **4**(3-4), 315–344, 2018
https://hal.archives-ouvertes.fr/hal-02045588/
94. Y. Pu, R.L. Pego, **D. Dutykh** and D. Clamond. *Weakly singular shock profiles for a non-dispersive regularization of shallow-water equations*, **Comm. Math. Sci.**, **16**(5), 1361–1378, 2018
https://hal.archives-ouvertes.fr/hal-01794005/
93. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *An adaptive simulation of nonlinear heat and moisture transfer as a boundary value problem*, **Int. J. Therm. Sci.**, **133**, 120–139, 2018
https://hal.archives-ouvertes.fr/hal-02021293/
92. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *Advanced reduced-order models for moisture diffusion in porous media*, **Transport in Porous Media**, **124**(3), 965–994, 2018
https://hal.archives-ouvertes.fr/hal-01522319/
91. A. Rashidi, Z.H. Shomali, **D. Dutykh** and N.K. Farajkhah. *Evaluation of tsunami wave energy generated by earthquakes in the Makran subduction zone*, **Ocean Eng.**, **165**(1), 131–139, 2018
https://hal.archives-ouvertes.fr/hal-01719939/
90. **D. Dutykh**, M. Hoefer and D. Mitsotakis. *Solitary wave solutions and their interactions for fully nonlinear water waves with surface tension in the generalized Serre equations*. **Theor. Comp. Fluid Dyn.**, **32**(3), 371–397, 2018
https://hal.archives-ouvertes.fr/hal-01465356/
89. J. Berger, H. Le Meur, **D. Dutykh**, D. M. Nguyen and A.-C. Grillet. *Analysis and improvement of the VTT mold growth model: application to bamboo fiberboard*, **Building and Environment**, **138**, 262–274, 2018
https://hal.archives-ouvertes.fr/hal-01759837/
88. **D. Dutykh** and J.-G. Caputo. *Wave dynamics on networks: method and application to the sine–Gordon equation*, **Appl. Numer. Math.**, **131**, 54–71, 2018
https://hal.archives-ouvertes.fr/hal-01160840/
87. D. Mitsotakis, **D. Dutykh** and Q. Li. *Asymptotic nonlinear and dispersive pulsatile flow in elastic vessels with cylindrical symmetry*, **Computers & Mathematics with Applications**, **75**, 4022–4047, 2018
https://hal.archives-ouvertes.fr/hal-01756720/
86. D. Clamond and **D. Dutykh**. *Accurate fast computation of steady two-dimensional surface gravity waves in arbitrary depth*. **J. Fluid Mech.**, **844**, 491–518, 2018
https://hal.archives-ouvertes.fr/hal-01465813/
85. J. Berger, S. Gasparin, **D. Dutykh** and N. Mendes. *On the solution of coupled heat and moisture transport in porous material*, **Transport in Porous Media**, **121**(3), 665–702, 2018
https://hal.archives-ouvertes.fr/hal-01770386/

84. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *An improved explicit scheme for whole-building hygrothermal simulation*, **Building Simulation**, **11**(3), 465–481, 2018
https://hal.archives-ouvertes.fr/hal-01495737/
83. V. Liapidevskii, **D. Dutykh** and M. Gisclon. *On the modelling of shallow turbidity flows*, **Adv. Water Resour.**, **113**, 310–327, 2018
https://hal.archives-ouvertes.fr/hal-01514095/
82. G. Khakimzyanov and **D. Dutykh**. *Numerical modelling of surface water wave interaction with a moving wall*, **Commun. Comput. Phys.**, **23**(5), 1289–1354, 2018
https://hal.archives-ouvertes.fr/hal-01546833/
81. S. Gasparin, J. Berger, **D. Dutykh** and N. Mendes. *Stable explicit schemes for simulation of nonlinear moisture transfer in porous materials*. **J. of Build. Perf. Sim.**, **11**(2), 129–144, 2018
https://hal.archives-ouvertes.fr/hal-01404578/
80. J. Berger, T. Busser, **D. Dutykh** and N. Mendes. *On the estimation of moisture permeability and advection coefficients of a wood fibre material using the optimal experiment design approach*, **Experimental Thermal and Fluid Science**, **90**, 246–259, 2018
https://hal.archives-ouvertes.fr/hal-01498638/
79. L. Vuillon, **D. Dutykh** and F. Fedele. *Some special solutions to the Hyperbolic NLS equation*, **Comm. Nonlinear Sci. Numer. Simulat.**, **57**, 202–220, 2018
http://hal.archives-ouvertes.fr/hal-00846801/
78. G. Khakimzyanov, **D. Dutykh** and O. Gusev. *Dispersive shallow water wave modelling. Part IV: Numerical simulation on a globally spherical geometry*, **Commun. Comput. Phys.**, **23**(2), 361–407, 2018
https://hal.archives-ouvertes.fr/hal-01558680/
77. G. Khakimzyanov, **D. Dutykh** and Z. Fedotova. *Dispersive shallow water wave modelling. Part III: Model derivation on a globally spherical geometry*, **Commun. Comput. Phys.**, **23**(2), 315–360, 2018
https://hal.archives-ouvertes.fr/hal-01552229/
76. G. Khakimzyanov, **D. Dutykh**, O. Gusev and N. Shokina. *Dispersive shallow water wave modelling. Part II: Numerical simulation on a globally flat space*, **Commun. Comput. Phys.**, **23**(1), 30–92, 2018
https://hal.archives-ouvertes.fr/hal-01551075/
75. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and D. Mitsotakis. *Dispersive shallow water wave modelling. Part I: Model derivation on a globally flat space*, **Commun. Comput. Phys.**, **23**(1), 1–29, 2018
https://hal.archives-ouvertes.fr/hal-01547833/
74. D. Clamond and **D. Dutykh**. *Non-dispersive conservative regularisation of nonlinear shallow water and isothermal Euler equations*, **Commun. Nonlinear Sci. Numer. Simulat.**, **55**, 237–247, 2018
https://hal.archives-ouvertes.fr/hal-01509445/

**— 2017 —**

73. G. Khakimzyanov and **D. Dutykh**. *On supraconvergence phenomenon for second order centered finite differences on non-uniform grids*, **J. Comput. Appl. Math.**, **326**, 1–14, 2017
https://hal.archives-ouvertes.fr/hal-01223522/
72. M. Chhay, **D. Dutykh**, M. Gisclon and Ch. Ruyer-Quil. *New asymptotic heat transfer model in thin liquid films*, **Appl. Math. Model.**, **48**, 844–859, 2017
https://hal.archives-ouvertes.fr/hal-01224182/

71. D. Mitsotakis, **D. Dutykh**, A. Assylbekuly and D. Zhakebayev. *On weakly singular and fully nonlinear travelling shallow capillary-gravity waves in the critical regime*, **Phys. Lett. A**, **381**(20), 1719–1726, 2017
https://hal.archives-ouvertes.fr/hal-01388481/
70. J. Berger, S. Gasparin, **D. Dutykh** and N. Mendes. *Accurate numerical simulation of moisture front in porous material*, **Building and Environment**, **118**, 211–224, 2017
https://hal.archives-ouvertes.fr/hal-01419018/
69. E. Dinvay, D. Moldabayev, **D. Dutykh** and H. Kalisch. *The Whitham equation with surface tension*, **Nonlinear Dynamics**, **88**(2), 1125–1138, 2017
https://hal.archives-ouvertes.fr/hal-01504047/
68. D. Mitsotakis, **D. Dutykh** and J. Carter. *On the nonlinear dynamics of the traveling-wave solutions of the Serre equations*, **Wave Motion**, **70**, 166–182, 2017
http://hal.archives-ouvertes.fr/hal-00984035/
67. **D. Dutykh**, D. Clamond and M. Chhay. *Serre-type equations in deep water*, **Math. Model. Nat. Phenom.**, **12**(1), 23–40, 2017
https://hal.archives-ouvertes.fr/hal-01340379/
66. D. Clamond, **D. Dutykh** and D. Mitsotakis. *Conservative modified Serre–Green–Naghdi equations with improved dispersion characteristics*, **Comm. Nonlinear Sci. Numer. Simul.**, **45**, 245–257, 2017
https://hal.archives-ouvertes.fr/hal-01232370/
65. J. Berger, **D. Dutykh** and N. Mendes. *On the optimal experimental design for heat and moisture parameter estimation*, **Experimental Thermal and Fluid Science**, **81**, 109–122, 2017
https://hal.archives-ouvertes.fr/hal-01334414/

**— 2016 —**

64. **D. Dutykh** and D. Clamond. *Modified Shallow Water Equations for significantly varying seabeds*. **Appl. Math. Modelling**, **40**(23-24), 9767–9787, 2016
http://hal.archives-ouvertes.fr/hal-00675209/
63. D. Clamond, **D. Dutykh** and A. Galligo. *Algebraic method for constructing singular steady solitary waves: A case study*, **Proc. R. Soc. A.**, **472**(2191), 20160194, 2016
https://hal.archives-ouvertes.fr/hal-01290471/
62. D. Clamond and **D. Dutykh**. *Multi-symplectic structure of fully-nonlinear weakly-dispersive internal gravity waves*, **J. Phys. A: Math. Theor.**, **49**, 31LT01, 2016
https://hal.archives-ouvertes.fr/hal-01296552/
61. **D. Dutykh**, D. Clamond and A. Durán. *Efficient computation of capillary-gravity generalized solitary waves*, **Wave Motion**, **65**, 1–16, 2016
https://hal.archives-ouvertes.fr/hal-01218989/
60. **D. Dutykh** and D. Ionescu–Kruse. *Travelling wave solutions for some two-component shallow water models*, **J. Diff. Equations**, **261**(2), 1099–1114, 2016
https://hal.archives-ouvertes.fr/hal-01294603/
59. **D. Dutykh** and O. Goubet. *Derivation of dissipative Boussinesq equations using the Dirichlet-to-Neumann operator approach*, **Math. Comput. Simul.**, **127**, 80–93, 2016
http://hal.archives-ouvertes.fr/hal-00596804/

58. F. Carbone, **D. Dutykh** and G. El. *Macroscopic dynamics of incoherent soliton ensembles: soliton-gas kinetics and direct numerical modeling: Incoherent soliton ensembles*, **EPL**, **113**, 30003, 2016
https://hal.archives-ouvertes.fr/hal-01248680/
57. M. Chhay, **D. Dutykh** and D. Clamond. *On the multi-symplectic structure of the Serre–Green–Naghdi equations*, **J. Phys. A: Math. Theor.**, **49**, 03LT01, 2016
https://hal.archives-ouvertes.fr/hal-01221356/
56. G. Khakimzyanov, N. Shokina, **D. Dutykh** and D. Mitsotakis. *A new run-up algorithm based on local high-order analytic expansions*, **J. Comput. Appl. Math.**, **298**, 82–96, 2016
http://hal.archives-ouvertes.fr/hal-01084811/

**— 2015 —**

55. A. Rafiee, **D. Dutykh** and F. Dias. *Numerical simulation of wave impact on a rigid wall using a two-phase compressible SPH method*. **Procedia IUTAM**, **18**, 123–137, 2015
http://hal.archives-ouvertes.fr/hal-00830054/
54. D. Clamond, **D. Dutykh** and A. Durán. *A plethora of generalised solitary gravity-capillary water waves*, **J. Fluid Mech.**, **784**, 664–680, 2015
http://hal.archives-ouvertes.fr/hal-01081798/
53. H. Nersisyan, **D. Dutykh** and E. Zuazua. *Generation of 2D water waves by moving bottom disturbances*. **IMA J. Appl. Math.**, **80**(4), 1235–1253, 2015
http://hal.archives-ouvertes.fr/hal-00752364/
52. D. Moldabayev, H. Kalisch and **D. Dutykh**. *The Whitham equation as a model for surface water waves*, **Phys. D**, **309**, 99–107, 2015
http://hal.univ-savoie.fr/hal-01136855/
51. **D. Dutykh**, D. Clamond and D. Mitsotakis. *Adaptive modeling of shallow fully nonlinear gravity waves*, **RIMS Kôkyûroku**, **1947**, 45–65, 2015
https://hal.archives-ouvertes.fr/hal-01088440/
50. **D. Dutykh**, D. Clamond and M. Chhay. *Numerical study of the generalised Klein–Gordon equations*, **Phys. D**, **304–305**, 23–33, 2015
http://hal.archives-ouvertes.fr/hal-00851030/
49. **D. Dutykh** and E. Tobisch (Kartashova). *Direct dynamical energy cascade in the modified KdV equation*, **Phys. D**, **297**, 76–87, 2015
http://hal.archives-ouvertes.fr/hal-00990724/
48. T. Stefanakis, S. Xu, **D. Dutykh** and F. Dias. *Run-up amplification of transient long waves*. **Quart. Appl. Math.**, LXXIII(1), 177–199, 2015
http://hal.archives-ouvertes.fr/hal-00768597/

**— 2014 —**

47. I. S. Gandzha, Yu. V. Sedletsky and **D. Dutykh**. *High-order nonlinear Schrödinger equation for the envelope of slowly modulated gravity waves on the surface of finite-depth fluid and its quasi-soliton solutions*. **Ukr. J. Phys.**, **59**(12), 1201–1215, 2014
https://hal.archives-ouvertes.fr/hal-01084747/

46. **D. Dutykh** and E. Pelinovsky. *Numerical simulation of a solitonic gas in some integrable and non-integrable models*, **Phys. Lett. A**, **378**(42), 3102–3110, 2014
http://hal.archives-ouvertes.fr/hal-00913960/
45. J.-G. Caputo and **D. Dutykh**. *Nonlinear waves in networks: a simple approach using the sine–Gordon equation*, **Phys. Rev. E**, **90**, 022912, 2014
http://hal.archives-ouvertes.fr/hal-00951705/
44. D. Mitsotakis, B. Ilan and **D. Dutykh**. *On the Galerkin / finite-element method for the Serre equations*. **J. Sci. Comp.**, **61**(1), 166–195, 2014
http://hal.archives-ouvertes.fr/hal-00834064/
43. **D. Dutykh** and E. Tobisch (Kartashova). *Observation of the Inverse Energy Cascade in the modified Korteweg–de Vries Equation*, **Eur. Phys. Lett.**, **107**, 14001, 2014
http://hal.archives-ouvertes.fr/hal-00991944/
42. Z.I. Fedotova, G.S. Khakimzyanov and **D. Dutykh**. *Energy equation for certain approximate models of long-wave hydrodynamics*, **Russian Journal of Numerical Analysis and Mathematical Modelling**, **29** (3), 167–178, 2014
https://hal.archives-ouvertes.fr/hal-00998927/
41. M. Hunt and **D. Dutykh**. *Visco-potential flows in electrohydrodynamics*, **Phys. Lett. A**, **378**, 1721–1726, 2014
http://hal.archives-ouvertes.fr/hal-00908724/
40. F. Dias, **D. Dutykh**, L. O'Brien, E. Renzi and T. Stefanakis. *On the modelling of tsunami generation and tsunami inundation*, **Procedia IUTAM**, **10**, 338–355, 2014
http://hal.archives-ouvertes.fr/hal-00728725/
39. C. Viotti, **D. Dutykh** and F. Dias. *The conformal-mapping method for surface gravity waves in the presence of variable bathymetry and mean current*, **Procedia IUTAM**, **11**, 110–118, 2014
http://hal.archives-ouvertes.fr/hal-00855780/
38. **D. Dutykh**. *Evolution of random wave fields in the water of finite depth*. **Procedia IUTAM**, **11**, 34–43, 2014
http://hal.archives-ouvertes.fr/hal-00864549/
37. F. Fedele and **D. Dutykh**. *Camassa–Holm equations and vortexons for axisymmetric pipe flows*. **Fluid Dyn. Res.**, **46**, 015503, 2014
http://hal.archives-ouvertes.fr/hal-00759444/
36. **D. Dutykh** and D. Clamond. *Efficient computation of steady solitary gravity waves*. **Wave Motion**, **51**, 86–99, 2014
http://hal.archives-ouvertes.fr/hal-00786077/


**— 2013 —**


35. F. Fedele and **D. Dutykh**. *Camassa–Holm type equations for axisymmetric Poiseuille pipe flows*. **Procedia IUTAM**, **9**, 16–24, 2013
http://hal.archives-ouvertes.fr/hal-00752999/
34. F. Carbone, **D. Dutykh**, J. Dudley and F. Dias. *Extreme wave runup on a vertical cliff*. **Geophys. Res. Lett.**, **40**(12), 3138–3143, 2013
http://hal.archives-ouvertes.fr/hal-00799118/

33. A. Durán, **D. Dutykh** and D. Mitsotakis. *On the Galilean invariance of some dispersive wave equations*, **Stud. Appl. Math.**, **131**(4), 359–388, 2013
http://hal.archives-ouvertes.fr/hal-00664143/
32. **D. Dutykh**, D. Clamond, P. Milewski and D. Mitsotakis. *Finite volume and pseudo-spectral schemes for the fully nonlinear 1D Serre equations*, **European Journal of Applied Mathematics**, **24**(5), 761–787, 2013
http://hal.archives-ouvertes.fr/hal-00587994/
31. D. Clamond and **D. Dutykh**. *Fast accurate computation of the fully nonlinear solitary surface gravity waves*. **Computers & Fluids**, **84**, 35–38, 2013
http://hal.archives-ouvertes.fr/hal-00759812/
30. C. Viotti, **D. Dutykh**, J. Dudley and F. Dias. *Emergence of coherent wave groups in deep-water random sea*. **Phys. Rev. E**, **87**(6), 063001, 2013
http://hal.archives-ouvertes.fr/hal-00795397/
29. **D. Dutykh** and H. Kalisch. *Boussinesq modeling of surface waves due to underwater landslides*, **Nonlin. Processes Geophys.**, **20**, 267–285, 2013
http://hal.archives-ouvertes.fr/hal-00654386/
28. F. Fedele and **D. Dutykh**. *Vortexons in axisymmetric Poiseuille pipe flows*. **Eur. Phys. Lett.**, **101**(3), 34003, 2013
http://hal.archives-ouvertes.fr/hal-00756531/
27. **D. Dutykh**, F. Fedele and M. Chhay. *Geometric numerical schemes for the KdV equation*. **Computational Mathematics and Mathematical Physics**, **53**(2), 221–236, 2013
http://hal.archives-ouvertes.fr/hal-00694896/
26. **D. Dutykh**, D. Mitsotakis, X. Gardeil and F. Dias. *On the use of finite fault solution for tsunami generation problems*. **Theor. Comput. Fluid Dyn.**, **27**, 177–199, 2013
http://hal.archives-ouvertes.fr/hal-00509384/
25. **D. Dutykh**, Th. Katsaounis and D. Mitsotakis. *Finite volume methods for unidirectional dispersive wave models*. **Int. J. Num. Meth. Fluids**, **71**, 717–736, 2013
http://hal.archives-ouvertes.fr/hal-00538043/

**— 2012 —**

24. F. Fedele and **D. Dutykh**. *Special solutions to a compact equation for deep-water gravity waves*. **J. Fluid Mech.**, **712**, 646–660, 2012
http://hal.archives-ouvertes.fr/hal-00687325/
23. **D. Dutykh**, D. Mitsotakis, L. Chubarov and Yu. Shokin. *On the contribution of the horizontal sea-bed displacements into the tsunami generation process*. **Ocean Modelling**, **56**, 43–56, 2012
http://hal.archives-ouvertes.fr/hal-00530999/
22. F. Fedele and **D. Dutykh**. *Solitary wave interaction in a compact equation for deep-water gravity waves*. **JETP Letters**, **95**(12), 703–706, 2012
http://hal.archives-ouvertes.fr/hal-00687668/
21. S.A. Beisel, L.B. Chubarov, **D. Dutykh**, G.S. Khakimzyanov and N.Yu. Shokina. *Simulation of surface waves generated by an underwater landslide in a bounded reservoir*. **Russ. J. Numer. Anal. Math.**

**Modelling**, **27**(6), 539–558, 2012
http://hal.archives-ouvertes.fr/hal-00788284/

20. D. Clamond and **D. Dutykh**. *Practical use of variational principles for modeling water waves*. **Physica D**, **241**(1), 25–36, 2012
http://hal.archives-ouvertes.fr/hal-00456891/

**— 2011 —**

19. F. Fedele and **D. Dutykh**. *Hamiltonian form and solitary waves of the spatial Dysthe equations*. **JETP Letters**, **94**(12), 921–925, 2011
http://hal.archives-ouvertes.fr/hal-00633389/

18. **D. Dutykh**, C. Labart and D. Mitsotakis. *Long wave runup on random beaches*. **Phys. Rev. Lett.**, **107**, 184504, 2011
http://hal.archives-ouvertes.fr/hal-00606627/

17. T. Stefanakis, F. Dias and **D. Dutykh**. *Local Run-Up Amplification by Resonant Wave Interactions*. **Phys. Rev. Lett.**, **107**, 124502, 2011
http://hal.archives-ouvertes.fr/hal-00605461/

16. **D. Dutykh** and D. Clamond. *Shallow water equations for large bathymetry variations*, **J. Phys. A: Math. Theor. 44**, 332001, 2011
http://hal.archives-ouvertes.fr/hal-00580310/

15. **D. Dutykh**, C. Acary-Robert and D. Bresch. *Mathematical modeling of powder-snow avalanche flows*. **Studies in Applied Mathematics**, **127**(1), 38–66, 2011
http://hal.archives-ouvertes.fr/hal-00354000/

14. **D. Dutykh**, R. Poncet and F. Dias. *The VOLNA code for the numerical modelling of tsunami waves: generation, propagation and inundation*. **European Journal of Mechanics B/Fluids**, **30**(6), 598–615, 2011
http://hal.archives-ouvertes.fr/hal-00454591/

13. **D. Dutykh**, Th. Katsaounis and D. Mitsotakis. *Finite volume schemes for dispersive wave propagation and runup*. **J. Comp. Phys.**, **230**, 3035–3061, 2011
http://hal.archives-ouvertes.fr/hal-00472431/

**— 2010 —**

12. Y. Meyapin, **D. Dutykh** and M. Gisclon. *Velocity and energy relaxation in two-phase flows*. **Studies in Applied Mathematics**, **125(2)**, 179 – 212, 2010
http://hal.archives-ouvertes.fr/hal-00440852/

11. **D. Dutykh** and D. Mitsotakis. *On the relevance of the dam break problem in the context of nonlinear shallow water equations*. **Discrete and Continuous Dynamical Systems**, **13**(4), 799–818, 2010
http://hal.archives-ouvertes.fr/hal-00369795/

10. **D. Dutykh** and F. Dias. *Influence of sedimentary layering on tsunami generation*, **Computer Methods in Applied Mechanics and Engineering**, **199**, 1268–1275, 2010
http://hal.archives-ouvertes.fr/hal-00288696/

9. F. Dias, **D. Dutykh** and J.-M. Ghidaglia. *A two-fluid model for violent aerated flows*, **Computers and Fluids**, 39(2), 283–293, 2010
http://hal.archives-ouvertes.fr/hal-00285037/

**— 2009 —**

8. **D. Dutykh**. *Group and phase velocities in the free-surface visco-potential flow: new kind of boundary layer induced instability*. **Physics Letters A**, **373**, 3212–3216, 2009
http://hal.archives-ouvertes.fr/hal-00334440/
7. **D. Dutykh**. *Visco-potential free-surface flows and long wave modelling*. **European Journal of Mechanics B/Fluids**, **28**(3), 430–443, 2009
http://hal.archives-ouvertes.fr/hal-00270926/
6. **D. Dutykh** and F. Dias. *Energy of tsunami waves generated by bottom motion*. **Proc. R. Soc. A** (2009) **465**, 725–744
http://hal.archives-ouvertes.fr/hal-00311752/
5. **D. Dutykh** and F. Dias. *Tsunami generation by dynamic displacement of sea bed due to dip-slip faulting*. **Math. Comp. Sim.**, **80**(4), 837-848, 2009
http://hal.archives-ouvertes.fr/hal-00174439/

**— 2007 —**

4. **D. Dutykh** and F. Dias. *Viscous potential free-surface flows in a fluid layer of finite depth*, **C. R. Acad. Sci. Paris**, Ser. I, **345**, 113–118, 2007
http://hal.archives-ouvertes.fr/hal-00145315/
3. **D. Dutykh** and F. Dias. *Dissipative Boussinesq equations*, **C. R. Mcanique**, **335**, 559–583, 2007
http://hal.archives-ouvertes.fr/hal-00137633/
2. Y. Kervella, **D. Dutykh** and F. Dias. *Comparison between three-dimensional linear and nonlinear tsunami generation models*, **Theor. Comput. Fluid Dyn.**, **21**:245–269, 2007
http://hal.archives-ouvertes.fr/hal-00113909/

**— 2006 —**

1. **D. Dutykh**, F. Dias and Y. Kervella. *Linear theory of wave generation by a moving bottom*, **C. R. Acad. Sci. Paris**, Ser. I **343**, 499–504, 2006
http://hal.archives-ouvertes.fr/hal-00114954/

### 8.5.4 Book chapters

**— 2025 —**

34. R. Matar, N. Abcha, N. Lecoq, E.-I. Turki, I. Abroug and **D. Dutykh**. *Exploring Extreme Wave Propagation in Coastal Zones: A Combined Physical and Numerical Modeling Study*, In A.-C. Bennis, J.-C. Dauvin, E. Feunteun, T. Komatsu, O. Matsuda and P. Prouzet (Eds.) Constraints and Adaptations to Global Change at the Land-Sea Interface: For a Shared Ecological and Energy Transition, Proceedings of the 19$^{th}$ French–Japanese Oceanography Symposium, COAST 2023, Springer, Cham, pp. 3–18, 2025

**— 2024 —**

33. M. Zafar, **D. Dutykh**, P. Sabatier, M. Banjan and J. Kim. *Uncovering a Paleotsunami Triggered by Mass-Movement in an Alpine Lake*. In: P. Gourbesville, G. Caignaert (Eds.) Advances in Hydroinformatics — SimHydro 2023 Volume 2. SimHydro 2023. Springer Water. Springer, Singapore, pp. 47–64, 2024
https://hal.science/hal-04261487/

**— 2022 —**

32. T.S.B. Abd Manan, T. Khan, W.H.M. Wan Mohtar, A. Machmudah, **D. Dutykh**, S. Qazi, A. Ahmad and N. Wan Rasdi. *Bioremediation of wastewater using algae for potential renewable bioenergy cogeneration*, In A. Ahmad, F. Banat and H. Taher (Eds.) Algal Biotechnology: Integrated Algal Engineering for Bioenergy, Bioremediation, and Biomedical Applications, Elsevier, Amsterdam, pp. 47–62, 2022

**— 2021 —**

31. V. Roubtsov and **D. Dutykh**. *Poisson and Symplectic structures, Hamiltonian action, momentum and reduction*, In M. Ulan and E. Schneider (Eds.) Differential Geometry, Differential Equations, and Mathematical Physics. Birkhäuser, Cham, pp. 1–29, 2021
https://hal.archives-ouvertes.fr/hal-02522156/
30. S. Lespinats, B. Colange and **D. Dutykh**. *Data Science Context*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 1–30, 2021
https://doi.org/10.1007/978-3-030-81026-9_1
29. S. Lespinats, B. Colange and **D. Dutykh**. *Intrinsic Dimensionality*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 31–44, 2021
https://doi.org/10.1007/978-3-030-81026-9_2
28. S. Lespinats, B. Colange and **D. Dutykh**. *Map Evaluation*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 45–66, 2021
https://doi.org/10.1007/978-3-030-81026-9_3
27. S. Lespinats, B. Colange and **D. Dutykh**. *Map Interpretation*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 67–88, 2021
https://doi.org/10.1007/978-3-030-81026-9_4
26. S. Lespinats, B. Colange and **D. Dutykh**. *Stress Functions for Unsupervised Dimensionality Reduction*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 89–118, 2021
https://doi.org/10.1007/978-3-030-81026-9_5
25. S. Lespinats, B. Colange and **D. Dutykh**. *Stress Functions for Supervised Dimensionality Reduction*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 119–156, 2021
https://doi.org/10.1007/978-3-030-81026-9_6
24. S. Lespinats, B. Colange and **D. Dutykh**. *Optimization, Acceleration and Out of Sample Extensions*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 157–176, 2021
https://doi.org/10.1007/978-3-030-81026-9_7

23. S. Lespinats, B. Colange and **D. Dutykh**. *Applications of Dimensionality Reduction to the Diagnosis of Energy Systems*, In: Nonlinear Dimensionality Reduction Techniques: A Data Structure Preservation Approach, Springer, Cham, pp. 177–192, 2021
https://doi.org/10.1007/978-3-030-81026-9_8

**— 2020 —**

22. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and O. Gusev. *Model Derivation on a Globally Flat Space*, In: Dispersive Shallow Water Waves: Theory, Modeling, and Numerical Methods, Lecture Notes in Geosystems Mathematics and Computing, Birkhäuser, Cham, pp. 1–43, 2020
https://doi.org/10.1007/978-3-030-46267-3_1
21. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and O. Gusev. *Numerical Simulation on a Globally Flat Space*, In: Dispersive Shallow Water Waves: Theory, Modeling, and Numerical Methods, Lecture Notes in Geosystems Mathematics and Computing, Birkhäuser, Cham, pp. 45–134, 2020
https://doi.org/10.1007/978-3-030-46267-3_2
20. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and O. Gusev. *Model Derivation on a Globally Spherical Geometry*, In: Dispersive Shallow Water Waves: Theory, Modeling, and Numerical Methods, Lecture Notes in Geosystems Mathematics and Computing, Birkhäuser, Cham, pp. 135–190, 2020
https://doi.org/10.1007/978-3-030-46267-3_3
19. G. Khakimzyanov, **D. Dutykh**, Z. Fedotova and O. Gusev. *Numerical Simulation on a Globally Spherical Geometry*, In: Dispersive Shallow Water Waves: Theory, Modeling, and Numerical Methods, Lecture Notes in Geosystems Mathematics and Computing, Birkhäuser, Cham, pp. 191–237, 2020
https://doi.org/10.1007/978-3-030-46267-3_4

**— 2019 —**

18. T. Torsvik, A. Abdalazeez, **D. Dutykh**, P. Denissenko and I. Didenkulova. *Dispersive and non-dispersive nonlinear long wave transformations: numerical and experimental results*, In A. Berezovski and T. Soomere (Eds.) Applied Wave Mathematics II, Mathematics of Planet Earth, Vol. **6**. pp. 41–66, Springer Cham, 2019
https://hal.archives-ouvertes.fr/hal-02289778/
17. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *A Brief History of Diffusion in Physics*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 1–5, 2019
https://doi.org/10.1007/978-3-030-31574-0_1
16. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Heat and Mass Diffusion in Porous Building Elements*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 9–43, 2019
https://doi.org/10.1007/978-3-030-31574-0_2
15. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Finite-Difference Method*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 45–87, 2019
https://doi.org/10.1007/978-3-030-31574-0_3
14. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Basics in Practical Finite-Element Method*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 89–100, 2019
https://doi.org/10.1007/978-3-030-31574-0_4

13. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Explicit Schemes with Improved CFL Condition*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 103–120, 2019 https://doi.org/10.1007/978-3-030-31574-0_5
12. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Reduced Order Methods*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 121–152, 2019 https://doi.org/10.1007/978-3-030-31574-0_6
11. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Boundary Integral Approaches*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 153–166, 2019 https://doi.org/10.1007/978-3-030-31574-0_7
10. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Spectral Methods*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 167–209, 2019 https://doi.org/10.1007/978-3-030-31574-0_8
9. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Exercises and Problems*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 213–227, 2019 https://doi.org/10.1007/978-3-030-31574-0_9
8. N. Mendes, M. Chhay, J. Berger and **D. Dutykh**. *Conclusions*, In: Numerical Methods for Diffusion Phenomena in Building Physics, Springer, Cham, pp. 229–232, 2019 https://doi.org/10.1007/978-3-030-31574-0_10

**— 2018 —**

7. A. Durán, **D. Dutykh** and D. Mitsotakis. *Peregrine's system revisited*, In N. Abcha, E. Pelinovsky and I. Mutabazi (Eds.) Nonlinear Waves and Pattern Dynamics, pp. 3–43, Springer Cham, 2018 https://hal.archives-ouvertes.fr/hal-01703491/

**— 2016 —**

6. D. Clamond and **D. Dutykh**. *Modeling water waves beyond perturbations*, in E. Tobisch (Ed.): New Approaches to Nonlinear Waves, Lecture Notes in Physics (**908**), 197–210, Springer, 2016 https://hal.archives-ouvertes.fr/hal-01102120/

**— 2014 —**

5. D. Clamond and **D. Dutykh**. *A Non-Hydrostatic Non-Dispersive Shallow Water Model*, Ph. Goubersville *et al.* (Eds.): Springer Hydrogeology. Advances in Hydroinformatics: SIMHYDRO 2012 — New Frontiers of Simulation, 197–206, Springer Singapore 2014 https://hal.archives-ouvertes.fr/hal-00907110/
4. V. Guinot, D. Clamond and **D. Dutykh**. *Finite Volume Implementation of Non-Dispersive, Non-Hydrostatic Shallow Water Equations*, Ph. Goubersville *et al.* (Eds.): Springer Hydrogeology. Advances in Hydroinformatics: SIMHYDRO 2012 — New Frontiers of Simulation, 189–196, Springer Singapore 2014 https://hal.archives-ouvertes.fr/hal-00907108/

**— 2011 —**

3. Y. Meyapin, **D. Dutykh** and M. Gisclon. *Two-fluid barotropic models for powder-snow avalanche flows*, E. Krause *et al.* (Eds.): Computational Sci. & High Performance Computing IV, NNFM 115, pp. 219–232. Springer-Verlag Berlin Heidelberg 2011
http://hal.archives-ouvertes.fr/hal-00394437/

**— 2007 —**

2. **D. Dutykh** and F. Dias, *Water waves generated by a moving bottom*. In book: "Tsunami and Nonlinear Waves", A. KUNDU (Editor), pp. 65–95, Springer Verlag 2007, Approx. 325 p., 170 illus., Hardcover, ISBN: 978-3-540-71255-8
http://hal.archives-ouvertes.fr/hal-00115875/

**— 2006 —**

1. F. Dias and **D. Dutykh**. *Dynamics of tsunami waves*. In Book: "Extreme Man-Made and Natural Hazards in Dynamics of Structures", A. IBRAHIMBEGOVIC and I. KOZAR (Eds), Springer Dordrecht, NATO Advanced Research Workshop, Opatija, Croatia, May 28 — June 1, pp. 35–60, 2006
http://hal.archives-ouvertes.fr/hal-00113612/

### 8.5.5 Peer-reviewed conference proceedings

**— 2022 —**

14. S. Mustatea, M. Aupetit, J. Peltonen, S. Lespinats, **D. Dutykh**. *Supervised dimensionality reduction technique accounting for soft classes*, European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN), Bruges, Belgium, 5–7 October 2022
https://www.esann.org/sites/default/files/proceedings/2022/ES2022-26.pdf

**— 2020 —**

13. B. Colange, J. Peltonen, M. Aupetit, **D. Dutykh** and S. Lespinats. *Steering Distortions to Preserve Classes and Neighbors in Supervised Dimensionality Reduction*, 34$^{th}$ Conference on Neural Information Processing Systems (NeurIPS 2020), Vancouver, Canada, 2020
https://proceedings.neurips.cc/paper/2020/hash/99607461cdb9c26e2bd5f31b12dcf27a-Abstract.html
12. M. Loukili, K. Kotrasova and D. **Dutykh**. *A computational simulation of steady natural convection in an $H$ −form cavity*. In Silhavy R., Silhavy P., Prokopova Z. (Eds.) Software Engineering Perspectives in Intelligent Systems. CoMeSySo 2020. Advances in Intelligent Systems and Computing, Vol. **1295**, pp. 164-177, Springer International Publishing, Cham, 2020
https://hal.archives-ouvertes.fr/hal-02615655/
11. A. Abdalazeez, I. Didenkulova and **D. Dutykh**. *Dispersive effects during long wave run-up on a plane beach* in "Advances in Natural Hazards and Hydrological Risks: Meeting the Challenge", Proceedings of the 2$^{nd}$ International Workshop on Natural Hazards (NatHaz'19), Pico Island – the Azores 2019, F. Fernandes, A. Malheiro, H.I. Chaminé (Eds.), pp. 143–146, Springer International Publishing, Cham, 2020
https://hal.archives-ouvertes.fr/hal-02366937/

**— 2019 —**

10. B. Colange, L. Vuillon, S. Lespinats and **D. Dutykh**. *Interpreting Distortions in Dimensionality Reduction by Superimposing Neighbourhood Graphs*. 2019 IEEE Visualization Conference (VIS), 20 – 25 October, Vancouver, BC, Canada, 2019
https://hal.archives-ouvertes.fr/hal-02290875/
9. M. Abdykarim, J. Berger, **D. Dutykh** and A. Agbossou. *An efficient numerical method for a long term simulation of heat and mass transfer: the case of an insulated rammed earth wall*. V. Corrado, E. Fabrizio, A. Gasparella and F. Patuzzi (Eds.), Proceedings of the 16$^{th}$ IBPSA Conference, pp. 1444 – 1451, Rome, Italy, September 2 – 4, 2019, ISBN: 978-1-7750520-1-2
https://hal.archives-ouvertes.fr/hal-02288858/
8. H. Geoffroy, J. Berger, B. Colange, S. Lespinats, **D. Dutykh**, C. Buhé and G. Sauce. *Use of Multidimensional Scaling for Fault Detection or Monitoring Support in A Continuous Commissioning*. V. Corrado, E. Fabrizio, A. Gasparella and F. Patuzzi (Eds.), Proceedings of the 16$^{th}$ IBPSA Conference, pp. 877 – 884, Rome, Italy, September 2 – 4, 2019, ISBN: 978-1-7750520-1-2
https://hal.archives-ouvertes.fr/hal-02411240/

**— 2012 —**

7. T. Stefanakis, F. Dias and **D. Dutykh**. *Resonant long-wave run-up on a plane beach*. ISOPE-2012 Rhodes Conference Proceedings, 2012
http://hal.archives-ouvertes.fr/hal-00728747/
6. L. O'Brien, P. Christodoulides, E. Renzi, **D. Dutykh** and F. Dias. *The force of a tsunami on a wave energy converter*. ISOPE-2012 Rhodes Conference Proceedings, 2012
http://hal.archives-ouvertes.fr/hal-00728888/

**— 2011 —**

5. **D. Dutykh**, D. Mitsotakis, S. Beisel and N. Shokina. *Dispersive waves generated by an underwater landslide*. Numerical methods for Hyperbolic Equations. Eds: Vazquez-Cendon *et al.*, Taylor & Francis Group, London, pp. 245–250, 2013. Proceedings of the Numerical Methods for Hyperbolic Equations: Theory and Applications. International Conference to honour Professor E.F. Toro in the month of his 65th birthday. Santiago de Compostela, Spain, July 4 – 8, 2011
http://hal.archives-ouvertes.fr/hal-00637102/
4. **D. Dutykh**, Th. Katsaounis and D. Mitsotakis. *Dispersive wave runup on non-uniform shores*. In: Fort J., Fürst J., Halama J., Herbin R., Hubert F. (Eds.) Finite Volumes For Complex Applications VI — Problems & Perspectives. Springer Proceedings in Mathematics, 2011, Volume **4**, Part 1, pp. 389–397
http://hal.archives-ouvertes.fr/hal-00553762/

**— 2009 —**

3. **D. Dutykh** and F. Dias. *How does sedimentary layering affect the generation of tsunamis?* Proceedings of OMAE 2009, 28th International Conference on Ocean, Offshore and Arctic Engineering, May 31 – June 5, 2009, Honolulu, USA

— 2008 —

2. **D. Dutykh**. *Visco potential free-surface flows*, XXII ICTAM, 25 – 29 August 2008, Adelaide, Australia
1. F. Dias, **D. Dutykh** and J.-M. Ghidaglia. *Simulation of Free Surface Compressible Flows Via a Two Fluid Model*, OMAE 2008, Nick NEWMAN's Symposium on Marine Hydrodynamics, **6**, 43–50, 2008. Estoril, Portugal, 15 – 20 June 2008
http://hal.archives-ouvertes.fr/hal-00258161/

### 8.5.6 Conference proceedings

— 2025 —

32. S. Mukherjee, L. Vuillon, L. Bou-Nassif, S. Giroux-Julien, H. Pabiou, **D. Dutykh** and I. Tsanakas. *Temporal Graph Neural Networks for Early Anomaly Detection and Performance Prediction via PV System Monitoring Data* in Proc. 42nd European Photovoltaic Solar Energy Conference and Exhibition (EU PVSEC 2025), Bilbao, Spain, 020338, pp. 001–003, 2025
https://arxiv.org/abs/2512.03114/
31. M. Zafar, P. Sabatier, **D. Dutykh**, H. Jomard, W. Rapuc, P. Lajeunesse and E. Chapron. *Modeling earthquake-induced seiche process and subsequent homogenite deposits in lacustrine setting*, EGU General Assembly 2025, Vienna, Austria, 27 Apr – 2 May 2025, EGU25-2461, 2025
https://doi.org/10.5194/egusphere-egu25-2461/

— 2024 —

30. A. Rashidi, **D. Dutykh**, Ch. Beck and M. Najaftomaraei. *Unraveling Tsunami Hazards along the Lesser Antilles, Eastern Caribbean Plate*, Proceedings of the 21st Iranian Geophysical Conference, December 10-11, Tehran, Iran, pp. 1318–1323, 2024
29. S. Mukherjee, L. Vuillon, **D. Dutykh** and I. Tsanakas. *Dimensionality Reduction of Environmental Data for Long-Term PV Performance Analysis Using Graph Based Methods*, EU PVSEC 2024, **020411**, pp. 001–004, 2024
https://userarea.eupvsec.org/proceedings/EU-PVSEC-2024/4CV.1.27/
28. C. Cárdenas-Bravo, S. Lespinats and **D. Dutykh**. *A Practical Example of the Impact of Uncertainty on the One-Dimensional Single-Diode Model*, 2024 IEEE International Symposium on Systems Engineering (ISSE), Perugia, Italy, pp. 1–5, 2024
https://ieeexplore.ieee.org/document/10741111/
27. M. Zafar, **D. Dutykh**, P. Sabatier, M. Banjan and J. Kim. *Uncovering a Paleotsunami Triggered by Mass-Movement in an Alpine Lake*. In: P. Gourbesville, G. Caignaert (Eds.) Advances in Hydroinformatics — SimHydro 2023 Volume 2. SimHydro 2023. Springer Water. Springer, Singapore, pp. 47–64, 2024
https://hal.science/hal-04261487/
26. A. Rashidi, M. Mokhtari, **D. Dutykh**, M. Masoodi, P. Faridi and S. Kiani. *A glimpse into tsunami hazard research in Makran with a focus on complex tsunamis*. The 9$^{th}$ International Conference on Seismology and Earthquake Engineering, Tehran, Iran, 2024
25. F. Carbone and **D. Dutykh**. *Route to chaos and resonant triads interaction in a truncated Rotating Nonlinear shallow-water model*, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024,

EGU24–13180, 2024
https://doi.org/10.5194/egusphere-egu24-13180/

**— 2023 —**

24. M. Bowen, P. Samozino, M. Vonderscher, **D. Dutykh** and B. Morel. *Development of a force-endurance model able to describe the muscle fatigability in the severe domain and validation of the RACLET test*, 28th annual congress of the European College of Sport Science, Paris, France, 4–7 July 2023
https://hal.science/hal-04084767/

**— 2022 —**

23. C. Cardenas-Bravo, **D. Dutykh** and D.L. Ha. *Computation of Faulty IV Curves Based on a Distributed Solar Cell Algorithm*, 8th World Conference on Photovoltaic Energy Conversion, Milano, Italy, 26–30 September 2022, pp. 701–703
https://www.eupvsec-proceedings.com/proceedings?char=C&paper=50890
22. M. Cugnet, F. Gallois, A. Kirchev and **D. Dutykh**. *Understanding self-discharge or what's going on inside a lead-acid battery when nothing happens outside*, AABC Europe — 12th International Advanced Automotive Battery Conference, 13 – 15 June 2022, Mainz, Germany
https://hal.insa-toulouse.fr/LITEN/cea-03715092/
21. I. Didenkulova, A. Abdalazeez, **D. Dutykh** and P. Denissenko. *Effects of coastal roughness on long wave runup*, EGU General Assembly 2022, Vienna, Austria, 23 – 27 May 2022, EGU22-8520
https://doi.org/10.5194/egusphere-egu22-8520/

**— 2020 —**

20. A. Rashidi, **D. Dutykh**, Z.H. Shomali and N.K. Farajkhah. *Probabilistic Tsunami Hazard of Splay and Normal Faults in the Western Makran/Iran*, The Second Eurasian RISK-2020 Conference and Symposium, Tbilisi, Georgia
https://www.researchgate.net/publication/343336669_Probabilistic_Tsunami_Hazard_of_Splay_and_Normal_Faults_in_the_Western_MakranIran/
19. S. Rusconi, **D. Dutykh**, A. Zarnescu, D. Sokolovski and E. Akhmatskaya. *Population balance approach for predicting polymer particles morphology*. The 11th Conference on Dynamical Systems Applied to Biology and Natural Sciences (DSABNS), Trento, Italy, 4 – 7 February 2020. ISBN: 978-989-98750-7-4
https://hal.archives-ouvertes.fr/hal-02518323/

**— 2019 —**

18. A. Abdalazeez, I. Didenkulova and **D. Dutykh**. *Nonlinear deformation and run-up of tsunami waves of positive polarity: numerical simulations and analytical predictions*. The First Eurasian conference "Risk – 2019", Baku, Azerbaijan; 22 – 24 May 2019
https://hal.archives-ouvertes.fr/hal-02145105/

**— 2017 —**

17. I. Didenkulova, D. Senichev and **D. Dutykh**. *Statistics for long irregular wave run-up on a plane beach from direct numerical simulations*. European Geosciences Union, Vienna, Austria; 23 – 28 April 2017. Geophysical Research Abstracts, **19**, p. 19133 https://meetingorganizer.copernicus.org/EGU2017/EGU2017-19133.pdf
16. D. Senichev, I. Didenkulova and **D. Dutykh**. *Study of statistics of random wave runup on a beach using direct numerical simulations*. Proceedings of the 23rd International scientific conference "Information Systems and Technologies" (IST–2017), 21st April 2017, Nizhny Novgorod, Russia; pp. 1001–1006 (in Russian)
15. M. Gisclon, M. Chhay, **D. Dutykh** and C. Ruyer-Quil. *Modélisation mathématique des films minces avec applications aux transferts de chaleur*. Congrès Français de Mécanique, Lille, France; 28 August – 1 September 2017. Publisher: Association Française de Mécanique (AFM) https://cfm2017.sciencesconf.org/135353/template_cfm2017_vf.pdf

**— 2016 —**

14. I. Didenkulova, **D. Dutykh** and D. Senichev. *Statistics for long wave run-up on a slopping beach: theoretical predictions and experimental data*. 15th Plinius Conference on Mediterranean Risks, Giardini Naxos, Italy; 8–11 June 2016. Publisher: Plinius Conference Abstracts, **15**, p. 38.

**— 2015 —**

13. M. Chhay, **D. Dutykh**, M. Gisclon and C. Ruyer-Quil. *Modélisation mathématiques des films minces*. Congrès Français de Mécanique, Lyon, France; 24 – 28 August 2015. Publisher: Association Française de Mécanique (AFM)
12. D. Clamond, **D. Dutykh** and A. Galligo. *Computer Algebra Applied to a Solitary Waves Study*. In Proceedings of the 2015 ACM on International Symposium on Symbolic and Algebraic Computation – ISSAC'15, ACM Press, 125–132, 2015

**— 2014 —**

11. A. Galligo, **D. Dutykh** and D. Clamond. *On detection of solitary waves, using phase diagrams and real discriminant*. In J.E. Garcia, J. Fernández Sánchez and M. Sombra (Eds.), Proceedings of the Encuentros de Algebra Computacional y aplicaciones Conference, Barcelona, Spain, 18 – 20 June 2014, pp. 119–122
10. **D. Dutykh** and J.-G. Caputo. *Sine-Gordon dynamics on graphs*. In O.O. Kochubey *et al*. (Eds.), Applied problems of the fluid mechanics and heat and mass transfer (pp. 29–31). Dnipropetrovsk, Ukraine. Dnipropetrovsk National University. pp 192, 2014
9. **D. Dutykh** and D. Clamond. *Modified Shallow Water Equations for Mild-slope Seabeds*. IWWWFB29, 30 March – 2 April 2014, Osaka, Japan, IWWWFB, 2014
8. F. Carbone, **D. Dutykh**, J.M. Dudley and F. Dias. *Extreme wave run-up on a vertical cliff*. In Zbornik radova konferencije MIT 2013, pp. 99–103, Beograd, Serbia, 2014

**— 2013 —**

7. F. Carbone, **D. Dutykh**, J.M. Dudley and F. Dias. *Extreme wave run-up on a vertical cliff*. IWWWFB28, 7 – 10 April 2013, Marseille, France
https://hal.archives-ouvertes.fr/hal-00927313/

**— 2012 —**

6. **D. Dutykh**, D. Mitsotakis, S. Beisel and N. Shokina. *On waves generated by an underwater landslide*. In Proceedings of the IV Conference "Applied Problems of the fluid mechanics, heat and mass transfer", 1 – 3 November 2012, Dnepropetrovsk, Ukraine, pp. 75–79, 2012

**— 2010 —**

5. **D. Dutykh**, Th. Katsaounis and D. Mitsotakis. *Finite volume schemes for Boussinesq type equations*. Proceedings of Colloque EDP-Normandie, 28 – 29 October 2010, Caen, France, pp. 15–21
http://hal.archives-ouvertes.fr/hal-00553754/
4. D. Clamond and **D. Dutykh**. *Dispersive wave equation derivation from a relaxed variational formulation*. Applied problems of aerodynamics, heat and mass transfers, pp. 40–42, Dnepropetrovsk, Ukraine, November 4 – 6, 2010

**— 2008 —**

3. **D. Dutykh** and F. Dias. *Tsunami wave energy*, 4th Canadian Conference on Geohazards, Universit Laval, May 20 – 24, 7 p., 2008
https://hal.science/hal-04474849/

**— 2007 —**

2. **D. Dutykh** and F. Dias, *Fault dynamics and tsunami generation*, ECCOMAS Thematic Conference on Multi-scale Computational Methods for Solids and Fluids. A. Ibrahimbegovic, F. Dias, H. Matthies, P. Wriggers (eds.), 79–81, 2007
1. C. Kassiotis, F. Dias, A. Ibrahimbegovic and **D. Dutykh**, *A partitioned approach to model tsunami impact on coastal protections*, ECCOMAS Thematic Conference on Multi-scale Computational Methods for Solids and Fluids. A. Ibrahimbegovic, F. Dias, H. Matthies, P. Wriggers (eds.), 134–139, 2007

### 8.5.7 Research reports

**— 2020 —**

7. J. Berger, T. Busser, **D. Dutykh** and N. Mendes. Retraction notice to "*On the estimation of moisture permeability and advection coefficients of a wood fibre material using the optimal experiment design approach*" [Exp. Therm. Fluid Sci. 90 (2017) 246–259], Exp. Therm. Fluid Sci., **113**, 109808, 2020

**— 2016 —**

6. **D. Dutykh**. *How to overcome the Courant-Friedrichs-Lewy condition of explicit discretizations?*. Technical report, 20 pp, 2016
https://hal.archives-ouvertes.fr/hal-01401125/

**— 2015 —**

5. **D. Dutykh** and L. Gosse. *Some topics of the NumHyp–2015' discussion session*. 25 pp., 2015
https://hal.archives-ouvertes.fr/hal-02485301/

**— 2014 —**

4. Ch. Lalanne, A. Rafiee, **D. Dutykh**, M. Lysaght and F. Dias. *Enabling the UCD–SPH code on the Xeon Phi*, 12 pp., 2014
http://hal.archives-ouvertes.fr/hal-00927227/

**— 2011 —**

3. F. Fedele and **D. Dutykh**. *Hamiltonian description and traveling waves of the spatial Dysthe equations*, 2011
http://hal.archives-ouvertes.fr/hal-00632862/

**— 2009 —**

2. F. Dias, **D. Dutykh** and J.-M. Ghidaglia. *A compressible two-fluid model for the finite volume simulation of violent aerated flows. Analytical properties and numerical results*, 2009
http://hal.archives-ouvertes.fr/hal-00279671/

**— 2008 —**

1. **D. Dutykh**. *Visco-potential free-surface flows*. Research report of CMLA, 2008

### 8.5.8 Book reviews

1. M. S. Howe, *Hydrodynamics and Sound*, Cambridge University Press, Cambridge (2007). European Journal of Mechanics - B/Fluids, **27(2)**, 218, 2008
https://hal.science/hal-04474814/

### 8.5.9 Theses

5. **D. Dutykh**. *Mathematical modeling in the environment*, Habilitation à diriger des recherches, University of Savoie, 2010
http://tel.archives-ouvertes.fr/tel-00542937/
4. **D. Dutykh**. *Mathematical modeling of tsunami waves*, PhD thesis, CMLA, ENS de Cachan, 2007
http://tel.archives-ouvertes.fr/tel-00194763/
3. **D. Dutykh**. *Moving load on a layered floating ice sheet*, Master 2 MN2MC thesis, CMLA, ENS de Cachan, 2005
https://arxiv.org/abs/2402.16890/
2. **D. Dutykh**. *Harmonic oscillations of an inhomogeneous elastic layer under the action of a stamp* (in Ukrainian), Master thesis, Faculty of Applied Mathematics, Oles Honchar National University, Dnepropetrovsk, Ukraine, 2004

1. **D. Dutykh**. *Harmonic oscillations of an inhomogeneous elastic layer* (in Ukrainian), Bachelor thesis, Faculty of Applied Mathematics, Oles Honchar National University, Dnepropetrovsk, Ukraine, 2003

### 8.5.10 Various writings

5. **D. Dutykh**. *How to overcome the Courant-Friedrichs-Lewy condition of explicit discretizations?*. Technical report, 20 pp, 2016
https://hal.archives-ouvertes.fr/hal-01401125/
4. **D. Dutykh**. *A brief introduction to pseudo-spectral methods: application to diffusion problems*. Lecture notes, 38 pp, 2016
https://cel.archives-ouvertes.fr/cel-01256472/
3. **D. Dutykh**. *Introduction into Hydrodynamics. Variational point of view*. Lecture notes (work in progress), 142 pp, 2015
https://github.com/dutykh/hydro/
2. **D. Dutykh**. *My favourite books, papers and software libraries*. Informal notes (work in progress), 18 pp, 2015
https://github.com/dutykh/libs/
1. **D. Dutykh**. *Mathematical exercises (with solutions)*. Work in progress, 40 pp, 2015
https://github.com/dutykh/exos/

### 8.5.11 General audience articles

25. **D. Dutykh**. *How farming helped cause two medieval tsunamis in an Italian lake, and why the danger has returned*. Science X Dialog, Phys.org, October 22, 2026
https://phys.org/news/2026-10-farming-medieval-tsunamis-italian-lake.html
24. Press releases on the paper *Human-enhanced tsunami hazard in large lakes*, **Nat. Commun.**, 2026:

    *Quand les activités humaines amplifient le risque de tsunami dans les grands lacs*. Communiqué de presse, CNRS, 16 octobre 2026
    https://www.cnrs.fr/fr/presse/quand-les-activites-humaines-amplifient-le-risque-de-tsunami-dans-les-grands-lacs

    *Menschliche Eingriffe können Tsunamigefahr in Seen erhöhen*. Medienmitteilung, Universität Bern, 15. Oktober 2026
    https://mediarelations.unibe.ch/medienmitteilungen/2026/medienmitteilungen_2026/tsunamigefahr/index_ger.html
23. **D. Dutykh**. *Forecasting air drag with deep learning helps predict when uncontrolled satellites and rocket stages fall to Earth*. Science X Dialog, Phys.org, October 16, 2026
https://phys.org/news/2026-10-air-deep-uncontrolled-satellites-rocket.html
22. **D. Dutykh**. *Three quantum-inspired cores leave opposite fingerprints on the ringing of black holes*. Science X Dialog, Phys.org, October 9, 2026
https://phys.org/news/2026-10-quantum-cores-fingerprints-black-holes.html
21. **D. Dutykh**. *The 2025 Kamchatka tsunami kept the Pacific ringing for more than ten hours. Its rhythm could help tsunami warnings*. Science X Dialog, Phys.org, October 1, 2026
https://phys.org/news/2026-09-kamchatka-tsunami-pacific-ten-hours.html

20. *Smooth black hole cores alter the ringing tones of disturbed spacetime*. The Primary, 30 September 2026
https://theprimary.com/science/2026-09-30/hayward-black-hole-ringdown
19. E. Papadima. *Κρυφή συμμετρία στις ταλαντώσεις πολυδιάστατων μαύρων τρυπών και πώς τη σπάει η θεωρία χορδών*. FOXreport.gr, 26 September 2026
https://www.foxreport.gr/science/kryfi-symmetria-stis-talantoseis-polydiastaton-mavron-trypon-kai-pos-ti-spaei-theoria-chordon/85391/
18. **D. Dutykh**. *Higher-dimensional black holes hide an exact symmetry in their ringing, and string-inspired gravity breaks it*. Science X Dialog, Phys.org, September 24, 2026
https://phys.org/news/2026-09-higher-dimensional-black-holes-exact.html
17. **D. Dutykh**. *Catching the wave: Disciplines unite to unearth a lake's hidden history*. Khalifa University Science and Tech Review, 16 September 2026
https://kustreview.com/catching-the-wave/
16. Press releases on the paper *On the parameters domain of the single-diode model*, **Solar Energy**, 2024:

    *À l'USMB, des recherches sont menées pour optimiser la modélisation des systèmes photovoltaïques*. Université Savoie Mont Blanc, 25 juillet 2024
    https://www.univ-smb.fr/2024/07/25/a-lusmb-des-recherches-sont-menees-pour-optimiser-la-modelisation-des-systemes-photovoltaiques/

    *Breakthrough Algorithm to Enhance Solar Panel Efficiency*. Khalifa University News, 15 July 2024
    https://www.ku.ac.ae/breakthrough-algorithm-to-enhance-solar-panel-efficiency
15. Press releases on the paper *Numerical Reconstruction of Landslide Paleotsunami Using Geological Records in Alpine Lake Aiguebelette*, **J. Geophys. Res. Solid Earth**, 2024:

    *Mathematical Model Leads to Discovery of 12,000-Year-Old Ancient Tsunami in Alpine Lake*. Khalifa University News, 4 June 2024
    https://www.ku.ac.ae/mathematical-model-leads-to-discovery-of-12000-year-old-ancient-tsunami-in-alpine-lake

    *Un tsunami préhistorique dans les Alpes françaises : découverte et implications d'un glissement de terrain sous-lacustre de 11 700 ans*. Communiqué de presse, Université Savoie Mont Blanc, 30 mai 2024
    https://www.univ-smb.fr/wp-content/uploads/2024/06/cp-usmb-un-tsunami-prehistorique-dans-les-alpes-francaises-30-mai-2024.pdf
14. ClassNeRV, une nouvelle méthode extrêmement efficace pour l'analyse de données, Le Fil d'Actualité de l'Université Savoie Mont Blanc, December 9, 2020
13. A dive into tsunami — the terrifying energy it possesses! Scrivial.com, 2015
12. C. Acary-Robert, **D. Dutykh** and M. Gisclon. *Un modello per simulare numericamente le valanghe di neve*. Translation by Roberto Natalini, 22 March 2012
https://maddmaths.simai.eu/divulgazione/un-modello-per-simulare-numericamente-le-valanghe-di-neve-2/
11. C. Acary-Robert, D. Bresch and **D. Dutykh**. *Simulation d'avalanches de neige*. Actualités scientifiques de l'INSMI (CNRS), 14 mars 2012
http://www.cnrs.fr/insmi/spip.php?article441

10. C. Acary-Robert, **D. Dutykh** and M. Gisclon. *Une approche pour simuler des avalanches de neige*. Images des mathématiques, 28 décembre 2011 https://images.math.cnrs.fr/une-approche-pour-simuler-des-avalanches-de-neige/
9. *Tsunamis: gare aux "avalanches" et à la deuxième vague*. Le Monde, 1er Octobre 2011
8. *New research may explain high runup from tsunami waves*. PhysicsCentral, Blog of the American Physical Society. September, 19, 2011
7. *Tsunami Puzzle Explained*. Physical Review Focus, 16 September 2011 https://physics.aps.org/story/v28/st11
6. *Tuned into Earth*. CNRS International Magazine, **21**, April 2011
5. *Quelle est la différence entre un tsunami et un raz-de-marée?*, Slate.fr, 24 mars 2011 http://www.slate.fr/story/36093/difference-tsunami-raz-de-maree
4. *Les maths à l'écoute de la Terre*. Le Journal du CNRS, N° 245, juin 2010
3. *Simuler une avalanche*. La Recherche N428 – Avril 2009 https://flipbook.larecherche.fr/simuler-une-avalanche/
2. *Springy sediments may amplify tsunamis*. Issue 2662 of New Scientist magazine, 25 June 2008, page 20 https://www.newscientist.com/article/1894608-springy-sediments-may-amplify-tsunamis/
1. *Comment naît un tsunami?*, Le Mensuel de l'Université, N° 23, Février 2008

## 8.6 Special issues editor

- "*Ten Years after the 2011 Tohoku Tsunami: Social and Environmental Impacts, Lessons Learned, and New Perspectives*" at Geosciences (ISSN 2076-3263, MDPI). Co-editor: Dr. Amin RASHIDI (Institute of Geophysics, University of Tehran, Iran). Submission deadline: the 31st of August 2021. https://www.mdpi.com/journal/geosciences/special_issues/2011_Tohoku_tsunami/
- "*Mathematical Modeling of Sediment Transport in Coastal Areas*" at Water (ISSN 2073-4441, MDPI). Submission deadline: the 31st of August 2021. https://www.mdpi.com/journal/water/special_issues/math_model_sediment_transport/

## 8.7 Referee activities

### 8.7.1 Mathematical databases

- MathSciNet
- Zentralblatt-MATH

### 8.7.2 International Journals

- Chaos, Solitons & Fractals
- Physical Review Letters
- Communications Physics
- Scientific Reports
- Journal of Fluid Mechanics
- Journal of Computational Physics
- Journal of Engineering Mathematics
- Theoretical and Computational Fluid Dynamics
- Journal of Nonlinear Science
- Journal of Hydraulic Research
- Journal of Applied Analysis
- Annals of Physics
- Applied Ocean Research
- American Journal of Physics
- Journal de Mathématiques Pures et Appliquées
- Journal of Mathematical Analysis and Applications
- Journal of Waterway, Port, Coastal, and Ocean Engineering
- Water Waves: An interdisciplinary journal
- Environmental Fluid Mechanics

- Acta Applicandae Mathematicae
- Applied Mathematical Modelling
- Applied Mathematics and Computation
- ESAIM: Mathematical Modelling and Numerical Analysis
- Computer Physics Communications
- Computers and Mathematics with Applications
- Computational and Applied Mathematics
- Numerical Algorithms
- Applied Numerical Mathematics
- Communications in Nonlinear Science and Numerical Simulation
- Coastal Engineering
- AIMS Mathematics
- Analysis and Mathematical Physics
- Communications on Pure and Applied Analysis
- Pure and Applied Geophysics
- Comptes Rendus Mécanique
- European Journal of Mechanics - B/Fluids
- Mathematics and Computers in Simulation
- Journal of Computational and Applied Mathematics
- Mathematical Notes
- Numerische Mathematik
- Ocean Engineering
- Partial Differential Equations in Applied Mathematics
- Proceedings in Applied Mathematics and Mechanics
- Ocean Modelling
- Journal of Ocean Engineering and Marine Energy
- Journal of Building Performance Simulation
- Physica D: Nonlinear Phenomena
- Physics Letters A
- Physics of the Dark Universe
- SIAM Journal on Applied Mathematics
- SoftwareX
- Wave Motion
- Marine Pollution Bulletin
- Mathematical and Computational Applications
- Mathematical Methods in the Applied Sciences
- Atmosphere
- Geosciences
- Geohazards
- Oceans
- Optik
- Natural Hazards and Earth System Sciences
- Networks and Heterogeneous Media
- Building and Environment
- Earth, Planets and Space (EPS)
- ICE — Engineering and Computational Mechanics
- Journal of Advanced Chemical Engineering
- KSCE Journal of Civil Engineering
- Fundamental and Applied Hydrophysics
- Symmetry
- Kuwait Journal of Science & Engineering
- Zeitschrift für Naturforschung A
- Rendiconti del Circolo Matematico di Palermo
- Open Engineering
- Journal of Ocean Engineering and Science
- Journal of Engineering Research
- Polarforschung
- Nonlinear Dynamics
- Qualitative Theory of Dynamical Systems
- Journal of Marine Science and Engineering
- International Journal of Theoretical Physics
- Ocean Dynamics
- Mathematical and Computational Applications
- Biomimetics
- Frontiers in Physics
- Gulf Journal of Mathematics
- Frontiers in Plant Science
- Algorithms
- Control and Cybernetics
- Drones
- Fractal and Fractional
- Mathematics
- Electronics
- Energies
- Plants
- Sustainability
- Technologies
- Fire
- Prevention and Treatment of Natural Disasters
- Metascience in Aerospace
- AgriEngineering
- Agronomy
- Applied Sciences

- Computers in Industry
- Results in Physics
- Environmental Research
- Journal of Environmental & Earth Sciences
- World Electric Vehicle Journal
- Transportation Research Part E: Logistics and Transportation Review
- High Energy Density Physics
- Journal of Applied Physics
- Natural Hazards
- Ain Shams Engineering Journal

### 8.7.3 International Conferences

- OMAE 2026
- FVCA VII
- FVCA VI
- ISOPE 2010
- ICTAM 2008
- ISOPE 2007
- FVCA V

### 8.7.4 Book proposals

- Springer Science

### 8.7.5 Calls for proposals

- Expert for Crédit d'Impôt Recherche (CIR) in the field Engineering Mathematics (A4b3) at Ministère de l'Enseignement Supérieur et de la Recherche (MESR), France
- KU Leuven Industrial Research Fund Council
- Irish Research Council
- Israel Science Foundation
- KAUST Research Proposals
- Marie Curie COFUND
- Cluster Environnement Rhône-Alpes (Projet 2: Risques gravitaires, séismes)
- Service de coopération universitaire et scientifique, Ambassade de France en Ukraine
- Swiss National Science Foundation
- Oregon Sea Grant

## 8.8 Delivered talks

Most of the presentation slides can be downloaded from my home page:
http://www.denys-dutykh.com/talks.php

### 8.8.1 International conferences

- *Nonlinear dispersive wave propagation on planetary scales*, The Third Eurasian Risk Conference And Symposium, 7 – 9 December, 2021, Tbilisi, Georgia
- *On shallow capillary-gravity waves*, International Conference on Scientific Computation And Differential Equations (SciCADE–2015), 14 – 18 September, 2015, Potsdam, Germany
- *Fully nonlinear weakly dispersive travelling capillary–gravity waves*, 12$^{e}$ Colloque Franco-Roumain de Mathématiques Appliquées, 25 – 30 August 2014, Lyon, France
- *Fast and accurate computation of solitary waves of the free surface Euler equations*, 16 – 20 September 2013, International Conference on Scientific Computation and Differential Equations (SciCADE), Valladolid, Spain
- *Extreme wave run-up on a vertical cliff*, 5 – 8 September 2013, Mathematical and Informational Technologies (MIT), Vrnjacka Banja, Serbia
- *Fast and accurate computation of gravity solitary waves*, 25 – 28 March 2013. The Eighth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory. Athens (GA), USA
- *Relaxed Variational Principle for Water Wave Modeling*, June 13 – 16, 2012. SIAM Conference on Nonlinear Waves and Coherent Structures. The University of Washington, Seattle, USA
- *Dispersive wave runup and some related amplification phenomena*, 27 – 31 August 2011, International Conference "Mathematical and Informational Technologies", MIT-2011, Vrnjacka Banja, Serbia
- *Finite volume schemes for dispersive wave equations*, Numerical Methods for Hyperbolic Equations: Theory and Applications. International Conference to honour Professor E.F. Toro in the month of his 65th birthday, Santiago de Compostela, Spain, July 4 – 8, 2011
- *Dispersive wave runup on non-uniform shores*, Finite Volumes for Complex Applications VI, Prague, Czech Republic, June 6 – 10, 2011
- *Numerical simulation of a dispersive wave runup*, 4 – 7 April 2011, The Seventh IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory, Athens, Georgia, USA
- *Modeling and simulation of compressible two-phase flows*, 17th September 2010, NumAn 2010 Conference in Numerical Analysis, Crete, Greece
- *Tsunami wave modeling*, 6 April 2010, "Exploring structural controls on great earthquake rupture and architecture of the Sunda/Sumatran convergent margin: international collaboration, links to tsunami modeling and planning of future research activities", Fondation des Treilles, France
- *Visco potential free-surface flows*, XXII International Congress of Theoretical and Applied Mechanics, Adelaide, Australia, 24–30 August 2008
- *Tsunami wave energy*, SIAM Conference on Nonlinear Waves and Coherent Structures (NW08), Universit di Roma "La Sapienza", Rome, Italy, July 21–24, 2008

- *Simulation of Free Surface Compressible Flows Via a Two Fluid Model*, The 27th International Conference on OFFSHORE MECHANICS AND ARCTIC ENGINEERING (OMAE 2008), Estoril, Portugal, 15 – 20 June, 2008
- *Simulation of free surface motions via a two fluid model*, International conference "Trends in Numerical and Physical Modeling for Industrial Multiphase Flows", September 17 – 21, 2007, Cargèse, Corsica, France
- *On the generation of tsunamis by earthquakes*, The Fifth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory, Athens, GA, USA, April 16 – 19, 2007
- *Tsunami generation*, SIAM Conference on Nonlinear Waves and Coherent Structures, September 9 – 12, 2006, University of Washington, Seattle, Washington

### 8.8.2 Workshops

- *Nonlinear waves in $Y$ –junctions*, the 5$^{th}$ of July 2022, Nonlinear waves and networks (ONL2022), the 60$^{th}$ Birthday of Jean-Guy Caputo, INSA de Rouen, Rouen, France
- *On a supraconvergence phenomenon in finite difference schemes*, the 21$^{st}$ of December 2021, The Sixth International Talent Forum of Mathematical Sciences, (virtually) Nankai University, Tianjin, China
- *Nonlinear dispersive wave propagation on planetary scales*, the 8$^{th}$ of December 2021, Modèles asymptotiques et méthodes numériques pour les milieux continus et la biologie, Saint-Étienne, France
- *Water waves without tears*, 19 – 29 August 2019, Geometrical methods in nonlinear PDEs and critical phenomena, Wisla, Poland
- *Energy-consistent shallow water models derivation with improved dispersion relation*, 16 – 17 May 2016, French–Spanish Workshop on Evolution Problems (FSWEP16), Valladolid, Spain
- *Modelling of shallow dispersive water waves*, 14 – 20 June 2015, Numerical approximations of hyperbolic systems with source terms and applications (NumHyp–2015), Cortona, Italy
- *Application of variational principles to water wave modelling*, 2 – 4 July 2014, 16$^{th}$ RIMS Workshop "Mathematical Analysis in Fluid and Gas Dynamics", RIMS, Kyoto University, Kyoto, Japan
- *Capillary–gravity waves in nonlinear shallow water and full Euler equations*, 23 – 26 April 2014, Wave Interaction (WIN–2014), Linz, Austria
- *Wave propagation over rapidly varying bottoms. Excursion into variational methods*, 9 – 11 April 2014, Mathematical Modelling for Tsunami Early Warning Systems, Málaga, Spain
- *Relaxed variational principle for water wave modeling*, 6 – 10 May 2013, Workshop on Ocean Wave Dynamics, Fields Institute, Toronto, Canada
- *Extreme wave run-up on a vertical cliff*, 14 – 18 April 2013, IUTAM Symposium 2013: Nonlinear interfacial wave phenomena from the micro to the macro-scale, Limassol, Cyprus
- *The emergence of coherent wave groups in deep-water random sea*, 14 – 18 April 2013, IUTAM Symposium 2013: Nonlinear interfacial wave phenomena from the micro to the macro-scale, Limassol, Cyprus
- *Extreme wave run-up on a vertical cliff*. 2$^{nd}$ ERC MULTIWAVE Workshop, 22 March 2013, University College Dublin, Ireland
- *Modified shallow water equations for large bathymetry variations*, 8 – 13 October 2012, "Mathematical modeling and analysis of extreme sea waves" Fondation des Treilles, Tourtour, France

- *Dispersive and non-dispersive wave runup and some related phenomena*, "The Mathematics of Extreme Sea Waves: Tsunamis, Rogue Waves, And Flooding" held at Fields Institute, Toronto, June 13 – 16, 2011
- *Modeling of tsunami wave generation*, 21st September 2010, Summer school and workshop on "Numerical Methods for interactions between sediments and water", Paris 13 University, France
- *A generalized variational principle for water wave modeling*, 11 December 2009, Hydrodynamique des lacs et approximation de Saint-Venant, Institut Jean le Rond d'Alembert, Université Pierre et Marie Curie (Paris 6), Paris, France
- *Powder-snow avalanche flow modelling*, 12 – 16 October 2009, 4th Russian-German Advanced Research Workshop, Freiburg, Germany
- *Tsunami wave energy*, 22 September 2009, Session "Numerical methods for complex fluid flows", Wolfgang Pauli Institute, Vienna, Austria
- *Numerical simulation of tsunami waves. Presentation of VOLNA code*. 27 January 2009, Océanographie et Mathématiques, Ecole Normale Supérieure, Paris, France
- *Influence of the mud layer on sea-bed deformations*, 2nd FORTH Workshop on Tsunami generation, 12 and 13 February, 2008, Heraklion, Crete (Greece)
- *On the dynamic generation of tsunamis by a moving bottom*, TRANSFER Workshop "Numerical Models, Inundation Maps and Test Sites", June 12 – 14, 2007, Fethiye, Turkey
- *Derivation and numerical resolution of long wave equations*, Wolfgang Pauli Institute, Vienna, Working session "Dispersive nonlinear longwave PDE's and applications in physics" organized by Jean-Claude Saut, 21 – 25 May 2007
- Conference *Results of the Sumatra Earthquake and Tsunami Offshore Survey 2005*, October 19 – 24, 2005, Fondation des Treilles

### 8.8.3 National conferences

- *Numerical simulation of dispersive waves*, the 29$^{th}$ October 2010, Colloque EDP Normandie, University of Caen, Caen, France
- *Simulation numérique dans l'hydrodynamique côtière*, 39e Congrès National d'Analyse Numérique (CANUM 2008), 26 – 30 of May 2008, Vendée, France

### 8.8.4 Participation in summer schools

- Summer school Wisla–19 "*Differential Geometry, Differential Equations, and Mathematical Physics*", 19 – 29 August 2019, Wisla, Poland
  http://www.baltinmat.com/summer-school-workshop-wisla-19/
- XXII Summer Diffiety School on the Geometry of PDEs, 20 – 28 July 2019, Lizzano-in-Belvedere, Italy
  https://sites.google.com/site/levicivitainstitute/Activities/DiffietySchools/xxii-summer-diffiety-school/
- Summer school Zetas–2018: "*Zeta functions, polyzeta functions, arithmetical series: applications to motives and number theory*", 18 – 29 of June 2018, University Savoie Mont Blanc, Le Bourget-du-Lac, France
  https://etzetas2018.sciencesconf.org/
- XXI Summer Diffiety School on the Geometry of PDEs, 19 – 31 July 2018, Lizzano-in-Belvedere, Italy
  https://sites.google.com/site/levicivitainstitute/Activities/DiffietySchools/xxi-summer-diffiety-school/

### 8.8.5 Seminars

- *Implicit ODEs and capillary-gravity solitary waves*, the 24th of October 2024, Mathematics Seminar, Mathematics Department, Khalifa University of Science and Technology, Abu Dhabi, UAE
- *A Hamiltonian regularization of shallow water waves*, the 24th of January 2024, Mathematics Seminar, Mathematics Department, Khalifa University of Science and Technology, Abu Dhabi, UAE
- *A gentle introduction to the water wave problem*, the 6th of October 2022, CoSMiC Seminar, Mathematics Department, Khalifa University of Science and Technology, Abu Dhabi, UAE
- *A Hamiltonian regularization of shallow water waves*, the 7th of May 2021, Applied Mathematics virtual Seminar, Mathematics Institute, The University of Warwick, UK
- *A Hamiltonian regularization of shallow water waves*, the 12th of March 2021, Virtual Seminar of the team $\mathcal{A}^3$, LAMFA UMR 7352, Université de Picardie Jules Verne, Amiens, France
- *A Hamiltonian regularization of shallow water waves*, the 9th of March 2021, Virtual Seminar of Information and Computational Technologies, ICT SB RAS, Novosibirsk, Russia
- *A Hamiltonian regularization of shallow water waves*, the 4th of March 2021, Virtual Seminar of the PDEs team, LMA UMR 7348, University of Poitiers, Poitiers, France
- *Computation of solitary wave solutions*, the 3rd December 2019, Seminar at M2C UMR 6143, Morphodynamique Continentale et Côtière, Université Caen Normandie, Caen, France
- *Nonlinear dispersive water wave modelling: mastering the dispersion*, 18 November 2019, Seminar of the team $\mathcal{A}^3$, LAMFA UMR 7352, Université de Picardie Jules Verne, Amiens, France
- *Nonlinear dispersive water wave modelling. Part 2: mastering the dispersion*, the 27th of September 2019, EDF R&D Center & Saint-Venant Hydraulics Laboratory, Chatou, France
- *Nonlinear dispersive water wave modelling. Part 1: the variational approach*, the 26th of September 2019, EDF R&D Center & Saint-Venant Hydraulics Laboratory, Chatou, France
- *Variational approach to water wave modelling*, 2 November 2017, Marine Systems Institute seminar, Tallinn University of Technology, Estonia
- *Water waves without tears*, 11 May 2017, Colloquium of Mathematics, Laboratoire de Mathématiques Raphaël Salem, Université de Rouen, France
- *Water waves without tears*, 16 March 2017, Colloquium of Mathematics and Statistics, Victoria University of Wellington, Wellington, New Zealand
- *On the complete classification of shallow travelling capillary-gravity solitary waves*, 14 November 2016, Seminar of the team $\mathcal{A}^3$, LAMFA UMR 7352, Université de Picardie Jules Verne, Amiens, France
- *On the complete classification of shallow travelling capillary-gravity solitary waves*, 10 November 2016, Groupe de Discussions, LAMA UMR 5127, Université Savoie Mont Blanc, France
- *Numerical methods on moving grids: une histoire de* $\mathcal{Q}$, 18 February 2016, Groupe de Discussions, LAMA UMR 5127, Université Savoie Mont Blanc, France
- *Computation of travelling wave solutions*, 20 May 2015, Groupe de Discussions, LAMA UMR 5127, Université Savoie Mont Blanc, France
- *Families of steady fully nonlinear shallow capillary-gravity waves*, NUMERIWAVES Seminar, 25 February 2015, Basque Center for Applied Mathematics (BCAM), Bilbao, Spain
- *Families of shallow capillary-gravity waves*, GIR Análisis Numérico de Problemas de Evolución, 20 February 2015, Instituto de Matemáticas imUVa, Universidad de Valladolid, Spain

- *Relaxed variational principle for water wave modeling*, Kolloquium Angewandte Mathematik, Friedrich-Alexander Universität Erlangen-Nürnberg, 13 November 2014, Erlangen, Germany
- *Resonant wave run-up on sloping beaches and vertical walls*, Seminar: Conservation Laws and Invariants of PDEs of Hydrodynamic type (16h00), 24 October 2014, Institute of Computational Technologies SB RAS, Novosibirsk, Russia
- *Relaxed variational principle for water wave modelling*, Seminar: Conservation Laws and Invariants of PDEs of Hydrodynamic type (11h00), 24 October 2014, Institute of Computational Technologies SB RAS, Novosibirsk, Russia
- *Relaxed variational principle for water wave modelling*, Seminar of the Laboratory of Differential equations, 23 October 2014, Lavrentyev Institute of Hydrodynamics SB RAS, Novosibirsk, Russia
- *Algebraic geometry for shallow capillary-gravity waves*, Seminar: Computational Technologies, 21 October 2014, Institute of Computational Technologies SB RAS, Novosibirsk, Russia
- *Fully nonlinear weakly dispersive capillary-gravity waves*, 8 July 2014, Department of Mathematics, Keio University, Japan
- *Some resonance phenomena during the wave run-up*, 23 January 2013, Department of Applied Mathematics, University of Sevilla, Spain
- *A Variational Approach for Water Wave Modelling*, 18 January 2013, NUMERIWAVES Group Seminar, Basque Center for Applied Mathematics (BCAM), Bilbao, Spain
- *Some critical comments on the landslides modelling*, 26 October 2012, Wave Group Seminar, School of Mathematical Sciences, University College Dublin, Ireland
- *Relaxed variational principle for water wave modeling*, May, 25, 2012. Wave Group Seminar, School of Mathematical Sciences, University College Dublin, Ireland
- *Wave run-up on random and deterministic beaches*, April, 16, 2012. Mathematical Physics Seminar, School of Mathematics, Georgia Institute of Technology, Atlanta, GA, USA
- *Wave run-up on random and deterministic beaches*, March, 2, 2012. Basque Center for Applied Mathematics (BCAM), Bilbao, Spain
- *Relaxed variational principle for water wave modeling*, February, 7, 2012. imUVA Seminario, Universidad de Valladolid, Spain
- *Dissipative and resonant effects during the wave runup process*, February, 2, 2012. Séminaire d'Analyse Numérique et de Calcul Scientifique, Laboratoire de Mathématiques de Besançon, Université de Franche-Comté, France
- *Dissipative and resonant effects during a wave run-up*, October, 20, 2011. Fluid Mechanics Seminar, Department of Mathematics, University of Bergen, Norway
- *Relaxed variational principle for water wave modeling*, 14 October 2011, Seminar in Nonlinear Waves, Department of Mathematics, University of Bergen, Norway
- *Dispersive and non-dispersive wave runup on complex beaches*, 12 July 2011, Seminar of the Applied Mathematics Department, University of Valladolid, Valladolid, Spain
- *Mathematical modeling and numerical simulation of long water waves*, 21 March 2011, Séminaire d'Analyse Appliquée, LATP, Marseille, France
- *Relaxed variational principle for water wave modeling*, 13th March 2011, Séminaire d'analyse appliquée $A^3$, Laboratoire Amiénois de Mathématique Fondamentale et Appliquée, Amiens, France

- *Mathematical modelling of tsunami wave generation*, 12 November 2009, Institut Jean le Rond d'Alembert, Université Pierre et Marie Curie (Paris 6), Paris, France
- *Numerical simulation of powder snow avalanches*. 26 March 2009, Atelier VOR, Laboratoire 3S-R, Grenoble, France
- *Simulation of free surface compressible flows via a two fluid model*, 27 October 2008, Séminaire et Groupe de travail de Modélisation Mathématique, Mécanique et Numérique (M3N), Laboratoire de Mathématiques Nicolas Oresme, Université de Caen, Caen, France
- *Mathematical modelling of tsunami waves*, 23 October 2008, Séminaire EDP-MOISE, Laboratoire Jean Kuntzmann, Grenoble, France
- *Simulation of free surface compressible flows via a two fluid model*, 20 October 2008, Rencontres Niçoises de la Mécanique des Fluides, Laboratoire J.A. Dieudonné, Nice, France
- *Mathematical modelling of tsunami generation*, LAMA, Université de Savoie, 10 October 2008, Le Bourget-du-Lac, France
- *Numerical modelling of tsunami waves. VOLNA code presentation*, LAMA, Université de Savoie, 4th July 2008, Le Bourget-du-Lac, France
- *A two-fluid model for violent aerated flows*, LAMA, Université de Savoie, April, 24, 2008, Le Bourget-du-Lac, France
- *A two-fluid model for violent aerated flows*, Groupe de Travail Numérique, Université d'Orsay Paris-Sud, April, 16, 2008, Orsay, France
- *Simulation numérique des écoulements à surface libre*, Institut de Mécanique des Fluides de Toulouse, April, 11, 2008, Toulouse, France
- *Numerical modelling of tsunami generation and runup*, Groupe de Travail Mécanique des Fluides Réels, 18 February 2008, CMLA, ENS de Cachan, France
- *Simulation d'écoulements compressibles avec surface libre par un modèle bifluide*, CLAROM - Séminaire hydrodynamique et océano-météo, 29 novembre 2007, Institut Français du Pétrole
- *Viscous shallow water equations: potential approach and numerical methods*, 13 mars 2007, Institut de Mathématiques de Bordeaux, Groupe de travail Océanographie
- *Unstructured Finite Volume solver for dissipative shallow-water equations*, 12 février 2007, CMLA, ENS de Cachan, Groupe de travail mécanique des fluides réels
- *Génération des tsunamis*, Inauguration de LRC CMLA/CEA, 19 juin 2006

### 8.8.6 Short courses

**April 2016** Short course (8h) on Spectral methods at the PhD School on Numerical Methods for Diffusion Phenomena, Pontifical Catholic University of Parana, Curitiba, Brazil

- Lecture notes: https://cel.archives-ouvertes.fr/cel-01256472/

**April 2015** Short course (8h of Lectures) on "*A short introduction to Fluid Dynamics*", Basque Center for Applied Mathematics (BCAM). Course programme:

1. Review of (exterior) vector calculus
2. Eulerian description of fluid flows
3. Lagrangian description of fluids
4. Smoothed Particle Hydrodynamics

- Lecture notes: https://github.com/dutykh/hydro/

**December 2014** Short course (20h of Lectures + 15h of TDs) on "*Lagrangian and Eulerian approaches to water wave modelling*", Faculty of Mechanics and Mathematics, Al-Farabi Kazakh National University, Almaty, Kazakhstan.

- Lecture notes: https://github.com/dutykh/hydro/

**May 2013** *Numerical methods for fully nonlinear free surface water waves* (in collaboration with Dr. Claudio Viotti), 15 – 16 May 2013, Fields Institute, Thematic Program on the Mathematics of Oceans, Toronto, Canada
http://cel.archives-ouvertes.fr/cel-00825492/

### 8.8.7 Posters

6. S. Mustatea, S. Bolik, A. Amato, **D. Dutykh**, J. Jouhet, S. Lespinats and O. Bastien. *Phylogeny and sequence space: a combined approach to analyze the evolutionary trajectories of homologous proteins. The case study of Betain Lipid*, Poster presented at the (on site) International Symposia on Plant Lipids (ISPL 2022), Grenoble, France
https://ispl2020.sciencesconf.org/
5. A. Abdalazeez, I. Didenkulova, A. Kurkin, A. Rodin and **D. Dutykh**. *Runup of long waves on composite coastal slopes: numerical simulations and experiment*, Poster presented at the (virtual) EGU General Assembly, 6 May 2020, Vienna, Austria
https://www.researchgate.net/publication/341205844_Runup_of_long_waves_on_composite_coastal_slopes_numerical_simulations_and_experiment
4. A. Abdalazeez, I. Didenkulova, **D. Dutykh** and C. Labart. *Run-up of narrow and wide-banded irregular waves on a beach*, Poster presented at the (virtual) EGU General Assembly, 4 May 2020, Vienna, Austria
https://www.researchgate.net/publication/341175353_Run-up_of_narrow_and_wide-banded_irregular_waves_on_a_beach
3. A. Abdalazeez, I. Didenkulova and **D. Dutykh**. *Steepening and Run-up of Long Single Waves of Positive Polarity*, Poster presented at the Fourteenth International MEDCOAST Congress on Coastal and Marine Sciences, Engineering, Management and Conservation, 21 – 26 October 2019, Marmaris, Turkey
https://www.researchgate.net/publication/336482076_Steepening_and_Run-up_of_Long_Single_Waves_of_Positive_Polarity
2. A. Abdalazeez, T. Torsvik, **D. Dutykh**, P. Denissenko and I. Didenkulova. *Dispersive effects during long wave run-up on a beach*, Poster presented at EGU General Assembly, 7 – 12 April 2019, Vienna, Austria
https://www.researchgate.net/publication/332269054_Dispersive_effects_during_long_wave_run-up_on_a_beach
1. A. Abdalazeez, I. Didenkulova and **D. Dutykh**. *Nonlinear deformation and run-up of long single waves of positive polarity: numerical simulations and analytical predictions*, Poster presented at EGU General Assembly, 7 – 12 April 2019, Vienna, Austria
https://www.researchgate.net/publication/332268913_Nonlinear_deformation_and_run-up_of_long_single_waves_of_positive_polarity_numerical_simulations_and_analytical_predictions

### 8.8.8 General audience lectures

- *Astonishing Mathematics*. CMI students seminar at the University Savoie Mont Blanc, 12 December 2019, Le Bourget-du-Lac, France

- *Tsunamis: du terrain au modèle numérique*. General audience lecture with Professor Christian Beck (LGCA, University of Savoie) in the framework of the *Fête de la Science*, 21 November 2009, Cinéma Curial, Chambéry, France
- *What is applied mathematics?* Talk given for the general audience at École Normale Supérieure de Cachan, April, 27, 2007
- *Tsunami waves*. Talk given for the general audience at École Normale Supérieure de Cachan, December, 5, 2006

## 8.9 Software development

- Public `GitHub` repository, which contains most of the codes mentioned hereinbelow: https://github.com/dutykh/
- Responding to the need of a growing community of students and researchers who wants to get involved in the field of electrochemical storage systems, NEOLAB offers a new tool dedicated to a modeling domain where almost no open-source solutions exist. Physics-based models of batteries require extensive knowledge in thermodynamics, electro-chemistry, mathematics, material and computer sciences. Based on the idea that a minimum working example is the best way to learn gradually how to model a battery, NEOLAB provides a solution to simulate the behavior of the negative electrode of lead-acid batteries and a framework to investigate other primary and secondary technologies.
  - https://github.com/dutykh/NEOLAB/

  Useful reference:
  - M. Cugnet, F. Gallois, A. Kirchev and **D. Dutykh**. *NEOLAB: A Scilab tool to simulate the Negative Electrode of Lead-Acid Batteries*. SoftwareX, **22**, 101394, 2023
- The following repository contains a Fourier-type pseudo-spectral solver for the Dysthe–Lo–Mei equation as described in Equations (2.6) – (2.9) in Lo & Mei (JFM, 1985) paper mentioned below. A very high order Runge–Kutta scheme is used for time integration with the adaptive time stepping. We employ also the integrating factor technique to slightly remove the stiffness of second order derivatives. The work of this code is illustrated on a simple evolution of the ground state to the low order underlying NLS equation.
  - https://github.com/dutykh/DystheEq/

  Useful references:
  - E. Lo and C.C. Mei. *A numerical study of water-wave modulation based on a higher-order nonlinear Schrödinger equation*, J. Fluid Mech., **150**, 395–416, 1985
  - F. Fedele and **D. Dutykh**. *Hamiltonian form and solitary waves of the spatial Dysthe equations*. JETP Letters, **94**(12), 921–925, 2011
    http://hal.archives-ouvertes.fr/hal-00633389/
  - F. Fedele and **D. Dutykh**. *Hamiltonian description and traveling waves of the spatial Dysthe equations*, 2011
    http://hal.archives-ouvertes.fr/hal-00632862/
- This Matlab code computes irrotational 2D periodic steady surface pure gravity waves of arbitrary length in arbitrary depth. The formulation is based on the so-called Babenko equation and pseudo-

spectral discretization in the conformal domain. The resulting equation is solved using Petviashvili iteration method.

- https://github.com/dutykh/SSGW/

Useful reference:

- D. Clamond and **D. Dutykh**. *Accurate fast computation of steady two-dimensional surface gravity waves in arbitrary depth*. J. Fluid Mech., **844**, 491–518, 2018 https://hal.archives-ouvertes.fr/hal-01465813/

- This Matlab code solves the classical nonlinear sine-Gordon equation on graphs using a symplectic Euler scheme in time

  - https://github.com/dutykh/sineGordonGraph/

  Useful reference:

  - **D. Dutykh** and J.-G. Caputo. *Discrete sine-Gordon dynamics on networks*, Submitted, 2016 https://hal.archives-ouvertes.fr/hal-01160840/

- A simple Matlab code, which solves numerically 2D Navier–Stokes equations in vorticity formulation using a Fourier-type pseudo-spectral method

  - https://github.com/dutykh/NavierStokes2D/

- The present Matlab code is an implementation of the full Euler equations solver based on the method of conformal variables. The peculiarity here is that the solver works on general (but smooth) bottoms. The method is described in the reference given below. In a few words it is a Fourier-type pseudo-spectral solver. Standard Matlab time stepper is used to advance the solution in time. The solution is expected to be spectrally accurate

  - https://github.com/dutykh/Euler_bottom/

  Useful reference:

  - C. Viotti, **D. Dutykh** and F. Dias. *The conformal-mapping method for surface gravity waves in the presence of variable bathymetry and mean current*, Procedia IUTAM, **11**, 110–118, 2014 http://hal.archives-ouvertes.fr/hal-00855780/

- This function computes the steady irrotational surface solitary (classical and generalized) capillary-gravity wave solutions of the full Euler equations (homogeneous, incompressible and perfect fluids). The full Euler system is recast under the form of the Babenko equation using the conformal mapping technique. The wave is defined by its initial Froude and Bond numbers (Fr, Bo) and the result is about twelve digits accurate. The method works for all but the highest waves.

  - https://github.com/dutykh/BabenkoCG/

  Useful reference:

  - **D. Dutykh**, D. Clamond and A. Durán. *Efficient computation of capillary-gravity generalized solitary waves*, Wave Motion, **65**, 1–16, 2016 https://hal.archives-ouvertes.fr/hal-01218989/

- Fourier-type pseudo-spectral solver of the full Euler equations with the free surface on a fluid layer of infinite depth. The time-dependent fluid domain is transformed into a strip using the conformal

mapping technique. Time discretization is done using the embedded Cash-Karp method of the order 5(4). The time integration is improved using the integrating factor technique (i.e. exact integration of linear terms). The solver is initialized to simulate the celebrated Peregrine breather evolution in the full Euler.

- https://github.com/dutykh/ConformalEulerDeepWater/

- `SerreGravityWave.m`: This Matlab script is a pseudo-spectral solver for the Serre-Green-Naghdi equations which model the propagation of long gravity waves. Here, for the sake of simplicity, we restrict our attention to the case of the flat bottom. The numerical scheme is described in the following publication:
  - **D. Dutykh**, D. Clamond, P. Milewski and D. Mitsotakis. *Finite volume and pseudo-spectral schemes for the fully nonlinear 1D Serre equations*, European Journal of Applied Mathematics, **24**(5), 761–787, 2013
    http://hal.archives-ouvertes.fr/hal-00587994/
  - https://github.com/dutykh/SerreGravityWave/

- `sG\_solver.epd`: This script allows to solve numerically the sine-Gordon equation in a $Y$–junction geometry using the Finite Element Method (FEM). The scheme is of 2$^{nd}$ order in space and time. The implixit-explicit time stepping method is of the Crank–Nicolson type and it possesses excellent energy conservation properties.
  - https://github.com/dutykh/sineGordon_FreeFem/

- Participation in the PRACE DECI-9 project "*High-end computational modelling of wave energy converters*" (1$^{st}$ November 2012 - 31 December 2013). The final report is available at:
  - Ch. Lalanne, A. Rafiee, **D. Dutykh**, M. Lysaght, F. Dias. *Enabling the UCD-SPH code on the Xeon Phi*, 2014
    http://hal.archives-ouvertes.fr/hal-00927227/

- `SolitaryWave.m`: this script computes in ultra-fast way and potentially to the arbitrary accuracy the solitary waves to the full free-surface Euler equations. The method is based on the conformal map technique and the Petviashvili iteration. Some more technical details and numerical results can be found in the following papers:
  - D. Clamond and **D. Dutykh**. *Fast accurate computation of the fully nonlinear solitary surface gravity waves*. Computers & Fluids, **84**, 35–38, 2013
    http://hal.archives-ouvertes.fr/hal-00759812/
  - **D. Dutykh** and D. Clamond. *Efficient computation of steady solitary gravity waves*. Wave Motion, **51**, 86–99, 2014
    http://hal.archives-ouvertes.fr/hal-00786077/
  - https://github.com/dutykh/BabenkoSolitaryWave/

  www.mathworks.com/matlabcentral/fileexchange/39189-solitary-water-wave/

- `OkadaSol.m`: this script computes co-seismic displacements according to the classical Okada solution. For more details you can have a look at the original Okada (1985) paper or this freely available my publication:

- **D. Dutykh**, F. Dias, *Water waves generated by a moving bottom*. In Book:"Tsunami and Nonlinear Waves", Kundu, A. (Editor), Springer Verlag 2007, Approx. 325 p., 170 illus., Hardcover, ISBN: 978-3-540-71255-8
  http://hal.archives-ouvertes.fr/hal-00115875/
- https://github.com/dutykh/Okada/

http://www.mathworks.com/matlabcentral/fileexchange/39819

- **VOLNA**: a finite volume code on triangular unstructured meshes for the simulation of the generation, propagation and runup of tsunami waves. Developed in collaboration with Raphaël Poncet and Frédéric Dias. Currently this code is maintained by Irish Centre for High-End Computing (ICHEC) and School of Mathematical Sciences, University College Dublin. The code is described and validation tests are given in this article:
  - **D. Dutykh**, R. Poncet, F. Dias. *The VOLNA code for the numerical modelling of tsunami waves: generation, propagation and inundation*. European Journal of Mechanics B/Fluids, **30**(6), 598–615, 2011
    http://hal.archives-ouvertes.fr/hal-00454591/

### 8.9.1 Water waves, fluid dynamics & nonlinear systems

- **1DWaveTank** — a numerical wave tank for long-wave models (dispersive and non-dispersive) built on the finite-volume method, designed to test various models and schemes
  https://github.com/dutykh/1DWaveTank/
- **FlowReconstruct** — potential-flow reconstruction from Dirichlet data with periodic boundary conditions
  https://github.com/dutykh/FlowReconstruct/
- **Babenko-EHD** — an electro-hydrodynamic version of the Babenko equation — manuscript and accompanying codes, shared for reproducibility
  https://github.com/dutykh/Babenko-EHD/
- **SerreCyl** — supplementary materials for our manuscript on long-wave weakly-dispersive models for blood flows
  https://github.com/dutykh/SerreCyl/
- **TsunamiTutorial** — companion repository to our tsunami modelling paper in *Geosciences*
  https://github.com/dutykh/TsunamiTutorial/
- **LorenzUPOs** — computation of unstable periodic orbits (UPOs) for the Lorenz system
  https://github.com/dutykh/LorenzUPOs/
- **markov-etat** — scripts accompanying our work with L. Vuillon on the Markov–Frobenius uniqueness conjecture
  https://github.com/dutykh/markov-etat/

### 8.9.2 Black-hole physics & quasi-normal modes

- **qnm-analyser** — an online tool to chase quasi-normal modes in gravitational physics
  https://github.com/dutykh/qnm-analyser/
- **qm-qnms** — spectral method for eigenvalue problems — codes behind our manuscript "Spectral Method for Eigenvalue Problems"
  https://github.com/dutykh/qm-qnms/

- **ASafeGravity** — codes and materials for our papers on quasi-normal modes of black holes in asymptotically-safe gravity
  https://github.com/dutykh/ASafeGravity/
- **KS-quantum** — data and codes for quasi-normal modes of the Kazakov–Solodukhin quantum-corrected black hole
  https://github.com/dutykh/KS-quantum/
- **massless-SdS** — quasi-normal modes of massless (scalar, electromagnetic, gravitational) perturbations of the non-extreme Schwarzschild–de Sitter metric
  https://github.com/dutykh/massless-SdS/
- **massive-scalar** — massive scalar perturbations of the Schwarzschild solution
  https://github.com/dutykh/massive-scalar/
- **GaussBonnet** — analysis of Schwarzschild–Tangherlini and Gauss–Bonnet black holes
  https://github.com/dutykh/GaussBonnet/
- **noncommschwarzschild** — quasi-normal modes of a non-commutative model for Schwarzschild black holes
  https://github.com/dutykh/noncommschwarzschild/
- **dirtybh** — quasi-normal modes of noncommutative dirty black holes
  https://github.com/dutykh/dirtybh/
- **morristhorne** — quasi-normal modes of Morris–Thorne wormholes
  https://github.com/dutykh/morristhorne/
- **EllisBronnikov** — quasi-normal modes of Ellis–Bronnikov wormholes via the spectral method
  https://github.com/dutykh/EllisBronnikov/
- **ncwh** — quasi-normal modes of noncommutative wormholes
  https://github.com/dutykh/ncwh/
- **LeeWick** — quasi-normal modes of Lee–Wick black holes
  https://github.com/dutykh/LeeWick/

### 8.9.3 Quantum gravity & mathematical physics

- **tduality** — spectral Green functions for T-duality-regularised electrostatics in finite and bounded planar geometries; archived release supporting our *Proc. R. Soc. A* manuscript
  https://github.com/dutykh/tduality/
  https://doi.org/10.5281/zenodo.22096405

## 8.10 Scientific meetings organization

- Scientific Committee member of the 5$^{th}$ International Conference on Structural and Physical Aspects of Construction Engineering (SPACE-2022), 12 – 14 October 2022, Kosice, Slovakia
  https://space.uis.svf.tuke.sk/
- Scientific Committee member of the Civil Engineering Conference, 9 – 10 February 2022, Kosice, Slovakia
  https://cec.svf.tuke.sk/
- Program Committee member of the "2$^{nd}$ International Workshop on Advanced Information and Computation Technologies and Systems" (AICTS 2021), December 6 – 10 2021, Irkutsk, Russia
  https://aicts.icc.ru/

- Program committee member of the "3rd International Workshop on Information, Computation, and Control Systems for Distributed Environments" (ICCS-DE 2021), July 5 – 9 2021, Irkutsk, Russia
  https://iccs-de.icc.ru/en/
- Organizing committee member of the "Scientific Solar Summer School" (4Sun), 8 – 12 June 2020, Le Bourget-du-Lac, France
  https://www.univ-smb.fr/solaracademy/2020/02/28/4sun/

  *Organizing committee*: David Bailleul (Centre Antoine Favre/USMB), Fanck Barruel (PFE/INES), Lamia Berrah (LISTIC/USMB), Florence Besson (LOCIE/USMB), Denys Dutykh (CNRS/LAMA/USMB), Christophe Ménézo (LOCIE/USMB), Sébastien Monnet (LISTIC/USMB), Aude Pommeret (IREGE/USMB), Emilie Planes (LEPMI/USMB), Ioannis Tsanakas (INES/CEA), Monika Woloszyn (LOCIE/USMB), Etienne Wurtz (INES/CEA).

  *Invited speakers*: Matheus Bassani (Universidade Federal do Rio Grande do Sul, Brasil), David Martineau (Solaronix, Switzerland), Victoria Timchenko (University of New South Wales, Sydney, Australia), Sadok Bendkhil (Dracula Technologies, France), Solenn Berson (CEA, France), Romain Cariou (CEA, France), Louis De Fontenelle (Université de Pau et des Pays de l'Adour, France), Philippe Jacques (Université Savoie Mont Blanc, France), Laurent Vuillon (Université Savoie Mont Blanc, France).
- Member of the organizing committee of the "Second Eurasian Risk-2020 Conference, Symposium and Spring school", 12 – 19 April 2020, Tbilisi, Georgia
  http://www.eurasianrisk2020.ge/
- Programme committee member of "the 1st International Workshop on Information, Computation, and Control Systems for Distributed Environments" (ICCS-DE), July 8 – 9 2019, Irkutsk, Russia
  https://iccs-de.icc.ru/ws2019.php
- Program committee member and co-organizer of the WIN–2014 "*Wave interactions*" workshop, 23–26 April 2014, Linz, Austria (with C.C. Mei, E. Pelinovsky, E. Kartashova and M. Onorato)

  *List of invited speakers*: Shalva Amiranashvili, Lushuai Cao, Amin Chabchoub, Walter Craig, Antonio Degasperis, Ira Didenkulova, Eric Falcon, Roger Grimshaw, Zaher Hani, Timothée Jamin, Shijun Liao, Kiori Obusé, Miguel Onorato, Efim Pelinovsky, Davide Proment, Stephane Randoux, Lev Shemer, Victor Shrira, Alexey Slunyaev, Pierre Suret, Tatiana Talipova, Elena Tobisch, Takuji Waseda
- Scientific Committee member of the Conference "*Finite Volumes for Complex Applications VII*", 16 – 20 June 2014, Berlin, Germany
  http://www.wias-berlin.de/fvca7/
- Member of the Organizing Committee of the Program "*The Mathematics of Oceans*", May – June 2013, The Fields Institute, Toronto, Canada (along with W. Craig, D. Henderson, K. Lamb, M. Onorato, E. Pelinovsky, H. Segur and C. Sulem)

  *List of participants:* More than 110 persons. The complete list is available here: http://www.fields.utoronto.ca/programs/scientific/12-13/mathofoceans/participants.html
- Co-organisation with Paul Milewski (University of Bath) of the Workshop "*Mathematical modeling and analysis of extreme sea waves*" at Fondation des Treilles, France, 8 – 13 October 2012.

*List of invited speakers*: Ricardo Barros, Oliver Bühler, Wooyoung Choi, Didier Clamond, Frédéric Dias, Angel Durán, Denys Dutykh, Francesco Fedele, Serge Guillas, Christian Kharif, Chiang C. Mei, Paul Milewski, Marie Nguyen, Themistoklis Stefanakis, Esteban Tabak, Jon Wilkening.

- Scientific Committee member of the Conference "*Finite Volumes for Complex Applications VI*", 6 – 10 June 2011, Prague, Czech Republic
  http://fvca6.fs.cvut.cz/
- Co-organisation of the Workshop *MathOcéan* held at LAMA, University of Savoie, 31 January – 1 February 2011

  *List of participants*: Céline Acary-Robert, Ricardo Barros, Philippe Bonneton, Afaf Bouharguane, Christian Bourdarias, Didier Bresch, Mathieu Cathala, Frderic Charve, Florent Chazel, Anne-Laure Dalibard, Thierry Dauxois, Laurent Debreu, Jérémie Demange, Denys Dutykh, Mehmet Ersoy, Stéphane Gerbi, Marguerite Gisclon, Boris Haspot, Christophe Lacave, David Lannes, Vincent Legat, Yong Lu, Carine Lucas, Fabien Marche, Pascal Noble, Jean Rajchenbach, Miguel Rodrigues, Antoine Rousseau, Chantal Staquet, Benjamin Texier, Marion Tissier, Jean Zabsonré
- Co-organisation (with Didier Bresch and Marguerite Gisclon) of the session entitled "*Numerical models and methods for compressible and two-phase flows*" at the Wolfgang Pauli Institute (Vienna, Austria), 17 – 21 May 2010
  http://www.denys-dutykh.com/wpi10/

  *List of invited speakers*: Médéric Argentina, Marx Chhay, Catherine Choquet, Didier Clamond, Denys Dutykh, Ahmed Ossama Ghanem, Marguerite Gisclon, Theodoros Katsaounis, Valery Liapidevskii, Dimitrios Mitsotakis, Jean Rajchenbach, Jean-Claude Saut
- Co-organisation (with Didier Bresch and Céline Acary-Robert) of the session entitled "*Numerical methods for complex fluid flows*" at Wolfgang Pauli Institute (Vienna, Austria), 21 – 25 September 2009
  http://www.denys-dutykh.com/wpi09/

  *List of invited speakers*: Céline Acary-Robert, Médéric Argentina, Marx Chhay, Didier Clamond, Vassilios Dougalis, Denys Dutykh, Marc Francius, Marguerite Gisclon, Theodoros Katsaounis, Paul Milewski, Dimitrios Mitsotakis, Jean-Claude Saut
- Atelier Cargèse: "*Modélisation physico–numérique pour les fluides, les particules et le rayonnement. Confrontation modèles physiques et modèles numériques*". Institut d'Etudes Scientifique de Cargèse, Corsica, France, 24 – 30 September 2006

  *List of participants*: Céline Baranger, Daniel Bouche, Barbara Bouffandeau, Jean-Philippe Braeunig, Michel Brochard, Christophe Buet, Gilles Carre, Frédéric Chardard, Alain Decoster, Benoît Desjardins, Laurent Desvillettes, Florian De Vuyst, Frédéric Dias, Denys Dutykh, Cédric Énaux, Christophe Fochesato, Jean-Michel Ghidaglia, Laurence Gozalo, Olivier Heuzé, Gilles Kluth, Kim-Claire Le Thanh, Antoine Llor, Julien Mathiaud, Jérôme Metral, Michaël Montout, Hai Yen Nguyen, Frédéric Pascal, Thierry Pougeard-Dulimbert, Olivier Poujade, Agnès Pujols, Bernard Rebourcet, Motte Renaud, Jean-Michel Rovarch, Gérald Samba, Muriel Sesques, Vincent Siess

## 8.11 Research projects

### 8.11.1 Faculty Start-Up Grants

**February 2023 – January 2025** Project title: "*Nonlinear waves in geophysics and biomechanics*", project number: FSU-2023-014. Funded by the Khalifa University of Science and Technology. Total cost: **1 426 232.49 AED**

### 8.11.2 ANR projects

ANR = Agence Nationale de la Recherche

- L'École Universitaire de Recherche (EUR) "**Solar Academy**" created at the University Savoie Mont Blanc in the framework of the "Programme d'Investissement d'Avenir" (PIA3). Project leader: Prof. Monika Woloszyn (LOCIE/USMB)
- Project **FRAISE** (2016 – 2020): "*Absorbent falling film with free-surface instabilities: exploration*". Principal investigator: C. Ruyer-Quil (LOCIE, Polytech Annecy–hambéry and Université Savoie Mont Blanc, Chambéry, France)
- Project **MathOcéan** (2009 – 2012): "*Analyse mathématique en océanographie et applications*". Principal investigator: D. Lannes (DMA, ENS Paris, France)
- Project **HEXECO** (2007 – 2010): "*Hydrodynamique extrême du large à la côte*". Principal investigator: O. Kimmoun (Ecole Centrale Marseille, France)

### 8.11.3 International cooperation projects

- **AAP USMB** (2020) of the University Savoie Mont Blanc: "*Genesis and propagation of a tsunami wave on an accretionary prism*". Co-PI: Prof. Emeritus Christian Beck (ISTerre, USMB). Cooperation with Prof. Valery Liapidevskii from the Lavrentyev Institute of Hydrodynamics, SB RAS, Novosibirsk, Russia.
- **AAP USMB** (2018) of the University Savoie Mont Blanc: "*Nonlinear waves on adaptive moving meshes*". Cooperation with the Victoria University of Wellington (New Zealand) and the University of Valladolid (Spain)
- **Partnership Hubert Curien (PHC) – Parrot 2017** (French – Estonian cooperation). Project title: "The effect of beach roughness on sea wave run-up". Partner: Tallinn University of Technology and Marine Systems Institute, Tallinn, Estonia. French PI: D. Dutykh (LAMA, University Savoie Mont Blanc), Estonian PI: I. Didenkulova (Marine Systems Institute, TalTech, Estonia)
- **LEA Math Mode** (Laboratoire Européen Associé CNRS Franco–Roumain Mathé-matiques et Modélisation) (2015 – 2016) project "*A variational approach to water waves in shallow waters*". Cooperation with Dr. Delia Ionescu-Kruse (IMAR, Bucharest, Romania)
- **AAP Montagne** (2016) of the University Savoie Mont Blanc: "*Modelling and simulation of sliding masses*". Cooperation with Prof. Valery Liapidevskii from the Lavrentyev Institute of Hydrodynamics, SB RAS, Novosibirsk, Russia
- French–Russian cooperative project (**Convention d'échange**) N° EDC26179 (2014 – 2015) "*Interaction of waves with obstacles*". Cooperation with Prof. G. Khakimzyanov (Institute of Computational Technologies, SB RAS, Novosibirsk, Russia)
- Project **PICS CNRS** (2010 – 2012) "*Numerical simulation of higly nonlinear water waves*". Cooperation with the Institute of Computational Technologies, Siberian Branch of Russian Academy of Sciences

and Novosibirsk State University. French leader: D. Dutykh (LAMA, University of Savoie), Russian leader: Yu. Shokin (academician, director of Institute of Computational Technologies)

- **Partnership Hubert Curien (PHC) – ULYSSES 2010** (French – Irish cooperation). Project title: "*Numerical Models for Compressible and Incompressible Flows and Applications*". Partners: School of Mathematical Sciences (University College Dublin), CMLA (ENS de Cachan) and LAMA, University of Savoie. French leader: D. Dutykh (LAMA, University of Savoie), Irish leader: T. Cox (School of Mathematical Sciences, University College Dublin)
- **CNRS/Russian Academy of Sciences exchange program** (2009 – 2011). Project title "*Analytical and numerical solutions for the models of powder-snow avalanches*". French leader: D. Dutykh (LAMA, University of Savoie), Russian partner leader: V. Liapidevskii (Lavrentyev Institute of Hydrodynamics, Novosibirsk)

### 8.11.4 Other projects

- Project **PEPS CNRS Énergie** (2017) "*Innovative numerical methods for more energetically efficient buildings*". Partners: LOCIE UMR 5271 (Polytech Annecy–Chambéry). Project leader: D. Dutykh (LAMA, University Savoie Mont Blanc)
- Project **PEPS CNRS InPhyNiTi** (INSMI/INP) (2014 – 2015) "*Faraday instability in the Hele-Shaw cell*". Partners: Laboratory J.-A. Dieudonné (LJAD, University of Nice Sophia Antipolis), Laboratory of the Condensed Matter Physics (LPMC, University of Nice Sophia Antipolis), Laboratory of Mathematics (LAMA, University Savoie Mont Blanc). Project leader: D. Dutykh (LAMA, University Savoie Mont Blanc)
- Project **PEPS CNRS** (INP) (2010 – 2011) "*Numerical simulation of nonlinear waves in variable medium*". Partners: Laboratory J.-A. Dieudonné (LJAD, University of Nice Sophia Antipolis), Laboratory of the Condensed Matter Physics (LPMC, University of Nice Sophia Antipolis), Laboratory of Mathematics (LAMA, University of Savoie). Project leader: D. Dutykh (LAMA, University of Savoie)
- Contract with région Rhône-Alpes (Cluster Environnement): "*Numerical simulation of snow avalanches*" (2009 – 2010)
- Project **PEPS CNRS** (INS2I) (2009 – 2010) "*PML, l'arithmétique et le calcul: vers l'arithmétique et le calcul numérique efficace et élégamment certifié*". Partners: teams LIMD and EDPs[2] of LAMA, University of Savoie. Project leader: C. Raffalli (LAMA, University of Savoie)

# 9 Teaching and supervision activities

## 9.1 Teaching

**Summer 2026** Instructor of one section of the Calculus III (MATH231, 3 credits) class after J. Stewart's book at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Spring 2026** Instructor of one section of the graduate course "Introduction to Spectral Methods" (MATH794, 3 credits) at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2025** Research Seminar II Graduate Course (SCIE703, 0 credits) at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2025** Instructor for Calculus III (MATH231, 3 credits) after J. Stewart's book at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Summer 2025** Instructor of the Calculus III (MATH231) class (3 + 3 credits; after J. Stewart's book) at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Spring 2025** Instructor for Calculus III (MATH231, 3 credits) and Engineering Mathematics (MATH232, 3 credits) classes after J. Stewart's and E. Kreyszig books at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2024** Instructor for Calculus III (MATH231, 3 credits) and Engineering Mathematics (MATH232, 3 credits) classes after J. Stewart's and E. Kreyszig books at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Summer 2024** Instructor for Calculus III (MATH231, 3 credits) and Engineering Mathematics (MATH232, 3 credits) classes after J. Stewart's and E. Kreyszig books at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Spring 2024** Instructor of the Calculus III (MATH231) class (3 credits; after J. Stewart's book) at the College of Arts and Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2023** Instructor and coordinator of the Calculus III (MATH231) class ($3+3$ credits; after J. Stewart's book) at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Summer 2023** Instructor and coordinator of the Calculus III (MATH231) class ($3+3$ credits; after J. Stewart's book) at the College of Computing and Mathematical Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Spring 2023** Instructor and coordinator of the Calculus III (MATH231) class ($3+3$ credits; after J. Stewart's book) at the College of Arts and Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2022** Instructor of the Calculus III (MATH231) class ($3+3$ credits; after J. Stewart's book) at the College of Arts and Sciences, Khalifa University of Science and Technology, Abu Dhabi, UAE

**Fall 2021** Practical sessions (16h) of the class "Applied Analysis" taught by Prof. Marguerite GISCLON for the first year Master degree students (M1) in Applied Mathematics at the University Savoie Mont Blanc

**Fall 2020** "Introduction to C++" (INFO701) course at the Department of Computer Science, University Savoie Mont Blanc (32h: Lectures and practical sessions for the first year Master degree students (M1) in Applied Mathematics)

**Fall 2019** "Programming in MATLAB™" (INFO701_MATH, M1 level) at the Department of Computer Science, University Savoie Mont Blanc (32h: Lectures and practical sessions for the first year Master degree students (M1) in Applied Mathematics)

**Fall 2018** "Mathematical tools — III" (MATH302_MPC, L2 level) at the Department of Mathematics, University Savoie Mont Blanc (47h: Lectures, practical sessions and final examination for second year students in Mathematics, Physics and Chemistry). Approximate course programme:

- Functions of many variables
- Vector calculus
- Fundamental theorems of integral calculus
- Curvilinear coordinates
- Differential operators in orthogonal non-Cartesian coordinate systems

**Fall 2018** "Programming in C++" (INFO901_MATH, M2 level) at the Department of Computer Science, University Savoie Mont Blanc (32h: Lectures and practical sessions for the second year Master degree students (M2) in Applied Mathematics)

**Fall 2018** "Programming in MATLAB™" (INFO701_MATH, M1 level) at the Department of Computer Science, University Savoie Mont Blanc (32h: Lectures and practical sessions for the first year Master degree students (M1) in Applied Mathematics)

**Fall 2017** "Mathematical tools — III" (MATH302_MPC, L2 level) at the Department of Mathematics, University Savoie Mont Blanc (47h: Lectures, practical sessions and final examination for second year students in Mathematics, Physics and Chemistry). Approximate course programme:

- Functions of many variables
- Vector calculus
- Fundamental theorems of integral calculus
- Curvilinear coordinates
- Differential operators in orthogonal non-Cartesian coordinate systems

**Fall 2017** "Programming in MATLAB™" (INFO701_MATH, M1 level) at the Department of Computer Science, University Savoie Mont Blanc (32h: Lectures and practical sessions for the first year Master degree students (M1) in Applied Mathematics)

**April 2016** Short course (8h) on Spectral methods at the PhD School on Numerical Methods for Diffusion Phenomena, Pontifical Catholic University of Parana, Curitiba, Brazil

- Lecture notes: https://cel.archives-ouvertes.fr/cel-01256472/

**April 2015** Short course (8h of Lectures) on "*A short introduction to Fluid Dynamics*", Basque Center for Applied Mathematics (BCAM). Course programme:

1. Review of (exterior) vector calculus
2. Eulerian description of fluid flows
3. Lagrangian description of fluids
4. Smoothed Particle Hydrodynamics
5. Lecture notes: https://github.com/dutykh/hydro/

**December 2014** Short course (20h of Lectures + 15h of TDs) on "*Lagrangian and Eulerian approaches to water wave modelling*", Faculty of Mechanics and Mathematics, Al-Farabi Kazakh National University, Almaty, Kazakhstan.

- Lecture notes: https://github.com/dutykh/hydro/

**May 2013** Short course on "*Numerical methods for fully nonlinear free surface water waves*", Fields Institute, Thematic Program on the Mathematics of Oceans, Toronto, Canada (4h)

- Slides: http://cel.archives-ouvertes.fr/cel-00825492/
- Videos: http://www.fields.utoronto.ca/video-archive/event/223/2013

**2009 – 2010** Part-time tutor at the University of Savoie (16 hours)

- Practical classes for the 3rd year Math students on: "*Numerical solution of ODEs*"

**2007 – 2008** Teaching assistant at the Department of Mathematics, École Normale Supérieure de Cachan (64 hours)

- Students preparation to the national civil service competitive examination "Agrégation", option "Scientific computing and modeling"
- Practical classes under Matlab for the course "Numerical methods and Scientific Computing"
- Commission member for oral trial examinations for the "Agrégation"

**2006 – 2007** Teaching assistant at the Department of Mathematics, École Normale Supérieure de Cachan (64 hours)

- Students preparation to the national civil service competitive examination "Agrégation", option "Scientific computing and modeling"
- Practical classes under Matlab for the course "Numerical methods and Scientific Computing"
- Commission member for oral trial examinations for the "Agrégation"

**2005 – 2006** Teaching assistant at the Department of Mathematics, École Normale Supérieure de Cachan (64 hours)

- Preparing students to the national civil service competitive examination "Agrégation", option "Scientific computing and modeling"
- Practical classes under Matlab for the course "Numerical methods and Scientific Computing"
- Practical classes under Matlab for the course "Optimization"
- Commission member for oral trial examinations for the "Agrégation"

## 9.2 Organization of teaching activities

**4 – 5 November 2019** Organization of a mini-course (6h) entitled "*Introduction to derived geometries*" by Dr. Jacob Kryczka (LAREMA, University of Angers, France) delivered in LAMA UMR 5127, University Savoie Mont Blanc. **Audience**: researchers and PhD students.

**June – July 2019** Organization of a mini-course (8h) entitled "*Convergence acceleration methods*" by Prof. Angel Durán (University of Valladolid, Spain) delivered in LAMA UMR 5127, University Savoie Mont Blanc. **Audience**: researchers, PhD and Master students.

## 9.3 Students supervision

### 9.3.1 Post-docs

- Dr. **Ahmad Deeb** (July 2023 – February 2026). Research topic: "*Globalized numerical integrators in time using the resummation of divergent series*". Funded by KU FSU grant.
- Dr. **Muhammad Asjad** (July 2023 – May 2024). Research topic: "*Quantum optics and quantum information theory*". Funded by KU FSU grant.

- Dr. **Amin Rashidi** (March 2021 – March 2022). Research topic: "*Genesis and propagation of a tsunami wave on an accretionary prism*". Co-supervision with Prof. Emeritus Christian Beck (ISTerre, USMB). Financed by USMB.
- Dr. **Julien Berger** (April 2016 – January 2017): Co-supervision with N. Mendes (PUCPR, Curitiba, Brasil). Research topic: "*Techniques de réduction de modèle pour la résolution de problèmes en physique du bâtiment*". Financed by CAPES. Currently working at Laboratoire des Sciences de l'Ingénieur pour l'Environnement (LaSIE) — UMR CNRS 7356, Université La Rochelle, France
- Dr. **Claudio Viotti** (September 2012 – August 2013): Co-supervision with F. Dias, ERC MULTIWAVE Project funding. Research topic: "*Breathers under the Dysthe and full Euler dynamics*". Currently working as a software engineer at Miravex, Dublin, Ireland
- Dr. **Francesco Carbone** (September 2012 – August 2013): Co-supervision with F. Dias, ERC MULTIWAVE Project funding. Research topic: "*Wave focussing effect in various physical systems*". Currently working at CNR, IIA, Italy.

### 9.3.2 PhD students

- (2024 – 2027) **Mark Essa Sukaiti**: Co-supervision with Prof. Davide Batic (Khalifa University). PhD thesis title: "*Spinning manifolds: the effects of rotation in general relativity*". Defence is expected in June 2027.
- (2022 – 2026) **Srijani Mukherjee**: Co-supervision with Dr. Ioannis Tsanakas (CEA/INES) and Prof. Laurent Vuillon (University Savoie Mont Blanc). PhD thesis title: "*Advanced diagnostics of photovoltaic plants by data fusion and image processing*". Defended on the 9$^{th}$ of September 2026.
- (2022 – 2025) **Muhammad Naveed Zafar**: Co-supervision with Prof. Pierre Sabatier (EDYTEM/University Savoie Mont Blanc). PhD thesis title: "*Numerical modelling of lacustrine paleotsunamis constrained by their geological records*". Defended on the 25$^{th}$ of November 2025.
- (2022 – 2025) **Florian Gallois**: Co-supervision with Dr. Mikaël Cugnet (CEA/INES) and Angel Kirchev (CEA/INES). PhD thesis title: "*Online diagnosis of the states of charge (SOC) and health (SOH) for lead-acid batteries*". Defended on the 11$^{th}$ of June 2025.
- (2021 – 2026) **Noura Al Akkari**: Co-supervision with Dr. Sylvain Lespinats (CEA Grenoble/INES) and Dr. Aurélie Foucquier (CEA Grenoble/INES). PhD thesis title: "*Supervised empirical decomposition of time signals of power consumption*". Defended on the 10$^{th}$ of July 2026.
- (2021 – 2026) **Sorina Mustatea**: Co-supervision with Dr. Sylvain Lespinats (CEA Grenoble/INES). PhD thesis title: "*Diagnostic and prognostic tools for inverters and PV modules using machine learning approaches*". Defended on the 10$^{th}$ of September 2026.
- (2020 – 2024) **Rim El Cheikh**: Co-supervision with Prof. Stéphane Gerbi (LAMA/USMB). PhD thesis title: "*Mathematical modelling of nonlinear waves in coastal and biological systems*". Defended on the 25$^{th}$ of December 2024
- (2021 – 2025) **Carlos Rodrigo Cardenas Bravo**: Co-supervision with Dr. Duy Long Ha (CEA Grenoble/INES). PhD thesis title: "*Coupling of Electrical and Thermal Models for the Diagnosis of Photovoltaic Modules*". Defended on the 25$^{th}$ of September 2025.
- (2018 – 2023) **Zhanat Karashbayeva**: Co-supervision with Prof. Bolatbek Rysbayuly and Prof. Abilmazhin Adamov (L.N. Gumilyov Eurasian National University, Astana, Kazakhstan). PhD thesis title: "*Development of methods for solving some coefficient-inverse problems of heat and mass transfer and computational experiments*". Defended on the 25$^{th}$ of August 2023.

- (2017 – 2021) **Benoît Colange**: Co-supervision with Dr. Sylvain Lespinats (CEA Grenoble/INES). PhD thesis title: "*Diagnostic de systèmes électriques par analyse intelligente de structures de données de grandes dimension*". Defended on the 26$^{th}$ of May 2021.
- (2017 – 2020) **Ahmed Alkarory Ahmed Abdalazeez**: Co-supervision with Prof. Ira Didenkulova (Marine Systems Institute, Tallinn University of Technology, Estonia). PhD thesis title: "*Influence of sea bed bathymetry and coastal topography on statistical characteristics of wave run-up on a beach*". Defended on the 26$^{th}$ November 2020.
- (2015 – 2019) Dr. **Suelen Gasparin**: Co-supervision with Prof. Nathan Mendes (PUCPR, Curitiba, Brasil). PhD thesis title: "*Numerical methods for predicting heat and moisture transfer through porous building materials*". Scholarship CAPES-COFECUB, projet N° 774/2013. Defended on the 3$^{rd}$ of June 2019.
- (2015 – 2018) Dr. **Amin Rashidi**: Co-supervision with Prof. Zaher Hossein Shomali (Institute of Geophysics, University of Tehran, Iran). PhD thesis title: "*Numerical simulation and hazard assessment of the effect of tsunamigenic scenarios on the Western Makran region using structural detection and restoration*". Defended on the 23$^{rd}$ October 2018.
- (2012 – 2017) **Aidar Assylbekuly**: Co-supervision with Prof. Dauren Zhakebaev (Faculty of Mechanics and Mathematics, Al-Farabi Kazakh National University). PhD thesis title: "*Modelling of the multifactor pulsed impact onto a multicomponent liquid*".
- (2007 – 2010) Dr. **Louis Stephan**: Co-supervision with Prof. Etienne Wurtz (INES-LOCIE, University of Savoie). PhD thesis title: "*Modélisation de la ventilation naturelle pour l'optimisation du rafraîchissement passif des bâtiments*". Defended on April 16, 2010.

### 9.3.3 Master 2 students

- **Aseel Alsaid Souliman** (Khalifa University of Science and Technology) (May 2025 – May 2026), subject: "*A framework combining optimal control and reduced order modeling for accelerated blood flow simulations*". Co-supervision with Prof. Aymen Laadhari (Khalifa University)
- **Joud Mohamad Mojahed Faraji** (Khalifa University of Science and Technology) (May 2025 – May 2026), subject: "*Applications of the spectral methods in quantum mechanics*". Co-supervision with Prof. Davide Batic (Khalifa University)
- **Florian Gallois** (University Savoie Mont Blanc) (March 2021 – August 2021), subject: "*Experimental validation of a two-electrode physical model and reduction of computational time vs. the empirical model*". Co-supervision with Dr. Mikaël Cugnet (CEA/INES)
- **Florian Gallois** (University Savoie Mont Blanc) (May 2021 – July 2021), CMI Project on the topic: "*Numerical solution of unsteady advection–diffusion–reaction equations*".
- **Michelle Lee** (University Savoie Mont Blanc) (April 2021 – September 2021), subject: "*Data analysis in the context of solar energy deployment in urban areas*". Co-supervision with Prof. Lamia Berrah (LISTIC/USMB), Dr. Aurélie Foucquier (CEA/INES) & Dr. Martin Thebault (LOCIE/USMB)
- **Yannick Meyapin** (March – July 2010) (University of Savoie), subject: "*Numerical simulation of single-velocity two-phase flows*". Co-supervision with M. Gisclon (LAMA, University of Savoie)
- **Ahmed Ossama Ghanem** (March – July 2010) (University of Haute Alsace), subject: "*Numerical simulation of the Faraday instability*". Co-supervision with M. Gisclon (LAMA, University of Savoie) and J. Rajchenbach (LPMC, University of Nice Sophia-Antipolis)
- **Xavier Gardeil** (March – September 2010) (University of Savoie): co-supervision with C. Beck (LGCA, University of Savoie), subject: "*Tsunami wave modeling at the North of Venezuela*"

- **Yannick Meyapin** (March – June 2009) (University of Savoie): Co-supervision with M. Gisclon (LAMA, University of Savoie), subject: "*Velocity and energy relaxation in two-phase flows*"
- **Youen Kervella** (March – July 2006) (Master 2 Physics of the Ocean and Atmosphere, University of Brest), subject: "*Comparison between linear and nonlinear models of tsunami generation*". Co-supervision with F. Dias

### 9.3.4 Senior Research Projects

- **Joudy Feras Jamal Beek** (Khalifa University ID 100061683) (September 2024 – May 2025), subject: "*A Spectral Approach for Quasi-normal Modes of Non-commutative Wormholes*". Co-supervision with Profs. Davide Batic (Mathematics Department, Khalifa University) and Fedor Kusmartsev (Physics Department, Khalifa University).
- **Zeinabou Ahmed Abou** (Khalifa University ID 100060187) (September 2024 – May 2025), subject: "*Quasinormal modes of noncommutative geometry-inspired dirty black holes*". Co-supervision with Prof. Davide Batic (Mathematics Department, Khalifa University).
- **Maria Benkhelifa** (Khalifa University ID 100059472) (September 2023 – May 2024), subject: "*Pushed-pulled front transitions in tumor growth: continuous modelling*". Co-supervision with Dr. Haralampos Hatzikirou (Mathematics Department, Khalifa University of Science and Technology).
- **Hana Herbawi** (Khalifa University ID 100059412) (September 2023 – May 2024), subject: "*Pushed-pulled front transitions in tumor growth: discrete modelling*". Co-supervision with Dr. Haralampos Hatzikirou (Mathematics Department, Khalifa University of Science and Technology).

### 9.3.5 Work-study students

- **Mathieu Vidal** (University Savoie Mont Blanc — NTN–SNR Group) (September 2022 – September 2023), subject: "*Integration of an optimization procedure into the existing computational tool to determine the preload value minimizing the dissipated power of a rolling*". Co-supervision with Cédric Burnet (NTN–SNR Group, Annecy).
- **Yacine Hedeoud** (University Savoie Mont Blanc — NTN–SNR Group) (September 2021 – September 2022), subject: "*Search of mathematical strategies allowing the reduction of the number of computational case studies*". Co-supervision with Cédric Burnet (NTN–SNR Group, Annecy).

### 9.3.6 Master 1 students

- **Léa Bucher** (University Savoie Mont Blanc) (May – August 2022), subject: "*Modeling of the self-discharge phenomenon of a lead-acid cell*". Co-supervision with Dr. Mikaël Cugnet (CEA/INES)
- **Mathieu Vidal** (University Savoie Mont Blanc) (May – August 2022), subject: "*Integration of an optimization procedure into the existing computational tool to determine the preload value minimizing the dissipated power of a rolling*". Co-supervision with Cédric Burnet (NTN–SNR Group, Annecy)
- **Yacine Hedeoud** (University Savoie Mont Blanc) (May – June 2021), subject: "*Search of mathematical strategies allowing the reduction of the number of computational case studies*". Co-supervision with Cédric Burnet (NTN–SNR Group, Annecy)
- **Florian Gallois** (University Savoie Mont Blanc) (May 2020 – August 2020), subject: "*Mathematical model of a porous lead electrode*". Co-supervision with Dr. Mikaël Cugnet (CEA/INES)
- **Magali Pollet** (University Savoie Mont Blanc) (December 2019 – May 2020), subject: "*Les exposants de Lyapunov dans des systèmes dynamiques à temps continu*"
- **Christoffer Stuart** (Chalmers University of Technology – University Savoie Mont Blanc) (January 2018 – May 2018), subject: "*A critical evaluation of modern constrained optimization solvers*"

- **Zakaria Ait Allal** and **Younès Mahri** (University Savoie Mont Blanc) (January 2018 – May 2018), subject: "*Méthodes variationnelles en mécanique analytique*"
- **Ariane Cotte** (April – July 2013) (École Polytechnique): Co-supervision with F. Dias (UCD), subject: "*Submarine landslide modelling on real-world 3D bathymetries*"
- **Lauranne Pellet** (March – July 2013) (École Centrale Marseille): Co-supervision with F. Dias (UCD), subject: "*Mathematical modelling of underwater microseisms*"
- **Mickaël Roulet** (March – May 2011) (University of Savoie, M1 Mathematics), subject: "*Finite volume schemes for Nonlinear Shallow Water Equations with wetting/drying processes*"
- **Mahmut Tuz** (May – June 2010) (University of Savoie, M1 Physics), subject: "*Numerical computation of the Dirichlet-to-Neumann map*"

### 9.3.7 Other students

- **Sofia Samir** (May – August 2024) (Undergraduate Applied Mathematics and Statistics major, KU ID 100061155): Khalifa University internship in the quality of a Teaching Assistant during Summer 2024 semester.
- **Cissy Boisdur** (January – May 2023) (L3 Maths/Info, University Savoie Mont Blanc): "*La méthode de quadrature de Gauß est-elle meilleure que celle de Clenshaw-Curtis?*"
- **Hugo Da Cunha** (May – June 2021) (ENS Lyon): "*Étude théorique et numérique d'un système couplé pour l'accoustique. Dérivées d'ordre non entier*". Co-supervision with Dr. Hervé Le Meur (CNRS/LAMFA/UPJV)
- **Reham Nassif** (January – June 2021) (CMI Mathématiques, USMB): "*Algorithmes d'approximation numérique de la mesure de Mahler*"
- **Martin Rialhe-Badet** (January – May 2020) (CMI Mathématiques, USMB): "*La théorie du quasi-déterminisme de Boccotti*"
- **Alizée Dubois** (June 2012) (L3 ENS Cachan-Bretagne) (co-supervision with F. Dias): "*Réflexion de la houle contre une paroi / Wave reflexion against a wall*"
- **Ianis Bernard** (March – June 2009) (Classes préparatoires, Nice). Participation in the supervision of a practical personal work (TIPE). Subject: "*Modeling of a hydraulic soliton*"

## 9.4 Habilitation thesis committees

- (referee) Dr. **Mehmet Ersoy**: Thesis title: "*From hydrostatic to non-hydrostatic models in fluid mechanics: modeling, mathematical and numerical analysis, and computational fluid dynamics*", the 1st December 2020, University of Toulon, France, 2020.
  - Composition of the committee: Didier Bresch, Denys Dutykh, Thierry Gallouët, Cédric Galusinski, Raphaële Herbin, Theodoros Katsaounis, Corrado Mascia, Antonin Novotny, Enrique Zuazua.
- (referee) **Vassili A. Gromov**: Dr.Sci. title: "Models, methods, and algorithms of bifurcation theory for nonlinear elliptic equations of von Karman type", October 5, 2017, Oles Honchar Dnipro National University, Dnipro, Ukraine.

## 9.5 PhD thesis committees

- (referee) Dr. **Alice Abbate**: PhD thesis title: "*On the trade-off between accuracy and efficiency in Probabilistic Tsunami Hazard Assessment*", March 28, 2025, University of Trieste, Italy. Advisor: Dr. Stefano Lorito (National Institute of Geophysics and Volcanology, Rome, Italy).

- (examiner) Dr. **Fatima-Zahra MIAMI**: PhD thesis title: "*Development of a phase-resolving computer model for operational nearshore wave assessment*", February 3, 2023, SIAME, Université de Pau et des Pays de l'Adour, France. Advisors: Prof. Volker ROEBER (HPC-Waves Chair, UPPA) and Dr. Denis MORICHON (UPPA).
- (examiner) Dr. **Iskander ABROUG**: PhD thesis title: "*Étude des vagues extrêmes se propageant d'une profondeur intermédiaire vers le rivage*", December 2, 2019, Laboratoire LOMC UMR 6294, Université Le Havre, France. Advisors: Prof. François MARIN (LOMC UMR 6294), Dr. Nizar ABCHA (M2C UMR 6143) and Dr. Armelle JARNO (LOMC UMR 6294).
- (referee) Dr. **Rémi CARMIGNIANI**: PhD thesis title: "*Redresseurs de vagues: vers une nouvelle stratégie d'extraction de l'énergie houlomotrice*", December 14, 2017, Laboratoire d'Hydraulique Saint-Venant, EDF Chatou, France. Advisors: Dr. Damien VIOLEAU (EDF Chatou, France) and Prof. Morteza GHARIB (Caltech, USA).
- (referee) Dr. **Marine LE GAL**: PhD thesis title: "*Influence des échelles de temps sur la dynamique des tsunamis d'origine sismique*", February 17, 2017, Laboratoire d'Hydraulique Saint-Venant, EDF Chatou, France. Advisor: Dr. Damien VIOLEAU.
- (referee) Dr. **Pauline ROBIN**: PhD thesis title: "*Hydrodynamique extrême en mer près des côtes*", July 18, 2013, Institut de Recherche sur les Phénomènes Hors Équilibre (IRPHE), Université de Provence – Aix-Marseille I, France. Advisors: Prof. Christian KHARIF and Dr. Olivier KIMMOUN.
- (examiner) Dr. **Brice EICHWALD**: PhD thesis title: "*Intégrateurs exponentiels modifiés pour la simulation des vagues non linéaires*", July 5, 2013, Laboratoire Dieudonné, Université de Nice Sophia-Antipolis, France. Advisors: Prof. Didier CLAMOND and Dr. Marc FRANCIUS.
- (examiner) Dr. **Georges SADAKA**: PhD thesis title: "*Étude Mathématique et numérique d'équations d'ondes aquatiques amorties*", November 25, 2011, LAMFA, University of Picardie Jules Verne, France. Advisor: Prof. Jean-Paul CHEHAB.
- (examiner) Dr. **Louis STEPHAN**: PhD thesis title: "*Modélisation de la ventilation naturelle pour l'optimisation du rafraîchissement passif des bâtiments*", April 16, 2010, INES-LOCIE, University of Savoie, France. Main advisor: Dr. Etienne WURTZ.
- (examiner) Dr. **Marx CHHAY**: PhD thesis title: "*Intégrateurs géométriques: Application à la Mécanique des Fluides*", December 16, 2008, LEPTIAB, University of La Rochelle, France. PhD advisors: Prof. Aziz HAMDOUNI and Prof. Pierre SAGAUT.

# 10 Responsibilities

## 10.1 Administrative Responsibilities

- Head of the Seminar Committee at Mathematics Department, Khalifa University of Science and Technology (2022 – present). Complete composition of the committee: Erkko LEHTONEN (Math Department, KU), Yerkin KITAPBAYEV (Math Department, KU).

- Member of the selection committee for the Master S3E – Solar Energy (2021 – 2022). Complete composition of the committee: David Bailleul (Centre Antoine Favre, USMB), Lamia Berrah (LISTIC EA 3703, USMB), Dorothée Charlier (IREGE, USMB), Gilles Fraisse (LOCIE UMR 5271, USMB), Christophe Ménézo (LOCIE UMR 5271, USMB), Emilie Planes (LEPMI UMR 5279, USMB), Aude Pommeret (IREGE, USMB), Simon Rouchier (LOCIE UMR 5271, USMB), Christian Ruyer-Quil (LOCIE UMR 5271), Bernard Souyri (LOCIE UMR 5271, USMB), Monika Woloszyn (LOCIE UMR 5271, USMB)
- In charge of communication ("Correspondant communication") with CNRS and USMB at LAMA UMR 5127 (2017 – 2022)
- Member of the consulting committee in Sections 25–26 at the University Savoie Mont Blanc (2015 – 2020)
- Video recording of some seminars taking place in LAMA (provided that the speaker accepts to be taped). (2017 – 2022) Recorded and processed videos are available at https://www.youtube.com/user/dutykh/
- Member of the UFR SFA Board (2009 – 2012)
- Member of the Research Board of the Laboratory of Mathematics LAMA, University of Savoie (2009 – 2012)
- In charge of innovation and knowledge transfer activities (chargé de valorisation) at LAMA (2008 – 2012)
- Representative of LAMA in **Fédération de recherche Vulnérabilité des Ouvrages aux Risques** (VOR) (2008 – 2012)
- Representative of LAMA in **International Center for Applied Computational Mechanics** (ICACM) (2008 – 2012)
- Member of the Hiring Committees in:
  - Applied Mathematics (section 26), Mechanics (section 60), Laboratoire J.-A. Dieudonné, University of Nice Sophia Antipolis
  - Applied Mathematics (section 26), LAMA, University of Savoie Mont Blanc
  - Thermics (section 62), LOCIE, Polytech Annecy Chambéry

## 10.2 Seminars

During past years I actively participated in running following seminars:

**October 2022 – present** Co-organizer and head of the Mathematics Seminar committee at the Khalifa University of Science and Technology, Abu Dhabi, UAE.

**January 2013 – July 2018** Seminar of the team EDPs[2] and Groupe de Discussion (GdD), LAMA, University of Savoie. Participation, invitation of speakers

**October 2008 – July 2012** Seminar of the team EDPs[2], LAMA, University of Savoie. Participation, invitation of speakers

**September 2004 – July 2008** Working group: *Mécanique des Fluides Réels*, Centre de Mathématiques et de Leurs Applications (CMLA), ENS de Cachan

## 10.3 General audience events

- Stand on tsunami waves (together with Professor Christian Beck, LGCA, University of Savoie) at the **Fête de la Science 2009**, Galérie Eureka, Chambéry, France

- General audience lecture: **Tsunamis: du terrain au modèle numérique** (with Professor Christian Beck, LGCA, University of Savoie) in the framework of the *Fête de la Science 2009*, 21 November 2009, Cinéma Curial, Chambéry, France
- Recurrent participation in **Open Doors Days** at the University of Savoie with public lectures on water and tsunami waves

# 11 ♥ Other interests

Sports cycling, skiing, bodybuilding, badminton
Hobbies reading, photography, tourism

# 12 Academic references

| Name and E-mail | Address |
|---|---|
| **Alexey Cheviakov**<br>shevyakov@math.usask.ca | University of Saskatchewan<br>Department of Mathematics and Statistics<br>Room 227 McLean Hall, 106 Wiggins Road<br>Saskatoon, SK, S7N 5E6 Canada |
| **Francesco Fedele**<br>fedele@gatech.edu | Georgia Institute of Technology<br>College of Engineering<br>School of Civil Environmental Engineering<br>Mason Building 3140A, Atlanta, USA |
| **Taras Lakoba**<br>tlakoba@uvm.edu | Department of Mathematics and Statistics<br>College of Engineering and Math. Sciences<br>The University of Vermont<br>Burlington, Vermont, USA |
| **Andrei Ludu**<br>Andrei.Ludu@erau.edu | Mathematics Department<br>Daytona College of Arts & Sciences<br>Embry-Riddle Aeronautical University<br>Daytona Beach, Florida, USA |
| **Dimitrios Mitsotakis**<br>dmitsot@gmail.com | Victoria University of Wellington<br>School of Mathematics and Statistics<br>PO Box 600 Wellington 6140<br>New Zealand |

## 12.1 Supplementary academic references

| Name and E-mail | Address |
|---|---|
| **Leonid Chubarov**<br>chubarov@ict.nsc.ru | Institute of Computational Technologies<br>Siberian Branch of the<br>Russian Academy of Sciences<br>Novosibirsk, Russia |
| **Didier Clamond**<br>didierc@unice.fr | Laboratoire J.-A. Dieudonné<br>Univ. Côte d'Azur, Parc Valrose<br>06108 Nice Cedex 02 France |
| **Olivier Goubet**<br>Olivier.Goubet@univ-lille.fr | Laboratoire Paul Painlevé<br>Université de Lille<br>Faculté des Sciences<br>33 Cité Scientifique<br>59655 Villeneuve d'Ascq Cedex |
| **Gayaz Khakimzyanov**<br>khak@ict.nsc.ru | Institute of Computational Technologies<br>Siberian Branchof the<br>Russian Academy of Sciences<br>Novosibirsk, Russia |
| **Nathan Mendes**<br>Nathan.Mendes@pucpr.br | Thermal Systems Laboratory<br>Pontifical Catholic University of Paraná<br>Rua Imaculada Conceição,<br>1155, CEP: 80215-901<br>Curitiba – Paraná, Brazil |
| **Paul Milewski**<br>P.A.Milewski@bath.ac.uk | Dept. of Mathematical Sciences<br>Univ. of Bath<br>Bath, BA2 7JX UK |